From Agriculture to Services: Development without Industrialization

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Abstract Does industrialization leave a lasting legacy after manufacturing declines? I argue that it does. Large manufacturing firms create demand for managerial and technical knowledge that remains valuable as economies transition toward services. I study this mechanism using cross-country data, U.S. county evidence from 1850 to 2000, and a quantitative model of structural transformation with endogenous skill formation and growth. Across countries, larger historical manufacturing peaks are associated with more skill-intensive service sectors and faster subsequent growth. Using a shift-share identification strategy across U.S. counties, I show that the lasting effects on service composition are driven by manufacturing industries characterized by large-scale production and intensive coordination demands. The calibrated model reproduces 98% of the observed cross-country relationship between historical manufacturing peaks and high-skill producer services, and 40% of the corresponding relationship with GDP per capita growth rates. Because manufacturing's lasting effects operate through the organizational knowledge created by production at scale rather than manufacturing employment itself, manufacturing subsidies are a poorly targeted way to recreate this legacy. Lasting gains instead arise when skill formation is combined with reforms that allow firms to grow and improve how production is organized.

Keywords: structural transformation, industrialization, firm scale, organizational capital, service-led growth.

JEL Classification: O14, O41, O33, O15, J24.

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1 Introduction

For most of economic history, development meant industrialization. Labor moved out of agriculture, manufacturing expanded, and services became dominant only after a substantial industrial phase.[1] That sequence is no longer typical. Many developing economies now move directly from agriculture into services, reaching lower manufacturing-employment peaks and deindustrializing at lower income levels than earlier industrializers (Rodrik, 2016). This raises a natural question: does a service economy that emerges directly from agriculture have the same foundations for sustained development as one that follows industrialization?

I argue that the answer depends on whether the development path passes through production at sufficient scale to make investment in organization worthwhile. Historically, industrialization concentrated production in large establishments, where coordinating workers, inputs, and information became an important part of production itself (Pollard, 1965; Chandler, 1990). Large firms responded by employing specialized managerial and technical workers and developing systematic solutions to recurring organizational problems. I refer to these functions as producer services and to the skilled workers who perform them as high-skill producer-service workers (HSPS). Because problems such as controlling costs, managing inventories, processing information, and assigning workers to tasks recur across sectors outside manufacturing, the knowledge developed to solve them can remain useful as the economy moves into services. Many foundational management practices were developed in and around large industrial firms, including scientific management at Midvale and Bethlehem Steel, return-on-investment accounting at DuPont, the multidivisional organizational structure at General Motors, and statistical process control at Western Electric. These methods later diffused across firms and sectors, often through managers, consultants, and other external sources of organizational knowledge (McKenna, 2006; Mol and Birkinshaw, 2014). The main argument of this paper is therefore that large-scale manufacturing can leave a lasting legacy by creating organizational knowledge that remains valuable as the economy moves toward services.

I provide evidence for this mechanism in two steps. First, across countries since 1990, larger historical manufacturing peaks predict a service transition tilted toward high-skill producer services rather than trade and other consumer-facing services, and faster subsequent GDP-per-capita growth even after manufacturing has declined. Second, using U.S. county data from 1850 to 2000, I show that counties with larger manufacturing peaks accumulated more high-skill producer-service employment within manufacturing and later developed service sectors more specialized in high-skill producer-service industries. A natural concern is that manufacturing simply made counties larger and more urban, independently generating subsequent demand for high-skill producer services. To address this concern, I draw on the historical distinction between high-throughput industries, in which large-scale plants and intensive coordination were technologically important, and low-throughput industries that could operate at smaller scale (Chandler, 1990). While both types of manufacturing predict population growth and urbanization, only high-throughput manufacturing predicts subsequent specialization in HSPS services. I then construct shift-share instruments that interact counties' 1880 manufacturing-industry composition with subsequent leave-county-out national industry growth. In the preferred IV specification, a ten-percentage-point increase in peak manufacturing employment induced by high-throughput national

1 See, among others, Kuznets (1973), Clark (2007), Gollin et al. (2002), and Herrendorf et al. (2014) for classic and modern accounts of structural transformation and economic development.

growth raises the high-skill producer-service share of private-service employment in 2000 by 2.83 percentage points, or about 45% of the cross-county average. By contrast, an instrument pooling national growth across all manufacturing industries also strongly predicts the manufacturing peak but produces no statistically detectable long-run effect on service composition.

The empirical evidence links large-scale manufacturing to the composition of services, but it cannot quantify how much this mechanism matters for aggregate development, how it propagates through skill accumulation and growth, or what would happen under alternative development paths. I therefore embed the mechanism in a quantitative model of structural change with endogenous skill formation and growth. Following Comin et al. (2021) I build a three-sector model where non-homothetic demand shifts expenditures from agriculture to manufacturing, and eventually to services as income rises. Starting with this framework I make two additions. First, within each sector, heterogeneous firms enter endogenously and operate under decreasing returns, generating variation in firm size across producers and sectors. Second, I introduce a scale-dependent organizational margin by letting the frequency with which firms encounter recurring organizational problems rise with employment. Each problem can be handled repeatedly as it arises or addressed systematically by paying a fixed skilled-labor cost to develop a reusable procedure (Chandler, 1990). Because developing a procedure entails a fixed cost while its value rises with how often it is used, larger firms develop more procedures and demand more high-skill producer-service labor. Newly developed procedures are initially proprietary but can later diffuse into a public stock of organizational knowledge, lowering the cost of organizational investment for firms in other sectors. A manufacturing phase can therefore leave the subsequent service economy with organizational knowledge that persists after manufacturing employment declines. Finally, I close the model by linking the organizational mechanism to long-run productivity growth. New cohorts choose whether to acquire skill in response to the skill premium, so greater demand for high-skill producer services raises the future skilled share. Productivity growth, in turn, rises with the mass of skilled workers, following the basic logic of Jones (1995). Because long-run growth would otherwise remain tied to an exogenous population path, on the household side I adopt the quantity–quality mechanism of Galor and Weil (2000) and Lagerlöf (2006) to stabilize growth and population. Faster technological change raises the return to educating children and lowers fertility. Education also becomes less costly as the economy becomes more skilled, linking the long-run skilled share to the demographic equilibrium. These feedbacks drive fertility to replacement and population to a finite endogenous level, leaving a positive long-run growth rate determined by the limiting skilled share rather than by exogenous population growth.

I calibrate the model to the U.S. transition from 1850 to 2019, initializing the economy in 1790 and disciplining structural change, firm scale, skill composition, producer services, fertility, and productivity growth. I then use the model to ask how economies with different peak manufacturing employment shares diverge in their subsequent development paths. The model reproduces 98 percent of the observed cross-country relationship between peak manufacturing employment share and the contemporary HSPS share of services, and 40 percent of the corresponding relationship with GDP-per-capita growth. Through the lens of the model, economies that pass through a larger manufacturing phase enter the service economy with more organizational knowledge, a more skill-intensive workforce, and a higher long-run growth rate. As an external validation of the knowledge-creation mechanism, I compare the model-implied annual flow of new organizational knowledge with the historical publication of books and manuals on business organization and management. The two series correlate 0.75 over 1906–1977, substantially more strongly than the model-implied knowledge flow correlates with 11 placebo publication categories.

I then use the model to ask which policies can reproduce these persistent effects in an economy that has not undergone a large manufacturing phase. I organize the experiments around the supply of skills and the firm-scale conditions that generate demand for them. I calibrate the model to representative lower-middle-income-country conditions and explore the persistent effects of policies which expand skills, remove barriers to firm growth, support the adoption of better management practices, or a combination of these instruments. Temporary support for skill formation reproduces the large contemporaneous reallocation toward organized employers documented in recent work (Cox, 2025; Engbom et al., 2025), but much of this response is transitory: with barriers to firm-growth in place, skilled employment is 6.58 percentage points above the no-policy path in the policy window and only 0.51 percentage points higher after the policy is removed. The picture changes when temporary support for skill formation and organizational investment is combined with the removal of distortions. With discounted fiscal expenditure of about 0.6 percent of discounted GDP during the intervention, the policy raises GDP per capita by 22.4 percent and leaves the skilled share 4.77 percentage points higher decades after the policy is withdrawn. In doing so, it creates a lasting stock of organizational knowledge that sustains skill demand and persistently raises the economy's growth rate. By contrast, temporary manufacturing wage subsidies leave essentially no organizational legacy because manufacturing employment is not itself the source of the externality. The policy implications therefore highlight that temporary skill expansions are most likely to have lasting effects when firms respond by creating organizational knowledge that persists, diffuses, and sustains skill demand by future firms.

The paper is most closely related to recent work connecting firm scale, skills, and the organization of production (Engbom et al., 2025; Cox, 2025; Gottlieb et al., 2025). In particular, Engbom et al. (2025) also build directly on Chandler's account of the modern enterprise, emphasizing how the supply of educated white-collar workers enables firms to delegate, reorganize, and grow. My mechanism runs in the complementary direction: large-scale production creates demand for managerial and technical workers who solve recurring organizational problems systematically. I then study the long-run consequence of this process: whether the organizational routines and specialized expertise developed inside manufacturing persist and support the emergence of producer services after manufacturing declines. The paper is also closely related to Peters et al. (2026), who study fast-growing developing economies that have expanded services without undergoing broad industrialization and argue that productivity growth in consumer services can sustain such a development path. I instead ask whether a substantial manufacturing phase changes the service economy that subsequently emerges, and show that large-scale manufacturing promotes high-skill producer services with persistent consequences for skill demand and growth.

More broadly, the paper connects structural transformation to firm organization. Existing work studies the forces behind sectoral reallocation and the changing role of manufacturing in development (Rodrik, 2016; Comin et al., 2021; Huneeus and Rogerson, 2024; Fujiwara and Matsuyama, 2024; Sposi et al., 2026); I instead ask whether the manufacturing phase itself changes the service economy that follows. On the firm side, I build on models of organization capital (Prescott and Visscher, 1980; Atkeson and Kehoe, 2005) by making the creation and diffusion of organizational knowledge endogenous to firm scale. The policy experiments also relate to recent evidence that temporary industrial interventions can have persistent effects (Juhász, 2018; Bustos et al., 2019; Incoronato and Lattanzio, 2025; Choi and Levchenko, 2025); here, persistence operates through organizational knowledge that remains valuable after manufacturing declines.

The remainder of the paper proceeds as follows. Section 2 presents the evidence, Section 3 develops the model, Section 4 describes its calibration and historical validation, and Section 5 reports the quantitative results. Section 6 studies policy implications and Section 7 concludes.

2 Empirical Evidence

This section presents the paper's empirical evidence in two complementary steps. First, I document two broad cross-country patterns: economies with larger historical manufacturing peaks develop service sectors more concentrated in high-skill producer services and grow faster, even decades after manufacturing has peaked. Second, I use U.S. counties from 1850 to 2000 to ask whether this legacy is tied specifically to the scale and organizational demands of industrialization rather than to a broader development pattern such as rising incomes or agglomeration driven by industrialization. Following Chandler (1990), I distinguish high-throughput industries—such as steel, chemicals, machinery, and processed foods—in which achieving low unit costs required large plants operating at high volume, from traditional industries—such as textiles, apparel, wood, and furniture—that could operate efficiently at smaller scale.[2] Both forms of industrialization predict population growth and urbanization, but only high-throughput manufacturing predicts the buildup of HSPS employment within manufacturing and, later, specialization in HSPS services. This differential response helps distinguish the organizational mechanism from broader consequences of industrialization, such as population growth and urbanization. A Bartik design based on counties' 1880 industry composition and national industry growth leads to the same conclusion: national manufacturing growth strongly predicts local manufacturing peaks, but a statistically detectable legacy for service composition emerges only when that growth is concentrated in high-throughput industries.

2.1 Cross-Country Facts

The two exercises use complementary country samples. The service-composition analysis uses the GGDC/UNU–WIDER Economic Transformation Database (ETD), which reports annual employment in 12 broad industries for 51 economies from 1990 through 2018. Motivated by Chandler's account of the managerial, technical, and information-processing functions that expanded with the modern industrial enterprise, I define high-skill producer-service (HSPS) industries as business services, finance, and real estate, the closest available ETD categories supplying these functions to firms. Generic market services comprise trade and other services outside the public sector. For these 51 economies, historical manufacturing peaks are constructed from the full available GGDC 10-Sector and ETD employment histories, with ETD taking precedence in overlapping years.[3] The growth exercise uses a broader 173-country sample, with service employment from ILOSTAT and GDP per capita from the World Bank via Gapminder. Appendix B describes the data, historical manufacturing series, concordances, and sample construction in detail.

***FACT 1**: Larger historical manufacturing peaks are associated with greater expansion of high-skill producer-service employment.*

2 The high- and low-throughput distinction follows Chandler's historical taxonomy of production technologies; the mapping to IND1950 industries is constructed independently of county outcomes. Appendix A.3 reports the complete concordance, the corresponding Chandler references, and the classification rationale for each industry.

3 The GGDC 10-Sector Database extends the historical employment series for 32 of the 51 economies; the remaining 19 use their full available ETD histories.

I begin by comparing service-employment trajectories across economies with different industrial histories. I split the 51-country ETD sample at the median historical manufacturing peak and compare the evolution of HSPS and generic market services as shares of total employment. To remove initial level differences, I rebase each country's employment share to zero in 1990 before computing unweighted group means. Figure 1 shows substantially greater HSPS expansion in economies with higher manufacturing peaks. Between 1990 and 2018, the HSPS share of total employment rises by 5.75 percentage points in the higher-peak group, compared with 2.77 percentage points in the lower-peak group. The 2.98-percentage-point difference is statistically significant in a two-sided Welch t-test ($p = 0.001$).

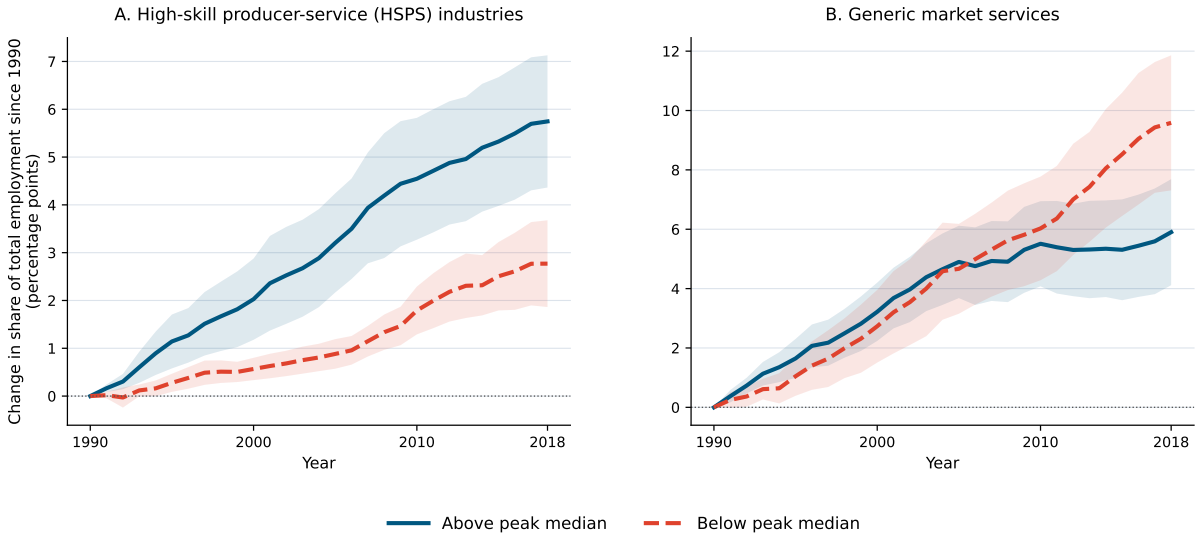


Figure 1: Manufacturing peaks and the subsequent service economy. The figure includes all 51 ETD countries observed annually over 1990–2018. Countries are grouped by the median historical peak manufacturing-employment share, calculated over their full available combined GGDC 10-Sector and ETD histories, with ETD taking precedence in overlapping years. The above-median group includes the country at the median and contains 26 countries; the below-median group contains 25. Panel A reports employment in high-skill producer-service (HSPS) industries—business services, finance, and real estate—as a share of total employment. Panel B reports employment in generic market services, defined as the ETD trade and other services categories, using the same denominator. Trade includes accommodation and food services. Transport and government services are excluded from both numerators. Each country's share is rebased to zero in 1990 before taking unweighted group means. Shaded regions show the mean plus or minus 1.96 cross-country standard errors.

The lower-peak group instead shows greater expansion in generic market services, which are more consumer-facing than HSPS industries. The generic market-service share rises by 9.59 percentage points in the lower-peak group, compared with 5.90 percentage points in the higher-peak group. Thus, the two groups differ in the service industries that absorb employment: higher-peak economies experience greater HSPS expansion, while lower-peak economies experience greater expansion of trade and other services.[4] Appendix B.2 shows that this relationship survives controls for contemporaneous GDP-per-capita growth, timing of the peak, income-group and region fixed effects, and the overall size of the service sector, and is driven primarily by business services growing more in industrialized economies.[5]

4 The ETD other-services category does not include government services.

5 In the ETD data, business services are the service activities most closely connected to firms' information and specialized professional needs—principally information and communication and professional, scientific, and technical activities.

***FACT 2**: Larger historical manufacturing peaks are associated with faster growth as economies transition into services.*

The rise of services is often associated with slower aggregate productivity growth. In the standard Baumol cost-disease logic, labor reallocates toward sectors with relatively slow productivity growth, reducing aggregate growth as the service share rises (Baumol, 1967; Duernecker and Sanchez-Martinez, 2023; Duernecker et al., 2024). Fact 1 suggests an additional dimension: service economies may differ in their internal composition depending on the development path through which they emerged. Figure 2 plots average annual PPP real GDP-per-capita growth over 2010–2019 against the average service-employment share over the same period for 173 countries. Countries are divided according to whether their historical manufacturing peak lies above or below the cross-country median. Service employment is negatively associated with growth in both groups. Conditional on the service-employment share, however, countries with above-median manufacturing peaks tend to grow faster.

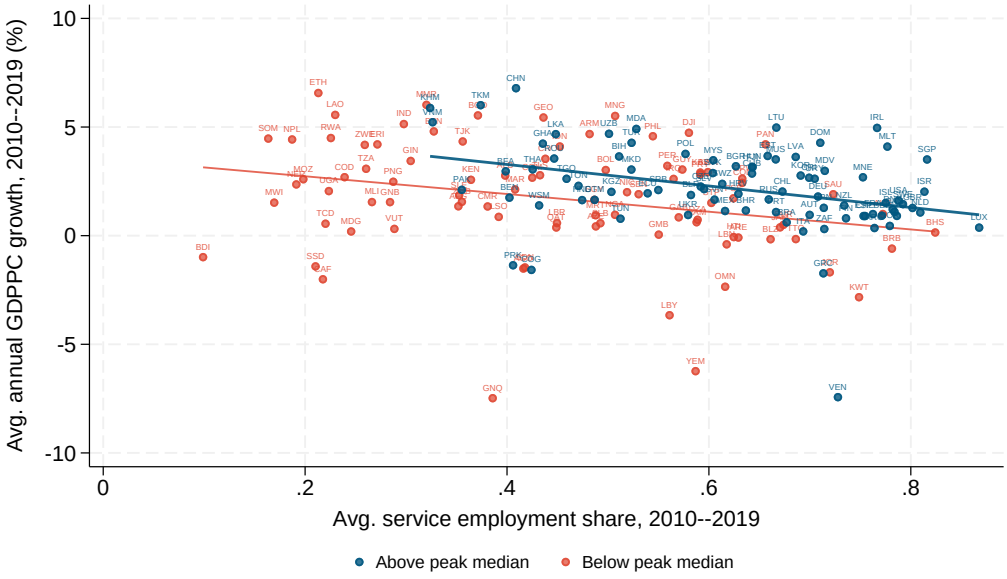


Figure 2: Growth in service economies by peak manufacturing exposure. The vertical axis is average annual growth in PPP real GDP per capita over 2010–2019; the horizontal axis is the average service-employment share over the same period. Countries are divided at the median historical manufacturing peak. Service employment is from ILOSTAT, GDP per capita is from the World Bank via Gapminder, and the historical manufacturing-peak series is described in Appendix B.

The relationship survives more demanding comparisons. Appendix Table B.2 shows that the manufacturing peak remains positively associated with subsequent GDP-per-capita growth after controlling for region and income-group fixed effects, institutional quality and the contemporaneous manufacturing employment share. The association also remains positive, with a slightly larger point estimate, among countries where services account for more than half of employment and peaked before the year 2000, at least a decade before GDP-per-capita growth is measured. These facts are consistent with a persistent legacy of industrialization rather than an effect operating only while manufacturing itself remains large.

The cross-country evidence does not, however, identify the mechanism behind these associations. Section 2.2 turns to the U.S. historical transition to test whether they reflect the scale and organizational demands of manufacturing and the managerial and technical capabilities generated in response.

2.2 U.S. County Evidence

The U.S. county evidence serves two purposes: to establish the timing implied by the mechanism and to sharpen identification. I first use a long panel of fixed-geography counties to trace managerial and technical employment within manufacturing and specialized HSPS services around county-specific manufacturing peaks. If industrialization builds capabilities that are later redeployed in services, we should observe HSPS workers emerge first within manufacturing and only later in specialized service industries. I then exploit counties' 1880 industry composition and national industry growth over 1880–1900 to isolate variation in the depth and type of local industrialization. This timing aligns with Chandler's dating of the beginnings of the modern industrial enterprise to the 1880s, so the 1880 industry shares capture local industrial composition at the outset of the organizational transformation studied in this paper. If the effect of manufacturing on the composition of services is driven by the organizational demands of large enterprises, we should see industrialization associated with the expansion of HSPS services only when it is driven by growth in high-throughput industries characterized by high fixed costs and large scale. If instead the relationship reflects broader consequences of industrialization, such as urbanization, population growth, or rising demand for skills, a similar long-run HSPS effect should also appear when manufacturing growth is driven by non-high-throughput industries.

2.2.1 Data and measurement

I combine county tabulations from the Census of Manufactures with full-count population censuses and postwar administrative employment data, harmonizing counties to fixed 1850 boundaries. The analysis distinguishes three objects: the depth of industrialization, measured by the peak manufacturing employment share over 1880-1940; managerial and technical capability within manufacturing, measured by HSPS employment in manufacturing relative to total classified employment; and the employment in specialized HSPS services, measured relative to the same employment base. For the long-run cross-sectional analysis, I additionally measure the HSPS share within private-service employment to isolate the association between industrialization and the composition of the service sector rather than its overall size.

The peak manufacturing employment share is manufacturing employment obtained from the Census of Manufactures relative to total classified employment obtained from the Census of Population. Let L_{ct}^{C} denote total classified employment, defined as the number of employed individuals whose industries can be classified as primary, manufacturing, or services.[6] I define county c's manufacturing peak as

$$M_c^{\text{peak}} = \max_{t \in \{1880,1900,1920,1930,1940\}} \left\{ \frac{L_{ct}^{M}}{L_{ct}^{C}} \right\}, \tag{1}$$

where L_{ct}^{M} is average manufacturing wage earners and L_{ct}^{C} is total classified employment. I let P_c denote the census year attaining this maximum. The peak window begins in 1880, before the national expansion of manufacturing and at the first major improvement in the quality of U.S. census enumeration.[7] The years 1890 and 1910 do not feature in the data because most of the 1890 population census was lost and the required county manufacturing tabulations are unavailable in 1910. Manufacturing scale is

6 More specifically, I define total classified employment as employment in primary industries (IND1950 codes 105, 116, and 126), manufacturing (IND1950 codes 306–499), and services (IND1950 codes 506–946). Mining, construction, unclassified employment, and industries outside these categories are excluded.

7 Before 1880, federal censuses were conducted by U.S. marshals; the 1880 Census was the first administered by a professional census office and enumerator force.

log manufacturing workers per establishment in the same peak year P_c, also measured in the Census of Manufactures.

To measure the organizational capabilities employed inside manufacturing, I identify workers performing the functions required to organize production at scale and construct

$$MHSPS_{ct} = 100 \times \frac{\text{HSPS-occupation workers in manufacturing}_{ct}}{L^C_{ct}}. \tag{2}$$

The manufacturing HSPS classification contains four functional groups: accounting and record keeping; purchasing, logistics, and coordination; personnel and labor relations; and technical and scientific work, including engineers, draftsmen, statisticians, surveyors, technicians, and inspectors. These occupations capture the information-processing, coordination, and technical functions that are necessary to operate large manufacturing enterprises (Chandler, 1977, 1990).[8] The service-side measure asks whether employment subsequently shifts toward industries that supply these capabilities on the market. I define

$$SHSPS_{ct} = 100 \times \frac{\text{employment in HSPS service industries}_{ct}}{L^C_{ct}}. \tag{3}$$

In the modern data, HSPS services comprise Information, Professional, Scientific and Technical Services, and Management of Companies—NAICS sectors 51, 54, and 55. These industries provide the closest service-sector counterpart to the organizational capabilities emphasized by Chandler (1977, 1990). NAICS 54 includes accounting, legal, engineering, scientific, technical, and consulting services that supply externally many functions historically organized within large manufacturers; NAICS 55 captures the management and coordination of enterprises; and NAICS 51 captures the production and transmission of information.[9] Because both the manufacturing occupations and the service industries used in these measures are relatively skill intensive, a natural concern is that the results capture generic skill upgrading rather than the organizational mechanism. Appendix C.2.3 addresses this by repeating the county event study for health care and education, two other skill-intensive service activities; neither exhibits the same sustained post-peak expansion as HSPS services. Appendix A reports the historical IND1950 and SIC concordances used to carry this functional definition backward in time. The county-industry panel within services uses the Population Census through 1950, County Business Patterns in 1960 and 1970, and the harmonized CBP panel of Eckert et al. (2020) thereafter; the manufacturing-HSPS measure uses the Population Census throughout because it requires occupation-by-industry information.[10] To measure the composition of services in 2000, I use the same HSPS industry definition but divide employment in NAICS 51, 54, and 55 by total private-service employment from the BLS Quarterly Census of Employment and Wages (QCEW), which directly reports the county-level private service-providing aggregate (U.S. Bureau of Labor Statistics, 2004).[11]

8 The core manufacturing HSPS set contains OCC1950 codes 0, 35, 41–49, 72, 83, 92, 95, 96, 200, 201, 280, 310, 321, 322, 341, 342, and 533. Appendix A reports the full labels and construction.

9 The cross-country ETD analysis necessarily uses a broader sectoral classification. Its business-services category includes information and communication and professional and technical activities, while finance and real estate are observed only as broad sectors.

10 Throughout, L^C_{ct} denotes the same broad classified-employment concept. For each outcome and data vintage, it is reconstructed from the same source as the corresponding numerator, avoiding comparisons between workplace-based postwar employment and residence-based Census employment.

11 Appendix C.2 describes the treatment of suppressed QCEW cells and reports corresponding results using alternative denominators, the harmonized CBP data, and a 2010 endpoint.

2.2.2 Manufacturing HSPS and the later rise of specialized services

I first trace how managerial and technical employment within manufacturing and in specialized HSPS service industries evolve around county-specific manufacturing peaks. Let M_c^{peak} denote the manufacturing employment share at county c's peak. I estimate by OLS the following specification separately for $MHSPS_{ct}$ and $SHSPS_{ct}$:

$$Y_{ct} = \alpha_c + \gamma_{s(c),t} + \lambda_t + \sum_{k \neq -40} \beta_k \left[\mathbf{1}\{t - P_c = k\} \times \frac{M_c^{\text{peak}}}{0.10} \right] + X_c' \delta \, Trend_t + \varepsilon_{ct}. \tag{4}$$

The specification absorbs county, state-by-calendar-year, and event-time fixed effects. The vector X_c contains 1850 log population, urbanization, colleges, school enrollment, and log enslaved population, together with 1870 shares of process, machinery, and traditional manufacturing. Each covariate is interacted with a linear calendar-time trend. Event time -40 is omitted. Because repeated county observations may be spatially correlated across neighboring local economies, the confidence intervals use a spatial covariance estimator allowing arbitrary covariance across all observations from counties whose centroids are within 100 kilometers.[12] This is a continuous-treatment event-study specification: each β_k measures the differential evolution of the outcome associated with a 10-percentage-point larger manufacturing employment share at the peak. The sample contains 1,116 fixed-geography counties peaking in 1900, 1920, or 1940 for which the peak treatment and a common forty-year pre-period are observed. Figure 3 establishes a clear sequence. Counties with larger manufacturing peaks begin accumulating more HSPS employment within manufacturing before reaching the peak, and this differential continues to widen for several decades thereafter. This timing is consistent with the national shift from traditional manufacturing toward high-throughput industries around the turn of the twentieth century (Chandler, 1990). As the manufacturing-HSPS differential subsequently recedes, HSPS service employment rises in counties that experienced greater industrialization.

Appendix C.2 shows that this sequence is not simply a consequence of general local development or skill upgrading. Larger manufacturing peaks, regardless of whether they occur in high-throughput or non-high-throughput industries, are associated with subsequent population growth and urbanization, but only high-throughput peaks are associated with a buildup of HSPS employment within manufacturing and its later expansion in services. The evolution of other skill-intensive service industries likewise exhibits no comparable post-peak increase or association with industrial history. The finding that long-run outcomes depend on the type of industrialization, rather than industrialization alone, also helps reconcile these results with evidence that earlier episodes of industrialization generated substantial short-run gains but adverse long-run effects (Franck and Galor, 2021).[13]

12 Clustering standard errors at the state level leaves results unchanged. I cluster spatially because this is the appropriate choice given industrialization historically exploited geographic features, such as the fall-line, coal mines, lakes and rivers, which transcend state boundaries.

13 Franck and Galor (2021) study the effects of the steam engine on French industrialization in 1860–65, before the late-nineteenth-century emergence of Chandlerian large industrial enterprise. They find initial income gains followed by a long-run reversal of the effects of industrialization on income and human capital. As a check on whether their findings conflict with the mechanism studied here, I repeat their exercise using French establishment-size information from 1896. The reanalysis yields a different pattern across manufacturing types: industrialization in scale-intensive Chandler-type industries predicts higher GDP per capita a century later, whereas industrialization in traditional industries predicts initial gains followed by long-run losses.

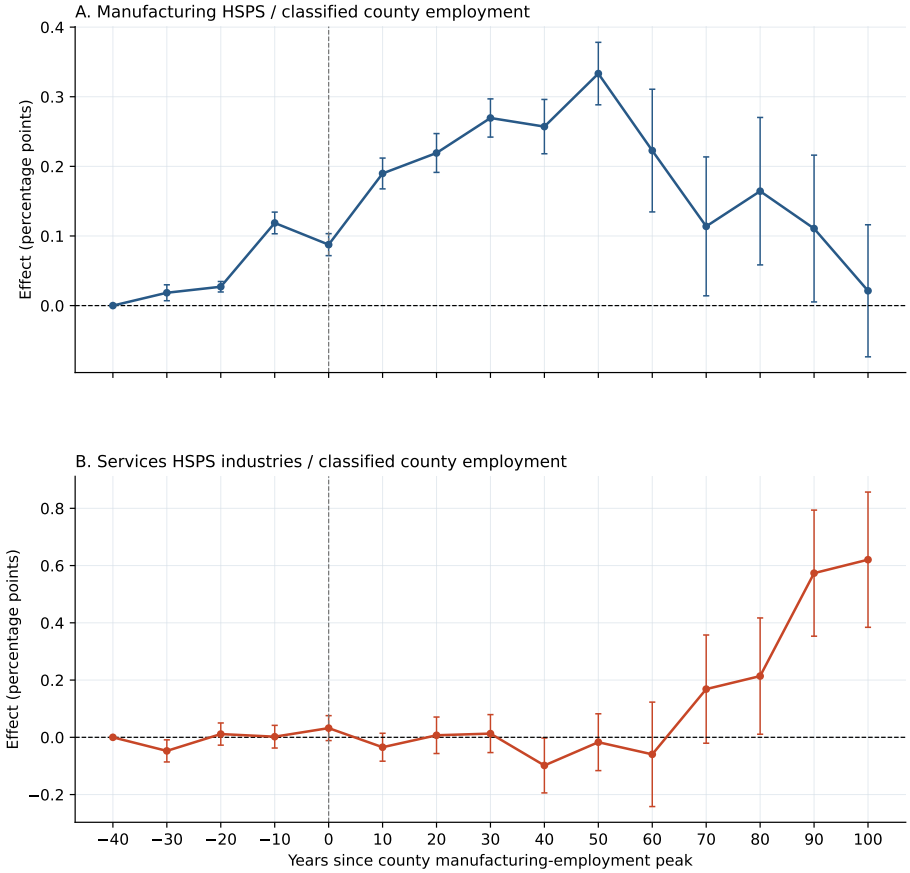


Figure 3: Manufacturing exposure at the peak and the capability sequence. The sample contains 1,116 fixed-geography counties peaking in 1900, 1920, or 1940. Each point is the coefficient associated with a 10-percentage-point larger manufacturing employment share at the county's peak; event time -40 is omitted. The upper outcome is manufacturing HSPS occupations divided by classified county employment, as defined in equation (2). The lower outcome is employment in HSPS service industries—the historical analogues of NAICS 51, 54, and 55—divided by the broad classified-employment base defined in equation (3). The lower-panel numerator and denominator are constructed from the same employment universe in each vintage: harmonized Census records through 1950, imputed early CBP in 1960 and 1970, and harmonized EFSY CBP thereafter. In the postwar years, both are therefore workplace-based employment counts rather than a workplace numerator divided by employed county residents. The regressions include county, state-by-calendar-year, and event-time fixed effects and baseline-covariate linear trends. Bars are 95-percent confidence intervals using a 100-kilometer spatial covariance estimator.

2.2.3 National industry shifts

The event-study evidence is consistent with a scale-dependent organizational legacy, but it remains descriptive. Counties that industrialized more deeply—and especially those specializing in high-throughput industries—may have differed beforehand in ways that also shaped the composition of their later service economies. To obtain variation in local industrialization driven by forces outside the county, I exploit counties' differential exposure to national manufacturing-industry growth over 1880–1900. I interact their 1880 industry composition with subsequent leave-county-out national industry growth. Because initial specialization may itself reflect pre-existing comparative advantage, identification is conditional on a rich set of observed covariates in 1850 and complete 1850–1870 histories of manufacturing employment, manufacturing HSPS, and HSPS services. The design therefore compares counties in the same

state and peak cohort that followed similar observed development paths before 1880 but were differently exposed to the national growth of the industries in which they were initially specialized.

The industry-level design also provides a direct test of whether the type of manufacturing expansion matters. I construct separate shifters using national growth in all manufacturing industries and in the high- and low-throughput groups introduced above. The all-industry shifter captures manufacturing expansion regardless of how production was organized, whereas the high-throughput shifter isolates expansion in industries where scale and coordination were technologically important. The mechanism predicts a persistent shift toward HSPS services for the latter, rather than for manufacturing expansion per se. For industry group $g \in \{A, H, L\}$, where A denotes all manufacturing, H high-throughput manufacturing, and L traditional low-throughput manufacturing, I construct

$$Z_c^g = \sum_{j \in g} \omega_{cj,1880}^g G_{j,-c,1880\text{–}1900}, \tag{5}$$

where $\omega_{cj,1880}^g$ is industry j's share of county c's 1880 manufacturing employment within group g, and $G_{j,-c,1880\text{–}1900}$ is national employment growth in industry j between 1880 and 1900 after removing county c. Each shifter is standardized within the common estimation sample. The within-group normalization makes the exposure shares sum to one and isolates which industries within a production regime a county initially specialized in rather than the overall size of the group in the local economy.

For each shifter Z_c^g, the first and second stages are

$$M_c^{\text{peak}} = \pi_g Z_c^g + X_{c,1850}'\Pi + \mathcal{H}_{c,<1880}'\Omega + \delta_{s(c)} + \lambda_{P_c} + u_c, \tag{6}$$

$$Y_{c,2000} = \beta_g \widehat{M}_c^{\text{peak}} + X_{c,1850}'\Gamma + \mathcal{H}_{c,<1880}'\Psi + \delta_{s(c)} + \lambda_{P_c} + \varepsilon_c, \tag{7}$$

Where the vector $X_{c,1850}$ contains the five baseline controls measured in 1850, while $\mathcal{H}_{c,<1880}$ collects the complete 1850–1870 histories of manufacturing employment, manufacturing HSPS employment, and HSPS-service specialization. The terms $\delta_{s(c)}$ and λ_{P_c} denote state and manufacturing-peak-cohort fixed effects. Table 1 reports the corresponding OLS benchmark and three just-identified 2SLS specifications on a common sample of 1,487 counties.

The all-industry and high-throughput shifters both strongly predict county manufacturing peaks, with Conley first-stage F statistics of 14.67 and 12.87 and Kleibergen–Paap statistics above 20. However, their long-run implications for the composition of services are different. Using national growth across manufacturing as a whole yields a 2SLS coefficient of 0.133, but neither the conventional estimate nor the weak-instrument-robust reduced form is statistically distinguishable from zero ($p_{AR} = 0.203$). By contrast, the high-throughput IV estimate is 0.283 with a Conley standard error of 0.118. A 10-percentage-point increase in the manufacturing peak induced by national high-throughput growth therefore raises the HSPS share of private-service employment in 2000 by 2.83 percentage points—approximately 45 percent of the 6.36-percent cross-county mean. The Anderson–Rubin test rejects zero at the 1-percent level ($p = 0.009$).

The traditional low-throughput shifter has essentially no first stage, so its 2SLS coefficient is not informative. The economically relevant contrast is between broad manufacturing growth, which strongly moves the manufacturing peak without producing a detectable shift in long-run service composition, and high-throughput growth, which moves both.

Table 1: National Manufacturing Industry Shifts and Persistent HSPS Services

	OLS	All industries IV	High-throughput IV	Low-throughput IV
Peak manufacturing employment share	0.029^{**}	0.133	0.283^{**}	-1.095
	(0.012)	(0.105)	(0.118)	(8.010)
First-stage coefficient		0.0150^{***}	0.0193^{***}	0.0006
		(0.0039)	(0.0054)	(0.0042)
First-stage Conley F statistic		14.67	12.87	0.02
Kleibergen–Paap F statistic		20.53	21.94	0.03
Anderson–Rubin p-value		0.203	0.009	0.596
State fixed effects	Yes	Yes	Yes	Yes
Peak-cohort fixed effects	Yes	Yes	Yes	Yes
1850 historical controls	Yes	Yes	Yes	Yes
Complete 1850–1870 histories	Yes	Yes	Yes	Yes
Observations	1,487	1,487	1,487	1,487

Notes: The dependent variable is the QCEW HSPS share of private-service employment described in Section 2.2.1. Peak manufacturing employment is Census of Manufactures average wage earners divided by total classified county employment and is defined over 1880, 1900, 1920, 1930, and 1940. Column 1 reports OLS. Columns 2–4 instrument the peak manufacturing employment share with standardized shift-share instruments based on, respectively, all manufacturing industries, Chandler high-throughput industries, and traditional low-throughput industries. The instruments combine 1880 county industry-employment shares within the relevant group with leave-county-out national industry growth from 1880 to 1900. All columns use the same 1,487 counties and include state and peak-cohort fixed effects, the five 1850 controls, and complete 1850–1870 histories of manufacturing employment and core manufacturing HSPS relative to total classified employment and of HSPS services relative to private-service employment. Coefficient and first-stage standard errors are Conley with a 100-kilometer cutoff. The first-stage Conley F statistic is the square of the Conley first-stage t statistic. The Kleibergen–Paap statistic is the heteroskedasticity-robust excluded-instrument Wald statistic. The Anderson–Rubin p-value is the corresponding Conley reduced-form test. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The contrast points to a scale-dependent organizational legacy of industrialization: persistent changes in service composition arise when manufacturing expansion occurs in production regimes where operating at scale creates intensive managerial, informational, and technical capabilities. Appendix C.2 subjects this result to the principal measurement and inference diagnostics.[14] The result is not driven by any single high-throughput industry, remains positive across alternative classification boundaries and endpoints, and survives substantially wider spatial-correlation cutoffs. Combined with the event-time evidence, these findings motivate the quantitative model in Section 3, which asks how this organizational legacy shapes structural transformation, skill accumulation, and long-run growth.

3 Model

3.1 Overview

Time is discrete and one period is one year. The economy contains three final-good sectors—agriculture, manufacturing, and consumer services—indexed by $j \in \mathcal{J} \equiv \{a, m, c\}$. Firms enter endogenously, draw permanent productivity upon entry, and face exogenous exit risk. Final goods are used for consumption and as intermediate inputs, while producer services help firms organize and coordinate production.

14 I report extensive robustness to alternative outcome construction, industry classifications, exposure definitions, geography, and inference procedures.

Nonhomothetic demand reallocates expenditure from agriculture toward manufacturing and services as income rises, generating the economy's structural transformation path. Households choose consumption, fertility, and general educational investment in their children, while entering cohorts separately choose whether to acquire the specialized skills required for high-skill producer-service work. General education affects the economy's capacity to develop new organizational procedures, whereas specialized skill determines the supply of workers able to operate a formalized organization. Both margins therefore evolve endogenously over the development path.

The central mechanism is that firm scale raises the return to formalizing organizational tasks. Every firm faces a unit measure of organizational tasks required to operate. Each task generates recurring organizational problems, with their frequency increasing linearly in firm size. These tasks are common across sectors: firms must coordinate workers, process information, manage inputs, and organize decision-making regardless of what they produce. Tasks can be handled in an ad-hoc manner, in which case the underlying problems must be resolved repeatedly, or formalized by installing a reusable procedure. The stock of formalized tasks constitutes the firm's *organizational capital*. Because a given task generates more problems in a larger firm, formalization avoids more repeated problem solving and is therefore more valuable at larger scale. Larger firms consequently formalize more tasks and demand more high-skill producer-service workers to do so.

Formalization can also create knowledge that extends beyond the firm. When a firm formalizes a task for which no established procedure exists, it must use skilled labor to develop a solution from scratch. The resulting method is initially proprietary, but may subsequently diffuse into a public stock of organizational knowledge, K_t, allowing other firms to formalize the same task at lower cost. The originating firm therefore does not capture the full value of the knowledge it creates. Because all firms, regardless of what they produce, face the same organizational problems, this externality is naturally cross-sectoral: methods developed inside one sector can later lower the cost of organizing production in another.[15]

Manufacturing matters in the model not because it uses intrinsically different organizational tasks, but because historically it operated at the scale at which formalization became privately worthwhile. A manufacturing-intensive development path therefore generates more organizational knowledge than an economy which moves from agriculture to a service sector composed of small firms. The service economy that follows manufacturing inherits organizational knowledge, formalizes more activities, and demands more HSPS labor. By raising demand for high-skill producer-service labor, this organizational legacy changes the economy's long-run skill composition and productivity growth.

The productivity-growth block follows Jones (1995). Unlike standard models in that framework, however, the long-run growth rate is independent of both population scale and population growth and is instead determined by the economy's endogenous share of skilled workers. To close the population side of the model, I combine the Jones-inspired productivity technology with the quantity–quality mechanism of Galor and Weil (2000). Fertility then converges to replacement and population to a finite level, while the limiting skilled *share* pins down sustained productivity growth. Manufacturing can therefore affect long-run growth insofar as it permanently changes the economy's skill composition. Proposition 2 characterizes this limiting equilibrium.

15 Historical work documents both sides of this mechanism. Large industrial firms developed new systems of cost accounting, return-on-investment control, statistical quality control, budgeting, workflow design, and personnel management as they confronted the organizational demands of operating at scale (Johnson, 1975; Kaplan, 1984; Chandler, 1977; Shewhart, 1931). Many of these methods were subsequently codified, professionalized, and applied outside manufacturing (McKinsey, 1922, 1924; Yates, 1993; McKenna, 2006; Mol and Birkinshaw, 2014). Appendix D.1 discusses several examples in detail.

The remainder of the section proceeds in four steps. I first characterize firms' organizational choices and show how scale determines formalization and HSPS demand. I then describe the creation, adaptation, and diffusion of organizational procedures and the entry margin that determines equilibrium firm scale. Next, I embed these decisions in the structural-transformation equilibrium and introduce the household choices that determine education and population growth. Finally, I characterize the limiting equilibrium, including the long-run composition of services, skill supply, and productivity growth. The main text emphasizes the economic mechanisms, while derivations, equilibrium definitions, and proofs are collected in Appendix D.

3.2 Production and organizational choice

Final-good production. A firm f in sector j has permanent ability $a_f \sim G$ and an inherited organizational state s_{fjt}, defined below. For a generic firm I suppress the f subscript and write its state as (a, s). Skilled and unskilled labor form a CES composite, and value added exhibits decreasing returns to firm scale:

$$\begin{aligned} \ell_{fjt} &= \left[\alpha_j (h_s L^s_{fjt})^{\frac{\sigma_L - 1}{\sigma_L}} + (1-\alpha_j)(h_u L^u_{fjt})^{\frac{\sigma_L - 1}{\sigma_L}}\right]^{\frac{\sigma_L}{\sigma_L - 1}}, \\ v_{fjt} &= A_{jt} a_f \ell^{\eta}_{fjt}, \qquad 0 < \eta < 1. \end{aligned} \tag{8}$$

Gross output y combines value added v, agricultural m^a and manufacturing m^m materials, and a producer-service composite P^p_{fjt}, defined below:

$$\begin{aligned} y_{fjt} &= Z_{jt} v^{\theta_{jt}}_{fjt} (m^a_{fjt})^{\gamma^a_{jt}} (m^m_{fjt})^{\gamma^m_{jt}} (P^p_{fjt})^{\gamma^P_{jt}}, \\ \theta_{jt} + \gamma^a_{jt} + \gamma^m_{jt} + \gamma^P_{jt} &= 1. \end{aligned} \tag{9}$$

Let w^v_{jt} denote the unit cost of the labor composite and let $n = \nu_{jt}\ell$ denote the associated employment headcount. Both follow from cost minimization in the CES labor nest. The distinction between ℓ and n matters because production depends on effective labor, whereas organizational problems arise from coordinating workers and therefore depend on employment headcount.

Organizational tasks. Every firm faces the same unit mass of recurring organizational tasks, such as accounting and record keeping, scheduling, inventory and quality control, procurement, and personnel management. Each instance of a task can be handled ad hoc, on a case-by-case basis, or formally by establishing an administrative system that structures how recurring instances are handled. Tasks that are formalized require less labor input to be handled, but must be managed by skilled workers. This abstraction captures the shift toward systematic administrative organization and high-skill professionals emphasized by Chandler (1990).[16]

Tasks differ in complexity $z \geq 0$, which governs how difficult they are to organize through such a formal system, and are exponentially distributed:

$$\mathcal{F}(z) = 1 - e^{-z}.$$

16 Engbom et al. (2025) similarly distinguish administrative tasks handled ad hoc from tasks assigned to professional workers, with professionals drawn from the higher-skill part of the workforce. Their ad-hoc task input is supplied by the entrepreneur, whereas here ad-hoc tasks use low-skill organizational labor and formalized tasks use skilled organizational labor.

A firm's organizational state is summarized by its formalization frontier s: all tasks with $z \leq s$ have been formalized. I refer to the share of tasks organized through formal systems as the firm's *organizational capital*,

$$q(s) \equiv \mathcal{F}(s).$$

The frequency with which each task must be carried out increases with firm size. Let

$$b(n) = \frac{n}{\bar{N}},$$

where n and $\bar{N}$ are measured in workers. Thus $b(n)$ is the number of normalized instances of each organizational task generated per period, with a firm of size $\bar{N}$ generating one instance of each task. I normalize a formalized instance to require one unit of high-skill organizational services and let an ad-hoc instance require $r > 1$ units of low-skill organizational services.

Let the superscript o denote producer services used to operate these recurring organizational functions. Conditional on (n, s), demand for high- and low-skill organizational services is therefore

$$H^o(n,s) = \frac{n}{\bar{N}} q(s), \qquad L^o(n,s) = \frac{n}{\bar{N}} r[1 - q(s)]. \tag{10}$$

Let p_t^H and p_t^L denote the corresponding competitive prices. Recurring organizational costs are

$$C_t^o(n,s) = \frac{n}{\bar{N}} \left[q(s) p_t^H + r[1 - q(s)] p_t^L \right]. \tag{11}$$

Formalization therefore lowers recurring organizational costs whenever $r p_t^L > p_t^H$. Because the frequency of each task rises with employment, the associated saving is larger in bigger firms.

Formalization also changes the composition of producer services used in gross output. A more formalized organization relies more heavily on high-skill professional services in production, so I let the same organizational state $q(s)$ govern the relative importance of high- and low-skill producer-service inputs. Letting the superscript p denote productive use,

$$P^p(s) = \left[q(s)^{1/\sigma_P} (H^p)^{\frac{\sigma_P - 1}{\sigma_P}} + [1 - q(s)]^{1/\sigma_P} (L^p)^{\frac{\sigma_P - 1}{\sigma_P}} \right]^{\frac{\sigma_P}{\sigma_P - 1}}. \tag{12}$$

Thus greater formalization shifts productive producer-service demand toward high-skill inputs. When $p_t^H > p_t^L$, this can raise the unit cost of producer services even as formalization lowers recurring organizational costs. The firm's organizational choice is therefore governed by a single tradeoff: formalization economizes on recurring problem solving but requires a more skill-intensive, and expensive, organization. The next subsection shows how the net return to this tradeoff varies with firm scale.

Static firm problem. After optimizing over labor composition, materials, and productive producer services, the firm's static operating profit can be written as

$$\pi_{jt}^*(a,s) = \max_{\ell \geq 0} \left\{ p_{jt}^v(q(s)) A_{jt} a \ell^\eta - [w_{jt}^v + c_{jt}^o(q(s))] \ell \right\}, \tag{13}$$

where

$$c_{jt}^o(q) = \frac{\nu_{jt}}{\bar{N}} \left[q p_t^H + r(1-q) p_t^L \right]$$

is recurring organizational cost per unit of effective labor, and $p^v_{jt}(q)$ is the shadow value of firm value added after materials and productive producer services have been chosen optimally. Its dependence on q captures the fact that using different organizational inputs changes the firm's unit cost in production. Appendix D derives both objects from the gross-output technology.

The firm's optimal effective labor and employment are

$$\ell^*_{jt}(a,s) = \left[\frac{\eta p^v_{jt}(q(s))A_{jt}a}{w^v_{jt} + c^o_{jt}(q(s))}\right]^{\frac{1}{1-\eta}}, \qquad n^*_{jt}(a,s) = \nu_{jt}\ell^*_{jt}(a,s). \tag{14}$$

Thus more productive firms are larger: employment rises with ability with elasticity $1/(1-\eta)$.

The key comparative static concerns the return to formalization. Combining the envelope condition with the firm's employment choice gives

$$\frac{\partial \pi^*_{jt}(a,q)}{\partial q} = n^*_{jt}(a,q)\Psi_{jt}(q), \tag{15}$$

where

$$\Psi_{jt}(q) \equiv \frac{w^v_{jt} + c^o_{jt}(q)}{\eta\nu_{jt}}\frac{\partial \log p^v_{jt}(q)}{\partial q} + \frac{rp^L_t - p^H_t}{\bar{N}}. \tag{16}$$

The first term is the marginal effect of formalization through the productive producer-service input. When high-skill producer services are more expensive than low-skill producer services, $p^H_t > p^L_t$, this term is negative: hiring skilled producer service workers raises unit cost and, in isolation, reduces desired firm scale. The second term is the recurring coordination saving: formalization replaces ad-hoc problem solving costing rp^L_t per problem with application of a formal procedure costing p^H_t. Thus $\Psi_{jt}(q)$ is the net marginal operating gain per worker from formalization, and the total operating gain is proportional to firm employment.

Proposition 1 (Firm scale and organizational formalization)**.** *Fix sector j, date t, and organizational state q. Conditional on equilibrium prices, if $\Psi_{jt}(q) > 0$, the marginal operating benefit of formalizing an additional task is strictly increasing in firm employment:*

$$\frac{\partial \pi^*_{jt}(a,q)}{\partial q} = n^*_{jt}(a,q)\Psi_{jt}(q).$$

Hence larger and more productive firms obtain a greater operating return from the same organizational procedure. Moreover, if $rp^L_t > p^H_t$, then $\Psi_{jt}(q) > 0$ implies that greater formalization also raises the firm's desired employment.

Proof. See Appendix D.

The scale dependence in Proposition 1 becomes consequential once formalization requires a one-time installation cost. Because this cost does not scale proportionally with firm employment, larger firms find it profitable to formalize a broader set of tasks. The next subsection introduces the one-time cost of formalization, characterizes firms' dynamic organizational investment decisions, and then shows how newly created procedures can diffuse into public organizational knowledge.

3.3 Organizational investment, diffusion, and entry

This subsection turns the static scale mechanism into a dynamic one. Firms pay a one-time cost to formalize organizational tasks, and the procedures they create can subsequently diffuse into a public stock of organizational knowledge, lowering the cost of formalization for other firms and sectors. Entry determines equilibrium firm size and hence whether these organizational investments become profitable.

Creation and adaptation. Let $K_t \in [0,1]$ denote the public stock of organizational knowledge, measured as the share of tasks for which a reusable procedure is publicly available. Because tasks are ordered by complexity, public knowledge takes the form of a frontier: a procedure for task z is publicly available whenever $\mathcal{F}(z) \leq K_t$. Formalizing a task requires a one-time organizational investment. If no public procedure exists, the firm must create one from scratch using high-skill producer-service labor; if a procedure is already contained in K_t, the firm can instead adapt it at a fraction of the creation cost:

$$C_t^C(z) = \frac{p_t^H}{\bar{e}_t^o} e^{z-1}, \tag{17}$$

$$C_{jt}^A(z) = \frac{\rho}{\xi_j} C_t^C(z), \qquad \xi_m = 1, \quad \rho \in (0,1). \tag{18}$$

Here $\bar{e}_t^o$ measures the average education embodied in the current workforce, inherited from the education choices made by households one generation earlier. For a given task, a more educated workforce requires fewer skilled-labor resources to produce a reusable procedure. Because education rises gradually over development, this technology also links the historical expansion of education to the timing at which increasingly complex organizational methods become economical to develop. The parameter ρ governs the common cost advantage of adapting an existing procedure, while ξ_j determines how adaptation costs vary across sectors.

A firm moving from inherited formalization frontier s to a new frontier $u \geq s$ therefore incurs the one-time investment cost

$$\mathcal{C}_{jt}^O(s,u;K_t) = \int_s^u \widetilde{C}_{jt}(z;K_t)\, d\mathcal{F}(z), \tag{19}$$

$$\widetilde{C}_{jt}(z;K_t) = \begin{cases} \min\{C_t^C(z), C_{jt}^A(z)\}, & \mathcal{F}(z) \leq K_t, \\ C_t^C(z), & \mathcal{F}(z) > K_t. \end{cases}$$

Thus a task beyond the public frontier must be formalized by creating a new procedure, whereas a task inside the frontier can exploit the accumulated organizational knowledge of previous firms.

A firm produces in period t using its inherited frontier s and chooses the frontier $u \geq s$ carried into period $t+1$. Because the economy is generally away from its limiting equilibrium, this is a nonstationary dynamic problem: the firm's choice depends on the entire perfect-foresight continuation path of aggregate states and prices. I suppress this dependence in the notation. Taking that continuation path as given, the firm's value satisfies

$$W_{jt}(a,s) = \pi_{jt}^*(a,s) + \max_{u \geq s} \left\{ -\mathcal{C}_{jt}^O(s,u;K_t) + \beta(1-\delta_f) W_{j,t+1}(a,u) \right\}. \tag{20}$$

Because the organizational investment cost is fixed and Proposition 1 shows that the benefit to formalization rises with firm scale, larger firms therefore find it profitable to formalize a broader range of tasks and incidentally create more organizational knowledge.

Diffusion. Newly created procedures are initially proprietary. Let $\mu_{jt}(a, s)$ denote the distribution of active firms in sector j, let

$$N_{jt} = \int n^*_{jt}(a, s)\, d\mu_{jt}(a, s), \qquad N_t = \sum_{j \in \mathcal{J}} N_{jt},$$

and let $c_{jt}(a, s; z) \in \{0, 1\}$ indicate that a firm creates at date t a procedure for task z that lies beyond the current public frontier. Economy-wide employment-weighted exposure to newly created proprietary procedures is

$$\mathcal{E}_t = \frac{1}{N_t} \sum_{j \in \mathcal{J}} \int n^*_{jt}(a, s) \left[\int_{\mathcal{F}(z) > K_t} c_{jt}(a, s; z)\, d\mathcal{F}(z) \right] d\mu_{jt}(a, s).$$

Thus $\mathcal{E}_t$ is larger when more procedures are created and when those procedures are created in firms employing more workers. Public organizational knowledge evolves according to

$$K_{t+1} = \min\{1, K_t + \delta_K \mathcal{E}_t\}, \tag{21}$$

where δ_K governs the rate at which proprietary procedures diffuse into the public stock. Because K_t is irreversible, a temporary period of large-scale production can therefore leave a persistent change in equilibrium skill-demand.

Entry. A potential entrant pays its setup cost before drawing permanent ability $a \sim G$ and enters with frontier $s = 0$. Let I_{jt} denote entry. Convex setup-input supply gives the marginal entry cost $F_j I_{jt}^{\phi_E}$. Free entry and the law of motion for establishment counts are summarized by

$$\begin{aligned} V^E_{jt} &= \int W_{jt}(a, 0)\, dG(a), & M_{jt} &= (1 - \delta_f) M_{j,t-1} + I_{jt}, \\ I_{jt} &\geq 0, & F_j I_{jt}^{\phi_E} - V^E_{jt} &\geq 0, \\ & & I_{jt} \left[F_j I_{jt}^{\phi_E} - V^E_{jt} \right] &= 0. \end{aligned} \tag{22}$$

The shifter F_j governs equilibrium entry and therefore average firm size at the sectoral level. Through Proposition 1, it consequently determines the strength of firms' incentives to invest in formal organization.

3.4 Structural transformation, skill supply, and demographic closure

The firm side determines how organizational knowledge affects demand for high-skill producer-service labor. I embed this mechanism in aggregate dynamics using three standard blocks. Nonhomothetic demand determines the path of structural transformation; entering cohorts choose whether to acquire specialized skill in response to the skilled-wage premium; and a demographic block determines population dynamics and maps the terminal skilled share into a long-run growth rate.

As introduced above, the model distinguishes general education from specialized skill acquisition. General educational investment, $e_{i,t}$, determines the broad education embodied in the workforce and thereby affects the cost of developing new organizational procedures through $\bar{e}_t^o$. The skilled share S_t, by contrast, measures the fraction of workers who acquire the specialized skills used in high-skill producer services. The first margin affects the economy's capacity to create organizational methods; the second determines the supply of workers who operate them.

Structural transformation and skill acquisition. Final consumption follows the nonhomothetic CES system of Comin et al. (2021):

$$\sum_{j\in\mathcal{J}} \omega_j^{1/\sigma} \left(\frac{C_{ij,t}}{C_{i,t}^{\varepsilon_j}}\right)^{\frac{\sigma-1}{\sigma}} = 1, \qquad 0 < \sigma < 1. \tag{23}$$

Income and relative-price effects shift expenditure away from agriculture, generate a hump-shaped manufacturing share, and eventually move activity toward services. This determines the sectoral path along which firm scale, organizational investment, and knowledge diffusion evolve.

Members of the entering cohort also choose whether to acquire skill. I summarize this decision by

$$p_t^S = \mathcal{S}(\Pi_{t+1}), \qquad \mathcal{S}'(\Pi) > 0, \tag{24}$$

where Π_{t+1} is the expected equilibrium skilled-wage premium next period. Greater formalization raises demand for high-skill producer-service labor and therefore increases the incentive of subsequent cohorts to acquire skill. In the quantitative model, $\mathcal{S}$ is parameterized as a smooth logistic choice probability; Appendix D reports the exact specification.

Demographic closure. To close population dynamics, I use the quantity–quality mechanism of Galor and Weil (2000) and Lagerlöf (2006). Households are indexed by employment origin $i \in \{a, m, s\}$, which affects only the fixed time cost of raising children. They choose fertility $n_{i,t}$ and general educational investment $e_{i,t}$ according to

$$U_{i,t} = (1-\gamma)\log C_{i,t} + \gamma \log[n_{i,t} h_{i,t}], \tag{25}$$

$$h_{i,t} = \frac{e_{i,t} + \tau_h}{e_{i,t} + \tau_h + g_t}, \qquad 0 < \tau_h \ll 1, \tag{26}$$

where g_t denotes aggregate productivity growth. Faster technological change raises the return to education, inducing households to invest more in their children and reduce fertility.

Raising and educating a child requires $\tau_i + \tau_e(S_t)e_{i,t}$ units of parental time, where

$$\tau_e(S) = 1 - S(1-\tau_e^s), \qquad \tau_e^s \in (0,1).$$

Hence educational investment is less costly in a more skilled economy. This reduced-form complementarity captures the idea that educated parents, teachers, peers, and institutions make further educational investment easier (Cunha and Heckman, 2007; Cunha et al., 2010). It is also the link through which the terminal skilled share affects the limiting growth rate. The household budget constraint and the fertility and education first-order conditions are reported in Appendix D.

Aggregate dynamics and productivity growth. Let Δ denote generation length and $\bar{n}_t$ average fertility. Population and the skilled share evolve according to

$$L_{t+1} = L_t \bar{n}_t^{1/\Delta}, \qquad S_{t+1} = \left(1 - \frac{1}{\Delta}\right) S_t + \frac{1}{\Delta} p_t^S. \tag{27}$$

Thus organizational change affects the future skilled share through the equilibrium wage premium, while fertility determines the evolution of population.

Sectoral value-added productivity growth increases with the mass of skilled workers:

$$g_{j,t}^A \equiv \frac{A_{j,t+1}}{A_{jt}} - 1 = \left[1 + \lambda_A (S_t L_t)^{\phi_j}\right]^{1/\Delta} - 1, \qquad \phi_j \in (0,1). \tag{28}$$

Consumer services and both producer-service industries share the service exponent ϕ_c, while gross-output productivity grows exogenously at rate g^{GO}.

Equation (28) contains a standard scale force: holding the skilled share fixed, a larger workforce raises productivity growth. The demographic block closes this force endogenously. A larger population raises productivity growth; faster technological change induces greater educational investment and lowers fertility; population therefore adjusts toward the level at which fertility is at replacement. Proposition 2 shows that, for a given terminal skilled share, this process converges to a finite population and a positive limiting growth rate. Because the cost of education falls with the skilled share, the limiting growth rate is itself increasing in that share.[17]

Equilibrium. Manufacturing is the numeraire. Given the initial state of the economy, a perfect-foresight competitive equilibrium is a sequence of prices, household choices, firm policies, entry, firm distributions, and aggregate states such that households optimize; firms solve (13) and (20); entry satisfies (22); final-good, producer-service, and labor markets clear; and the firm distribution, public organizational knowledge, population, skills, and sectoral productivities follow their respective laws of motion. Aggregate uncertainty is absent, although firms face idiosyncratic exit risk and entrants draw heterogeneous ability. A formal definition is provided in Appendix D.[18]

3.5 Limiting growth and the manufacturing legacy

I characterize the model's long-run behavior in two steps. First, I show that the terminal skilled share pins down the economy's limiting productivity growth rate, while population scale adjusts to the finite level required to sustain it. Second, I show how a temporary manufacturing phase can permanently raise the terminal skilled share, and thus long-run growth, by expanding the stock of public organizational knowledge inherited by the service economy. For the analytical results, I assume that broad services,

17 The dependence of productivity growth on skilled-worker mass follows the broad idea-based growth tradition of Jones (1995). The innovation in this model is to combine the idea-generating function of Jones (1995) with the utility function of Galor and Weil (2000) and its functional forms from Lagerlöf (2006); who provide the closure needed to make the long-run growth rate endogenous.

18 Computationally, I solve the perfect-foresight transition using the extended-path logic of Fair and Taylor (1983). I conjecture the aggregate path, solve firms' dynamic decisions backward along that path, propagate the firm distribution and aggregate states forward, and iterate until the conjectured and implied aggregate sequences coincide.

including consumer and producer services, absorb the workforce in the limit; in the calibrated economy they account for 92 percent of terminal employment.[19]

Proposition 2 (Endogenous population scale and limiting growth)**.** *Fix a terminal skilled share $S \in (0,1)$ and let*

$$\zeta \equiv 1 + \frac{d^{\infty}}{\bar{w}^{\infty}}$$

denote the full-income ratio in the terminal service equilibrium. Suppose $\gamma\zeta > \tau_s$ and, for this analytical result, $\tau_h = 0$. Population converges from every $L_0 > 0$ to a unique steady state $L^{\infty}(S;\zeta)$, and fertility converges to replacement. At that steady state,

$$g^{\infty} = \mathcal{G}(S;\zeta) \equiv \frac{(\gamma\zeta - \tau_s)^2}{\tau_s} \times \frac{1}{\tau_e(S)}, \tag{29}$$

$$L^{\infty}(S;\zeta) = \frac{1}{S}\left[\frac{[1+\mathcal{G}(S;\zeta)]^{\Delta} - 1}{\lambda_A}\right]^{1/\phi_c}. \tag{30}$$

Holding ζ fixed,

$$\frac{\partial \mathcal{G}(S;\zeta)}{\partial S} > 0.$$

Proof. See Appendix D.

To gain intuition, consider an economy starting from a constant growth path where fertility is exactly at replacement. Suppose the terminal skilled share rises. Holding population fixed, the mass of skilled workers SL increases, raising productivity growth. Faster technological change raises the return to educating children and lowers fertility, causing population to fall; this adjustment by itself pushes growth back down. The key additional force is that a higher skilled share also lowers the parental time cost of education, $\tau_e(S)$. Hence, at the original growth rate, the lower time cost of education reduces total parental time per child—even though households optimally choose more education—so fertility rises above replacement. Population can therefore be stationary only at a higher growth rate: faster technological change induces enough additional education per child to restore the parental time commitment consistent with replacement fertility. Hence a higher terminal skilled share implies a higher limiting growth rate, while population adjusts to the finite level required to sustain it.

I next ask when passing through manufacturing places the economy on a permanently higher growth path. Proposition 2 establishes that limiting productivity growth is determined by the terminal skilled share. Manufacturing therefore raises long-run growth only if it leaves the subsequent service economy sufficiently more skill intensive to raise S^{∞}. In the model, this occurs through organizational knowledge: large firms create reusable procedures, those procedures diffuse, and service firms later formalize more tasks and demand more high-skill producer-service labor than they would have without the manufacturing phase. To characterize when this chain of events occurs in the model, consider two economies beginning from the same state (S_1, L_1, K_1). Economy A passes through manufacturing before entering services, whereas economy B moves directly from agriculture into services. I isolate the mechanism using a three-stage comparison with symmetric static firms, common prices and operating-surplus terms,

19 The remaining nonservice employment in the calibrated economy reflects input-output linkages: even at high income levels, service production continues to require agricultural and manufacturing intermediates.

and a common terminal full-income ratio.[20] I additionally require that the service economy is not organizationally saturated, so that access to an additional public procedure induces some service firms to formalize an additional task. Appendix D states this condition formally.

Proposition 3 (Manufacturing and long-run growth)**.** *Under the preceding restrictions,*

$$g_A^\infty > g_B^\infty \quad \Longleftrightarrow \quad S_A^\infty > S_B^\infty \quad \Longleftrightarrow \quad K_{A,3} > K_{B,3} \quad \Longleftrightarrow \quad \underbrace{\frac{\tau_m}{\tau_s}}_{\text{education}} \underbrace{(S_1 L_1)^{\phi_m - \phi_c}}_{\text{productivity growth}} \underbrace{\left(\frac{F_m}{F_c}\right)^{\frac{2}{1+\phi_E}}}_{\text{firm scale}} > 1. \tag{31}$$

Proof. See Appendix D. Proposition 3 makes the model's central mechanism explicit. Manufacturing raises long-run growth when it leaves the subsequent service economy with a larger stock of public organizational knowledge: cheaper adaptation induces service firms to formalize more tasks, raising HSPS demand, the return to skill, and ultimately S^∞ and g^∞. Condition (31) shows when this occurs. The terms τ_m/τ_s and $(S_1 L_1)^{\phi_m - \phi_c}$ capture the education force along the transition, through differences in household education choices and sectoral productivity growth, respectively; both affect the availability of educated labor able to develop new procedures. The term $(F_m/F_c)^{2/(1+\phi_E)}$ captures the complementary scale force: through entry, it determines relative firm size and hence the private return to formalization. The condition therefore summarizes the interaction between the capacity to create organizational knowledge and the scale at which creating it becomes worthwhile. The quantitative model relaxes the simplifying restrictions behind the proposition and determines the strength of these forces in full general equilibrium.

4 Calibration

I calibrate the model at an annual frequency to the U.S. development path from 1850 to 2019. The economy is initialized in 1790, allowing the endogenous state variables to evolve before the first targeted observations, and is solved through a terminal horizon well beyond 2019. Parameters fixed by normalization, timing conventions, direct empirical evidence, or prior literature are assigned externally, while the remaining 19 parameters are estimated jointly by simulated method of moments (SMM). Appendix D.4 describes the data, external parameter choices, numerical solution, parameter estimates, and local identification analysis.

Let $m_{bk}(\Theta)$ and $\widehat{m}_{bk}$ denote the model and data values of moment k in block b. I estimate Θ by minimizing the block-weighted SMM objective

$$r_{bk}(\Theta) = \frac{m_{bk}(\Theta) - \widehat{m}_{bk}}{\max\{|\widehat{m}_{bk}|, 0.01\}}, \qquad \mathcal{L}(\Theta) = \frac{\sum_b \omega_b K_b^{-1} \sum_{k=1}^{K_b} r_{bk}(\Theta)^2}{\sum_b \omega_b}. \tag{32}$$

The normalization expresses discrepancies relative to the magnitude of each target, with the 0.01 floor preventing moments close to zero from receiving disproportionate weight. Averaging within each block prevents long time series from mechanically dominating scalar targets.[21]

20 For this analytical comparison I also set $\tau_h = 0$ and suppress the annualization.

21 I assign a larger weight to the manufacturing peak share, manufacturing peak timing, sectoral productivity growth, and sectoral HSPS shares. The additional weight is placed on moments that directly discipline the paper's central structural-transformation and organizational mechanisms. Appendix D.4 reports the complete weighting scheme.

The set of moment blocks disciplines the main quantitative margins of the model. Sectoral employment paths, normalized GDP per capita, and the height and timing of the manufacturing employment peak discipline structural transformation and aggregate development. Fertility paths and the agriculture-to-services fertility ratio discipline the demographic transition. The aggregate skilled share, skilled earnings premium, and sectoral value-added skill shares discipline skill supply and demand. Sectoral and aggregate high-skill producer-service (HSPS) shares, together with sectoral firm size, discipline the scale–formalization mechanism. Finally, sectoral value-added productivity growth disciplines the productivity-growth technology. Scalar targets are measured in 2019 in both the model and the data unless otherwise noted; the agriculture-to-services fertility ratio is measured over its historical window, while manufacturing peak timing is treated as an interval target, with dates between 1930 and 1960 receiving no penalty. Appendix D.4 reports the data sources, measurement windows, block weights, and construction of each target.

4.1 Measurement

Producer and consumer services. The model distinguishes services by their use rather than by the industry of the supplier. This distinction matters because transportation, wholesale trade, finance, insurance, real estate, legal services, and other business services sell to both firms and final consumers. I combine 2019 IPUMS USA microdata with the 2019 BEA Summary Use Table. Within agriculture, manufacturing, and consumer-service industries, occupations are divided between sectoral value-added work and organizational or technical producer-service tasks. Within each group workers are then divided by education, with high skill defined as at least a bachelor's degree.

Employment in industries that primarily supply intermediate services is allocated across destination sectors using BEA intermediate-use expenditure shares; the portion sold to final consumers is assigned to consumer-service value added. This construction yields sector-specific HSPS and skilled value-added employment shares without imposing a mutually exclusive producer-service industry classification. The IPUMS–BEA concordance is described in Appendix D.4. The allocation applies a common skill composition and labor requirement to an industry's sales across buyers, the identifying restriction required because worker records cannot be linked to the purchaser of the industry's output.

Firm scale. Modern firm-size targets come from the 2019 Census Statistics of U.S. Businesses and are measured as employees per employer firm. Average firm sizes are 7.435 workers in agriculture, 49.694 in manufacturing, and 20.454 in broad services. I calibrate the model's scale dimension using firm size rather than establishment size as an establishment size measure would disproportionately lower scale in services, where multi-establishment firms are especially common (Hsieh and Rossi-Hansberg, 2023). A common initial scale of 10.7 workers is anchored in both manufacturing and services to the 1880 Census of Manufactures; the subsequent historical manufacturing observations, excepting the 2019 targets, are used to assess the model's performance in generating the timing of firm size growth and decline over the development path.

4.2 Parameters

Parameters governing standard preferences, firm dynamics, substitution elasticities, and units are fixed before estimation using conventional values or normalizations; Appendix D.4 reports the complete list and their sources. I discuss here the assignments that are most specific to the mechanism.

The dispersion of idiosyncratic skill-acquisition costs, σ_{skill}, determines the elasticity of skill supply to the expected skilled earnings premium. With the normalization $\beta_\pi = 1$, the logit rule in equation (24) implies

$$\sigma_{\text{skill}} = \frac{\Delta \log \Pi}{\Delta \log\{p^S/(1-p^S)\}}.$$

I discipline this parameter using the long-run U.S. joint evolution of college attainment and the college earnings premium. Using bachelor-or-higher attainment and the corresponding skilled–unskilled earnings premium gives an implied value close to 0.10, which I use as the baseline. This follows the quantitative education-choice approach of Heathcote et al. (2010), who discipline heterogeneity in college costs using aggregate changes in educational attainment and the college wage premium; the logit random-utility structure follows McFadden (1974). The location parameter c^{skill} is instead estimated jointly and primarily disciplines the level of the skilled share.

The knowledge-diffusion parameter δ_K governs how quickly exposure to newly created proprietary procedures expands the public stock of organizational knowledge. I set $\delta_K = 0.09$. Mansfield (1961) reports a mean time of 7.8 years from first commercial use to 50 percent adoption across twelve industrial innovations, corresponding to an annual closing rate of $1 - 2^{-1/7.8} = 0.085$. Appendix D.4 reports the full mapping and shows that the headline results are robust to significant variations in δ_K.

Two additional organizational parameters are assigned rather than estimated directly. The scale normalization $\bar{N} = 25$ implies that recurring coordination absorbs approximately five percent of employment in the calibrated economy, an order of magnitude consistent with the organization-capital evidence in Atkeson and Kehoe (2005). The adaptation-cost ratio is set to $\rho = 0.40$, implying a substantial but incomplete cost advantage to adapting an existing procedure rather than developing it from scratch. This is consistent with evidence that established management practices can be transferred across firms through consulting, training, and managerial knowledge flows, and that informational barriers are an important obstacle to their adoption (Bloom et al., 2013; Giorcelli, 2019). Because these studies do not identify the adaptation-to-creation cost ratio directly, I treat ρ as a maintained benchmark and report sensitivity to it in Appendix D.4.

The remaining 19 parameters are chosen jointly by simulated method of moments. Table 2 reports their estimates and principal empirical disciplines. Fertility moments primarily discipline (τ, τ_a^n); sectoral productivity growth and the GDP-per-capita path discipline $(\phi_a, \phi_m, \phi_c, g_{GO})$; aggregate and sectoral skill shares discipline $(c_{\text{col}}, \alpha_a, \alpha_m, \alpha_c)$; structural-change paths discipline (ϵ_c, w_a, w_c);[22] sectoral HSPS shares discipline (R, ξ_a, ξ_c); and sectoral firm-size moments discipline (F_a, F_m, F_c). These mappings describe the principal sources of variation rather than one-parameter–one-moment identification: all 19 parameters are estimated jointly.

Two parameters lie close to economically meaningful bounds. The estimated service nonhomotheticity parameter, $\epsilon_c = 1.0105$, implies only a modest direct luxury-good effect for services. The estimate is lower than the service nonhomotheticity typically obtained in models using CLM-type preferences, but the comparison is not one-to-one: here ϵ_c is estimated jointly with sector-specific productivity growth,

22 I set ϵ_a to 0.01 consistent with significant subsistence constraints, Kongsamut et al. (2001); Herrendorf et al. (2014).

Table 2: Joint Calibration and Model Fit

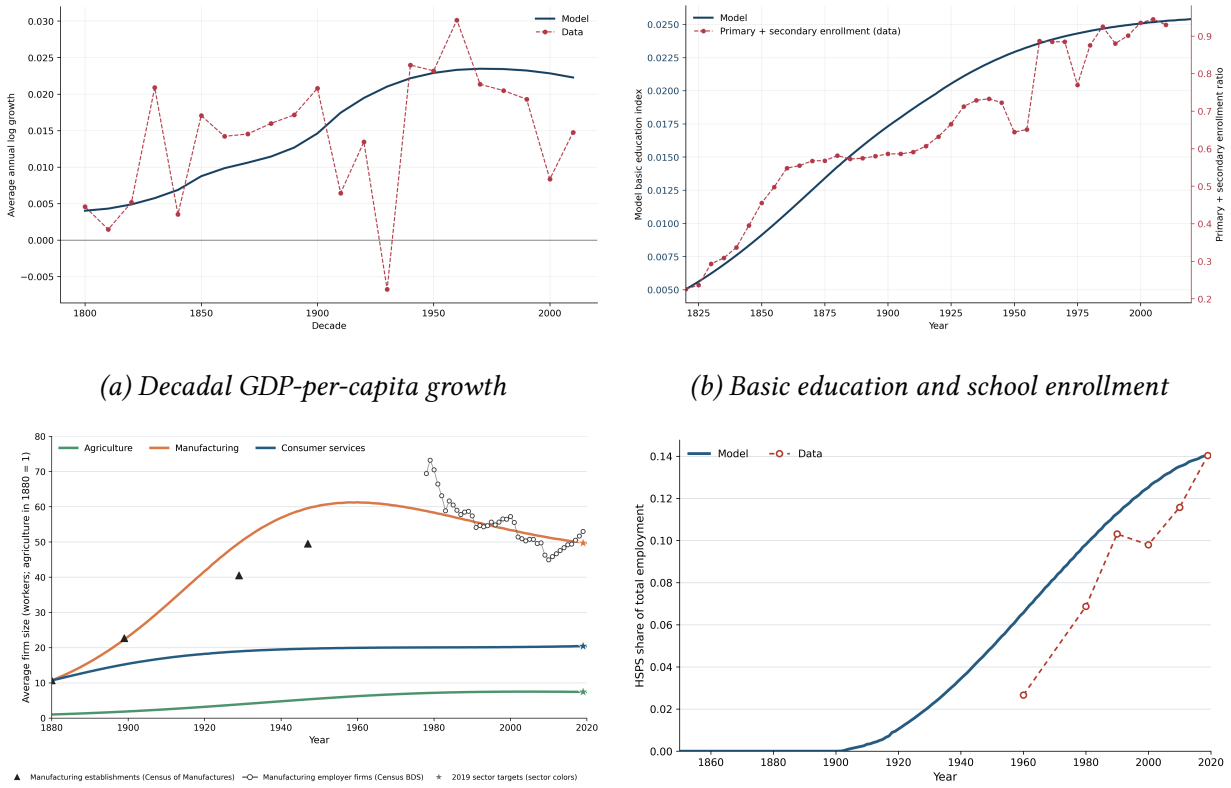
Block	Parameter	Value	Empirical discipline / moment	Data	Model
Fertility	τ	0.03236	Fertility path	$(+0.180)$	$(+0.180)$
	τ_a^n	0.44017	Agriculture/service fertility ratio	2.000	1.990
Productivity	ϕ_a	0.99840	Agricultural VA productivity growth	0.0371	0.0385
	ϕ_m	0.91919	Manufacturing VA productivity growth	0.0317	0.0298
	ϕ_c	0.69046	Consumer-service VA productivity growth	0.0142	0.0131
	g_{GO}	0.00318	Real GDP-per-capita path	(-0.134%)	(-0.134%)
Skills	c_{col}	0.56821	Aggregate skilled share, 2019	0.3299	0.3247
	α_a	0.24338	Within-VA skilled share, agriculture	0.1629	0.1627
	α_m	0.33001	Within-VA skilled share, manufacturing	0.2313	0.2293
	α_c	0.40995	Within-VA skilled share, consumer services	0.2926	0.2956
			Skilled earnings premium	1.600	1.645
Demand	ϵ_c	1.01048	Service employment transition	$(+3.168)$ pp	$(+3.168)$ pp
	w_a	0.65105	Agricultural employment transition	(-5.778) pp	(-5.778) pp
	w_c	3.82326	Service employment transition	$(+3.168)$ pp	$(+3.168)$ pp
			Peak manufacturing employment share	0.2828	0.2493
Organization	R	22.78696	Aggregate HSPS share	0.4232	0.4234
	ξ_a	3.56402	HSPS share, agriculture	0.3245	0.3248
	ξ_c	6.87739	HSPS share, consumer services	0.3989	0.3993
			HSPS share, manufacturing	0.4608	0.4608
Entry	F_a	25.872	Average firm size, agriculture	7.435	7.441
	F_m	187.625	Average firm size, manufacturing	49.694	49.717
	F_c	85.117	Average firm size, services	20.454	20.454

Notes: VA denotes value added. The calibration is joint, so the empirical discipline reported alongside each parameter should not be interpreted as a one-to-one estimation equation. Data and Model report scalar calibration moments where available. Parentheses indicate parameters disciplined primarily by an empirical path or transition rather than by a single scalar moment and report the corresponding deviation, with a positive number indicating that the model is above the data. Rows with blank parameter cells report additional moments targeted jointly by the corresponding block.

time varying input-output linkages, and endogenous firm organization. As emphasized by Herrendorf et al. (2013), the apparent importance of income effects in the structural change literature depends importantly on production structure of the model and how it is mapped to the data. In this model, the estimate therefore implies that the U.S. service transition is not being generated primarily by a large income effect. The agricultural growth elasticity, $\phi_a = 0.9984$, is similarly close to its upper bound and implies nearly linear sensitivity of agricultural productivity growth to skilled-worker mass. This reflects the high historical rate of agricultural productivity growth relative to manufacturing and services, which the common growth shifter must reconcile through sectoral elasticities.

4.3 Fit and untargeted validation

Figure 5 compares the model with the U.S. employment transition. The model reproduces the decline of agriculture, the rise of services, and the hump-shaped manufacturing path. Manufacturing peaks in 1928 at 24.93 percent of employment, compared with 28.28 percent in 1950 in the data. Table 2 reports all the calibrated parameters next to the fit associated with their principal empirical discipline. The moments



(a) Decadal GDP-per-capita growth

(b) Basic education and school enrollment

(c) Sectoral firm size

(d) High-skill producer-service employment

Figure 4: Historical-path validation. Panel (a) reports annualized GDP-per-capita growth by decade; the model line is the seven-decade quadratic low-frequency trend, with the unsmoothed series available in the replication files. Panel (b) compares the model's basic-education index with the primary-plus-secondary enrollment ratio. Because the two education measures have different units, panel (b) evaluates timing and shape. Panel (c) compares the model's sectoral firm-size paths with historical manufacturing establishment sizes, later manufacturing employer-firm sizes, and the 2019 sectoral calibration targets. Panel (d) compares the model-implied employment share of high-skill producer services with its empirical counterpart, constructed using harmonized IPUMS occupations and industries together with BEA input-use weights.

most directly tied to the paper's mechanism—sectoral firm size and HSPS shares—are matched almost exactly, with the largest deviation below 1 percent.

Four untargeted implications evaluate the calibrated transition. First, decadal GDP-per-capita growth is computed from the targeted GDP-level series but does not itself enter the objective. Across the 22 decades in Figure 4, the model matches the hump-shaped pattern from low growth in the early nineteenth century, to faster growth during the industrial expansion, and then to a gradual slowdown as the economy transitions toward services. Second, I compare the model's household education choice with the average of primary and secondary enrollment rates over 1825–2020. The correlation between the model's education choice and enrollment is 0.94, so the quantity–quality household block captures the secular expansion of education remarkably well despite the historical education path being untargeted. Third, the model generates substantial cross-firm dispersion in formalization and a secular increase in

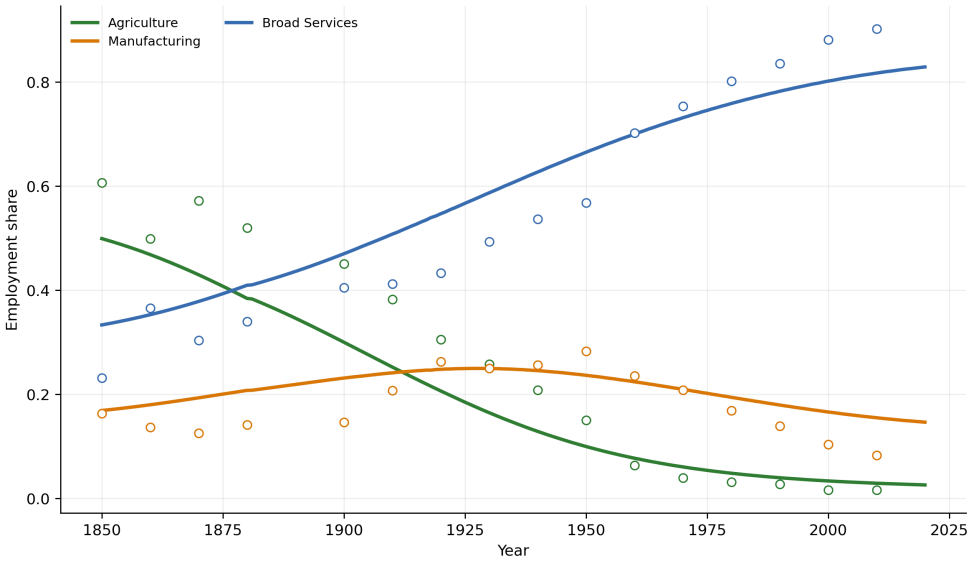


Figure 5: Fit to U.S. structural change. Lines are model outcomes and circles are historical employment shares. The sectoral employment paths enter the calibration objective.

the dispersion of firm size, both untargeted and well-known empirical features of firm dynamics.[23] Finally, the model captures the rise and subsequent decline of average manufacturing firm size and the growth of the HSPS employment share even though only the terminal observations of these series are targeted.[24] The next subsection provides the strongest untargeted validation of the model's novel mechanism by showing that the accumulation of public organizational knowledge in the model closely tracks the historical production of books and manuals codifying new methods of business organization and management in the data.

4.4 Organizational knowledge in the historical record

I assess the timing of public organizational knowledge using an independent historical series. The model's annual increment,

$$\Delta K_t \equiv K_{t+1} - K_t, \tag{33}$$

measures additions to the stock of public knowledge concerning business organization which firms can draw on to formalize production. This object has a natural empirical counterpart: the publication of books and manuals through which methods of business organization and management became codified and publicly available. I construct this series from digitized volumes of the Library of Congress *Catalog of Copyright Entries*. Individual titles are parsed and classified using terms associated with accounting and record keeping, office and business systems, management methods, and the organization of sales, personnel, and multi-unit firms. The classification includes general Taylorist concepts such as "scientific

23 See, for instance, Poschke (2018) for the secular rise in firm-size dispersion and Bloom et al. (2012) for dispersion in management practices across firms. The employment-weighted standard deviation of firm-level formalization in the model is 0.176 in agriculture, 0.187 in manufacturing, and 0.049 in services. I construct data-based proxies of 0.229, 0.205, and 0.211 using the DEFRA Farm Business Survey 2024/25 for agriculture and the ONS Management and Expectations Survey 2019 for manufacturing and services. None of these moments enters the calibration objective.

24 For firm size, the initial level is normalized to its value in the data, so the untargeted validation comes from the subsequent path, in particular the decline in manufacturing firm size after the manufacturing peak.

management," "time study," and "motion study," and excludes titles containing explicit manufacturing or industrial language.[25] This produces a measure of organizational knowledge that is not mechanically driven by references to manufacturing. Using the same underlying data source and classification method, I also construct analogous series for 11 different placebo topics ranging from children's literature to fields which are more plausibly connected to manufacturing such as chemistry and applied physics.

Because the number and size of digitized catalog volumes vary across years, I measure publication intensity rather than raw counts. For publication category g, define

$$P_{gt} \equiv \frac{\text{matched copyright records}_{gt}}{\text{OCR words}_t}. \tag{34}$$

All categories use the same annual support and OCR denominator. The common sample contains 72 annual observations from 1906 through 1977.

Figure 6 presents the comparison. Panel A compares ΔK_t with business-organization publication intensity. The displayed lines are centered five-year averages normalized by their respective maxima; the reported correlation uses the unsmoothed annual observations. The correlation is 0.752. Both series rise during the first half of the twentieth century, peak around mid-century, and subsequently decline.

Panel B evaluates the specificity of this relationship using eleven placebo categories spanning scientific, technical, household, and literary subjects. Business-organization publications rank first, with an annual correlation of 0.752. The closest placebo is household manuals, with a correlation of 0.486; all remaining correlations are 0.416 or lower. In one-sided comparisons using Newey–West standard errors with four annual lags, the business correlation exceeds each placebo correlation. The largest comparison p-value is 0.069. The ranking does not depend solely on the common hump: using five-year log changes, business organization remains first ($r = 0.543$, compared with 0.355 for fiction and novels), and the difference is significant using Newey–West standard errors with five lags (one-sided $p = 0.033$). During the period in which the model predicts rapid additions to public organizational knowledge, the historical record shifts disproportionately toward works about business organization. This pattern reflects the co-movement of manufacturing employment and organizational-knowledge production that appears in both the model and the data. To the best of my knowledge, this is the first study to document that the production of codified business-organization knowledge moved systematically with the rise and decline of manufacturing employment and to relate that historical comovement to the knowledge flow generated by a structural model.

5 Results

5.1 Manufacturing exposure and development outcomes

I use the calibrated model to quantify how the historical importance of manufacturing shapes subsequent development. Starting from the U.S. calibration, I vary only the manufacturing weight in final demand, w_m. This traces an equilibrium locus of otherwise identical economies with different endogenous peak manufacturing employment shares.[26] The eight model economies in Figure 7 span manufacturing peaks from 6.35 to 38.80 percent of employment. Larger manufacturing peaks are associated with

25 For instance titles including the words "industrial, manufacturing, factory, etc." are excluded.

26 The experiment also has a price-wedge interpretation. Suppose the manufacturing demand weight changes from w_m to $f_m w_m$, and let $k_j \equiv \varepsilon_j/\varepsilon_m$. Holding w_m fixed, the same relative final-demand schedules are obtained by applying wedges to

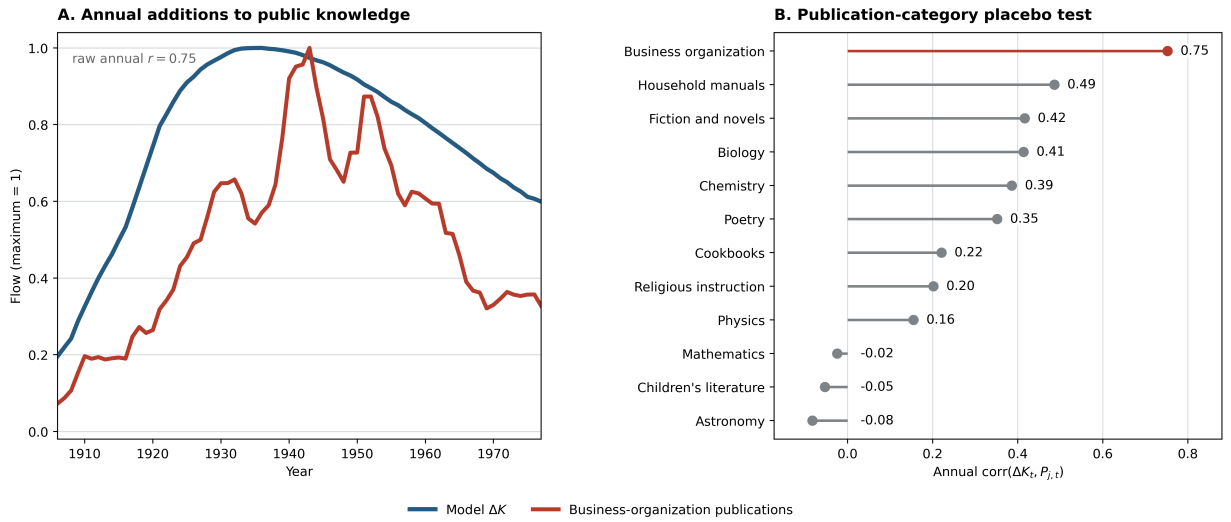


Figure 6: Public organizational knowledge and copyright publications. Panel A compares the annual increase in the model's public knowledge stock with business-organization publication intensity. The displayed series are centered five-year averages normalized by their own maxima; the reported correlation uses unsmoothed annual observations. Panel B reports the corresponding annual correlation for the business-organization category and eleven placebo publication categories. All publication series use identical processing and the common 1906–1977 support.

greater tertiary education, a composition of services tilted towards HSPS employment, higher 2019 GDP per capita, and faster GDP-per-capita growth over 2010–2019. They are also associated with lower population growth over 1850–2019, as in the data. In the model, deeper industrialization raises productivity growth, which increases the return to education and lowers fertility. The model performs best on the outcomes closest to the organizational mechanism: it accounts for 98 percent of the empirical HSPS gradient and 32 percent of the tertiary-education gradient. It also accounts for 40 percent of the 2010–2019 GDP-per-capita growth gradient, 17 percent of the 2019 GDP-per-capita gradient, and 17.6 percent of the population-growth gradient. Variation in manufacturing history therefore generates a substantial share of the observed cross-country differences in service composition, higher education, and growth in income per capita, while accounting for a smaller, but still substantial, share of differences in income levels and demographic change.

The cross-country accounting exercise describes differences observed around the end of the calibration period, but the model implies that the consequences of industrial history persist beyond that point. Because a deeper manufacturing phase leaves the service economy with a larger skilled share, Proposition 2 implies a permanently higher limiting productivity-growth rate. Table 3 illustrates this mechanism by comparing two otherwise identical economies with different manufacturing histories. The low-peak economy reaches a manufacturing employment share of 11.19 percent, close to the average peak among low- and middle-income economies, while the high-peak economy reaches 29.88 percent. The high-peak

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the prices of agriculture and consumer services such that

$$1+\tau_a = f_m^{-k_a/(1-\sigma)}, \qquad 1+\tau_c = f_m^{-k_c/(1-\sigma)}.$$

Varying w_m is therefore isomorphic, for household final demand, to varying relative price wedges on agricultural and service consumption. The exercise can be read as a parsimonious change in the relative pull of manufacturing, rather than as a structural change in preferences. Appendix D.4 derives this equivalence.

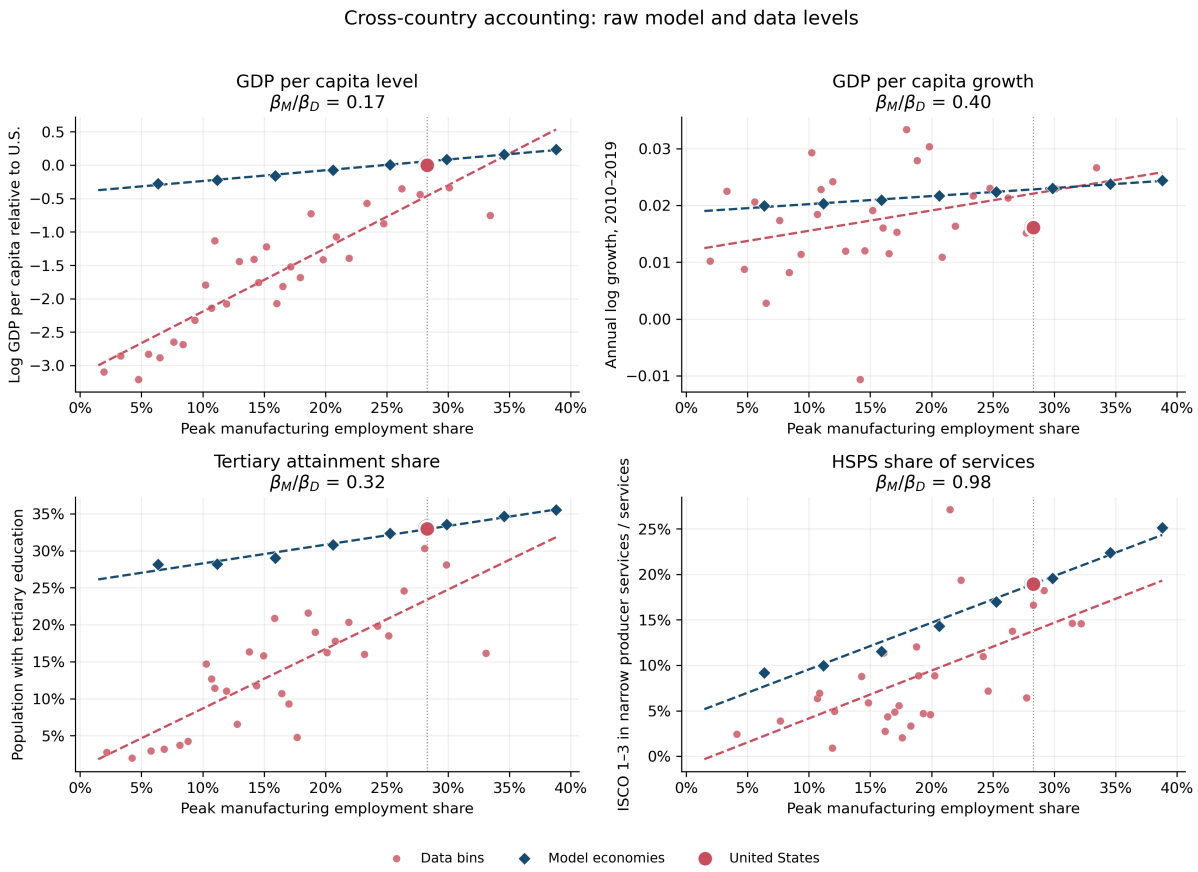


Figure 7: Manufacturing peaks and development outcomes. Red circles show means within 30 approximately equal-sized country bins; blue diamonds show model economies. Dashed lines are OLS fits to the underlying observations; β_M/β_D denotes the ratio of model to empirical slopes. The large red circle identifies the United States. Manufacturing peaks use the harmonized historical-employment series described in Appendix A.1. GDP levels (2019, in logs relative to the United States) and annual log growth (2010–2019) use Gapminder's PPP GDP-per-capita series. Completed tertiary attainment averages Barro–Lee observations for 2010 and 2015. HSPS is employment in ISCO-08 groups 1–3 within ISIC Rev. 4 divisions 62–66 and 69–75, divided by employment in market services (sections G–N and R–S), using ILOSTAT data averaged over 2010–2019. Samples comprise 172 countries for GDP, 137 for education, and 48 for HSPS. U.S. manufacturing and education observations use the empirical calibration targets.

economy enters the service era with a persistently larger skilled-employment share and therefore grows faster thereafter. The annual growth gap is modest—0.25 percentage points in 2030 and 0.17 points in 2070—but compounds over time: GDP per capita in the high-peak economy rises from 39 percent above the low-peak economy in 2030 to 51 percent above it by 2070. In the model, industrial history therefore affects not only the level of income reached by the end of industrialization, but also the subsequent growth path of the service economy.

5.2 Which mechanisms matter?

To understand which source of sector-heterogeneity is driving the results, I repeat the manufacturing-demand experiment after removing one source of heterogeneity at a time. Each experiment equalizes one of the following: entry barriers F_j, sectoral productivity growth ϕ_j, adaptability ξ_j, or value-added

Table 3: Manufacturing history and subsequent growth paths

Economy	Outcome	2030	2050	2070
High-peak economy	GDP per capita	50.36	78.43	121.80
	Skilled-employment share (%)	34.18	34.64	35.02
	GDP-per-capita growth (% p.a.)	2.24	2.22	2.20
Low-peak economy	GDP per capita	36.12	53.88	80.72
	Skilled-employment share (%)	28.74	29.37	29.83
	GDP-per-capita growth (% p.a.)	1.99	2.01	2.03
High–low comparison	GDP per capita (low peak = 100)	139.44	145.57	150.89
	Skilled-employment-share gap (pp)	5.44	5.27	5.19
	GDP-per-capita growth gap (pp p.a.)	0.25	0.21	0.17

Notes: The low-peak economy sets $w_m = 0.60$ and reaches a manufacturing-employment peak of 11.19 percent in 1941. The high-peak economy sets $w_m = 3.49$ and reaches a peak of 29.88 percent in 1927. GDP-per-capita levels equal nominal GDP divided by the consumption price index and population. The skilled-employment share is the model stock S_t. GDP-per-capita growth is the annualized log change in chain-weighted real GDP per capita over the preceding ten years. In the GDP-per-capita comparison, the low-peak economy is normalized to 100. The skill and growth comparisons report high-minus-low differences in percentage points.

skill intensity α_j. The final experiment sets $\delta_K = 0$, eliminating the diffusion of newly formalized routines into public organizational knowledge. All other parameters remain at their calibrated U.S. values. Because each intervention changes the mapping from w_m into structural transformation, every coefficient is estimated against the counterfactual economy's realized manufacturing peak. Table 4 shows the results.

The results in Table 4 isolate one source of cross-sector heterogeneity as the main force behind the paper's quantitative cross-country accounting results. Equalizing F_j leaves 49.0 percent of the GDP-level slope and 40.7 percent of the growth slope, *reverses* the tertiary-attainment slope, and reduces the HSPS slope to 12.7 percent of baseline. Thus, manufacturing exposure raises the skill-share and producer-service employment primarily when sectoral entry conditions allow manufacturing firms to operate at scale. Knowledge diffusion propagates this effect. Setting $\delta_K = 0$ preserves 71.3 percent of the GDP-level slope and 67.6 percent of the growth slope, but only 26.5 percent of the tertiary slope and 47.2 percent of the HSPS slope. Private formalization still depends on manufacturing exposure through the model's other channels, but the effect of the manufacturing legacy is substantially reduced. The population-growth relationship is driven more directly by sectoral productivity growth: equalizing ϕ_j reduces the population-growth slope to 1.5 percent of baseline. This is consistent with the demographic mechanism in the model, through which faster productivity growth raises the return to education and lowers fertility.

Equalizing value-added skill shares leaves all five slopes essentially unchanged or slightly larger on account of the large value-added skill-share of consumer services. Equal adaptability retains between 77.9 and 92.7 percent of the baseline slopes. Equal productivity growth reduces the GDP-level and tertiary slopes while increasing the growth and HSPS slopes. Sectoral entry conditions therefore generate the scale margin which makes manufacturing especially important and knowledge diffusion transmits its effects to the broader economy.

Table 4: Mechanism accounting for the model's cross-country slopes

Outcome	Baseline	Equal entry barriers	Equal productivity growth	Equal adaptability	Equal VA skill intensity	No knowledge spillover
GDP-per-capita level, 2019	1.720 [100.0]	0.843 [49.0]	1.314 [76.4]	1.594 [92.7]	1.871 [108.8]	1.225 [71.3]
GDP-per-capita growth, 2010–2019	0.015 [100.0]	0.006 [40.7]	0.018 [118.7]	0.013 [89.7]	0.015 [99.8]	0.010 [67.6]
Population growth, 1850–2019	−0.678 [100.0]	−0.244 [36.0]	−0.010 [1.5]	−0.630 [93.0]	−0.832 [122.8]	−0.460 [67.9]
Tertiary attainment, 2010–2019	0.301 [100.0]	−0.116 [−38.4]	0.246 [81.8]	0.235 [77.9]	0.340 [113.0]	0.080 [26.5]
HSPS share of services, 2010–2019	0.580 [100.0]	0.074 [12.7]	0.691 [119.1]	0.485 [83.6]	0.579 [99.7]	0.274 [47.2]

Notes: Each cell reports a model slope; brackets report its percentage of the baseline slope. Outcomes match Figure 7. All columns use the converged common support $w_m \in \{1.16, 1.8, 2.56, 3.49, 4.69\}$ and realized manufacturing peaks. Counterfactuals equalize F_j, ϕ_j, or α_j across sectors; set $\xi_a = \xi_c = \xi_m = 1$; or set $\delta_K = 0$, as indicated.

5.3 Robustness

The results are stable across a broad sensitivity battery. Across 18 economic variants all 90 cross-country slopes retain their benchmark sign. Median absolute changes are 2.9 percent for GDP levels, 8.5 percent for GDP growth, 10.2 percent for tertiary education, 2.4 percent for HSPS, and 4.3 percent for population growth. Excluding the two deliberately radical exercises—eliminating knowledge spillovers and replacing employment weights with equal sector weights—the largest absolute changes are 15.1, 17.5, 27.3, 16.4, and 13.7 percent, respectively. A separate exact grid over $\delta_K \in [0.01, 0.15]$ also preserves every sign. Appendix D.4 reports the complete sensitivity results.

6 Industrial Policy: Skill Supply and Firm Scale

The policy experiments separate the supply of skills from firms' demand for them. I consider interventions that operate on one or both of these margins and ask a simple question:

Can temporary development policies leave a persistent effect after they are withdrawn?

Recent work shows that expanding skill supply can induce firms to reorganize production and increase employment in larger, more organized firms (Engbom et al., 2025; Cox, 2025). Whether these effects persist after the expansion in skill supply ends is less clear. In general equilibrium, the skill premium adjusts as skilled workers become more abundant, so the initial increase in skills need not generate the demand required to sustain itself. The framework in this paper makes this distinction explicit. Proposition 2 implies that a temporary intervention can permanently raise the growth rate of the economy only if it changes the limiting skilled share. In the model, this can occur when the intervention induces firms to create organizational procedures that diffuse into the public stock of knowledge. The resulting organizational knowledge survives the policy itself, sustaining firms' demand for skilled labor after the original increase in skill supply has disappeared. If the skill expansion does not induce additional organizational investment, however, the increase in skilled employment and organized-firm activity is largely transitory: once the policy is withdrawn, both converge back toward their no-policy paths.

6.1 A LMIC policy laboratory

To study this interaction in a developing-country setting, I construct a lower-middle-income version of the benchmark economy. I introduce productivity-correlated firm distortions in the spirit of Bento and Restuccia (2017, 2021), which reduce the scale at which productive firms operate, and recalibrate

the model to representative LMIC moments. Specifically, I re-estimate the organizational-scale parameter, the manufacturing and service setup-cost shifters, the sectoral productivity-growth parameters, the manufacturing-demand shifter, and the skill-acquisition cost to match average firm size, sectoral productivity growth, the manufacturing-employment peak, the skilled share, and the HSPS share. The remaining structural parameters are inherited from the benchmark economy or fixed at their externally disciplined values. Appendix D.5 describes the distortion technology, parameterization, and recalibration in detail.

I first remove the productivity-correlated distortion without introducing any temporary skill or organizational policy. Beginning in 2026, the distortion is phased out linearly through 2035 while all other recalibrated LMIC parameters are held fixed. This experiment asks whether the increase in firm scale generated by reform is sufficient, on its own, to activate the organizational mechanism.

Table 5: Permanent distortion reform: scale without temporary policy

Outcome in 2089	Distortions remain	Full reform
Average manufacturing firm size	5.71	13.42
Average service firm size	1.89	4.48
GDP per capita relative to unreformed path (%)	0.00	13.64
Formalized task-share change (pp)	0.00	9.50
Skilled employment-share change (pp)	0.00	1.84
Public organizational knowledge, K	0.129	0.203

Notes: Full reform removes the excess productivity-correlated distortion linearly during 2026–2035. Firm-size and public-knowledge entries are levels. GDP, formalization, and skilled employment are measured relative to the unreformed LMIC path. No temporary policy is active in either economy. The institutional or administrative cost of the permanent reform is not modeled.

Table 5 shows that removing the distortion substantially raises firm scale and income. Average firm size more than doubles in manufacturing and services, and GDP per capita is 13.64 percent higher by 2089. The increase in scale also activates part of the organizational mechanism: the formalized-task share rises by 9.50 percentage points, the skilled-employment share by 1.84 percentage points, and public organizational knowledge increases from 0.129 to 0.203. Removing barriers to firm growth therefore does more than increase output; it also leaves the economy with greater organizational knowledge and a more skill-intensive workforce, persistently raising its growth rate.

6.2 Temporary policies

I next consider four temporary interventions operating from 2026 through 2045. A *college scholarship* lowers the private cost of acquiring skill. *Formalization support* reimburses part of the one-time cost of creating an organizational procedure. A *mixed package* combines the two instruments. Finally, a *manufacturing wage subsidy* reimburses part of the skilled and unskilled manufacturing wage bill and provides a benchmark for a policy targeted directly at manufacturing employment. Each intervention is evaluated both with the calibrated LMIC distortions left in place and together with the permanent distortion reform described above.

In the distorted LMIC economy, the scholarship raises the skilled share of recent cohorts by 7.30 percentage points by 2045 and increases the nonagricultural employment share of establishments with at least six workers that use skilled organizational tasks by 4.91 percentage points. The implied response,

$4.905/7.302 = 0.672$, is close to the 0.654 response obtained by Engbom et al. (2025) and the 0.730 estimate in Cox (2025).[27]

Policy rates are chosen so that discounted government expenditure corresponds to roughly 0.6 percent of GDP yearly while the programs are active over 2026–2045. Financing is done through contemporaneous lump-sum household taxes. To isolate the effect of the policy from any anticipatory behavior I consider an unexpected withdrawal design: through 2045 households and firms behave as if the policy will continue, and the subsidy is removed at the start of 2046. The post-withdrawal path therefore isolates persistence generated by the economic state accumulated while the intervention was in place rather than anticipation of a pre-announced termination date.

6.3 Persistence after policy withdrawal

Table 6 compares the effects of each intervention in 2045, the final year before withdrawal, with its long-run effects in 2089. All effects are measured relative to the no-temporary-policy path under the same distortion environment.

When the LMIC distortions remain, the scholarship reproduces the large organized-employer response documented by Cox (2025) and Engbom et al. (2025) while the policy is active, but most of that response subsequently disappears once the subsidy is withdrawn. Skilled employment is 6.58 percentage points above the no-policy path in 2045 and only 0.51 percentage points above it in 2089; employment inside firms with at least six workers similarly falls from 4.89 percent above control to 0.15 percent. A strong contemporaneous response to skill supply therefore need not imply a persistent change in firm structure. Direct support for formalization behaves differently, leaving public organizational knowledge 0.203 above control in 2089 and sustaining substantially larger effects on skilled employment and employment in large firms after withdrawal.

Removing the firm-growth distortion makes temporary policies more effective. The scholarship leaves skilled employment 3.36 percentage points and employment in six-plus-worker firms 5.26 percent above control in 2089. Formalization support produces the largest organizational legacy, whereas the mixed package generates the largest long-run GDP and welfare gains. Larger firms therefore do more than increase contemporaneous demand for skill: by increasing the return to reusable organizational procedures, they exploit a temporary expansion in skills to change the organizational technologies they employ, leaving behind a stock of knowledge which persists once the policy is withdrawn.

Manufacturing wage subsidies do not generate welfare gains or persistent changes in skill demand. They temporarily raise employment in larger firms but leave essentially no organizational knowledge and almost no long-run effect on skills or firm structure. The policy lesson is therefore that lasting gains come not from subsidizing manufacturing employment itself, but from policies that induce firms to reorganize production, changing their organizational technology and generating knowledge which leaves a persistent effect once support for skill expansion is withdrawn.

27 Cox estimates the change in the share of formal employment in firms with more than six workers per percentage-point increase in college attainment among 25–34-year-olds. Engbom et al. simulate the same education expansion and measure the corresponding change in employment in model firms with more than six workers. I measure the change in nonagricultural employment in establishments with at least six workers and a complete block of skilled organizational tasks per percentage-point increase in the recent skilled share. The three exercises therefore capture related employment reallocation measures. In my model, this reallocation is primarily driven by existing firms with more than six employees adopting the skilled organizational technology, rather than a move of employment from small to large firms.

Table 6: Temporary policies at withdrawal and in the long run

Temporary policy	CEV (%)	GDPpc 2089 (%)	Skilled emp. 2045 → 2089 (pp)	Emp. in 6+ firms 2045 → 2089 (%)	ΔK 2089
Panel A: LMIC distortions remain					
College scholarship	6.75	9.35	$6.58 \to 0.51$	$4.89 \to 0.15$	0.063
Formalization support	2.81	11.77	$3.59 \to 2.45$	$7.59 \to 2.92$	0.203
Mixed package	2.00	4.80	$5.16 \to 0.48$	$1.69 \to 0.39$	0.018
Manufacturing wage subsidy	-5.56	-3.69	$0.02 \to -0.02$	$5.75 \to 0.33$	0.000
Panel B: Full permanent distortion reform					
College scholarship	7.93	17.38	$8.34 \to 3.36$	$7.86 \to 5.26$	0.193
Formalization support	4.68	20.71	$5.57 \to 5.15$	$10.72 \to 8.12$	0.340
Mixed package	8.67	22.37	$9.19 \to 4.77$	$9.99 \to 7.53$	0.265
Manufacturing wage subsidy	-0.05	-0.21	$0.01 \to 0.09$	$0.88 \to 0.04$	0.003

Notes: Effects are measured relative to the no-temporary-policy path under the same distortion environment. Programs operate through 2045 and are withdrawn unexpectedly at the start of 2046. CEV is evaluated over 2026–2089. Policy rates are chosen to generate discounted government expenditure of approximately 0.6 percent of discounted GDP over 2026–2045. Formalization support under the permanent reform is evaluated at the highest stable reimbursement rate and spends 0.423 percent of GDP.

7 Conclusion

This paper asks whether economies that move directly from agriculture into services develop the same productive capabilities as economies that first pass through large-scale manufacturing. The evidence suggests that they do not. Across countries, larger historical manufacturing peaks are followed by service economies more concentrated in high-skill producer services and continue to grow faster after manufacturing declines. The same relationship between industrialization and the composition of services appears in U.S. counties, where larger manufacturing peaks are associated with greater managerial and technical employment within manufacturing around the peak and specialization in high-skill producer services decades later. The results are concentrated in manufacturing industries where production in large establishments was technologically important. These findings point to a persistent organizational legacy of production at scale: the capabilities needed to manage large manufacturing firms remain valuable as the economy transitions to services, sustaining demand for skills and supporting faster growth.

I formalize this mechanism in a growth model in which larger firms have stronger incentives to replace recurring organizational problems with reusable procedures, which can subsequently diffuse across firms and sectors. Calibrated to the U.S. transition, variation in manufacturing history generates almost all of the observed cross-country gradient in the HSPS share of services and substantial shares of the corresponding gradients in tertiary education and contemporary growth rates. Firm scale is central to the creation of organizational capabilities, while knowledge diffusion transmits its effects into the service economy that follows.

However, the fact that manufacturing historically generated these capabilities does not imply that manufacturing itself should be subsidized. Manufacturing wage subsidies are poorly targeted to the underlying externality. Lasting gains instead come from combining investments in skills with reforms that allow productive firms to grow and adopt better management practices. Development without industrialization is possible, but economies that bypass industrialization must still find other ways to build the capabilities that large-scale manufacturing historically created.

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A Data

This appendix defines the cross-country and U.S. county data, variables, and harmonizations. Data used only in the quantitative model are described in Appendix D.4.

A.1 Cross-country evidence

The facts in Section 2 use complementary samples. Service composition requires detailed sectors and uses the balanced Economic Transformation Database (ETD) panel; growth requires longer manufacturing histories and uses the broader combined GGDC–ILOSTAT sample.

A.1.1 Detailed sectoral employment and the composition of services

The service-composition evidence uses employment from the GGDC/UNU–WIDER ETD (de Vries et al., 2021; Kruse et al., 2022). The database reports annual persons engaged and value added in 12 sectors for 51 economies over 1990–2018: agriculture, mining, manufacturing, utilities, construction, trade, transport, business services, finance, real estate, government services, and other services.

High-skill producer-service (HSPS) employment comprises ETD business services, finance, and real estate; generic market services comprise trade and other services. Figure 1 plots

$$HSPS_{ct} = 100 \times \frac{EMP_{ct}^{Business} + EMP_{ct}^{Finance} + EMP_{ct}^{RealEstate}}{EMP_{ct}^{Total}}, \tag{35}$$

$$GenericServices_{ct} = 100 \times \frac{EMP_{ct}^{Trade} + EMP_{ct}^{Other}}{EMP_{ct}^{Total}}. \tag{36}$$

Both outcomes use total employment as the denominator. Trade includes accommodation and food services; transport and government services enter neither numerator.

For each of the 51 ETD economies, the historical manufacturing peak uses the full combined GGDC 10-Sector–ETD employment history, with ETD taking precedence in overlapping years. Splitting at the median peak places 26 economies, including the median country, in the above-median group and 25 in the below-median group.

The dynamic comparison subtracts each country's 1990 employment share before taking unweighted group means. Shaded bands in Figure 1 are the mean plus or minus 1.96 cross-country standard errors.

A.1.2 Income, growth, population, education, and institutions

PPP real GDP per capita and population come from Gapminder's harmonized series, combining modern World Bank estimates with historical income series. Average annual log GDP-per-capita growth over 2010–2019 is

$$g_c = \frac{100}{9} \left[\log y_{c,2019} - \log y_{c,2010}\right]. \tag{37}$$

The service-employment regressor is the mean ILOSTAT service share over 2010–2019; later observations are excluded. Income-quintile fixed effects use terminal GDP per capita, and years since the manufacturing peak equal $2019 - P_c$.

Institutional quality is the World Bank Worldwide Governance Indicators control-of-corruption percentile rank. Initial population and GDP per capita in the service-composition robustness regressions

are measured in 1990; initial tertiary education is World Bank gross tertiary enrollment in, or closest to, 1990.

A.2 U.S. county evidence

Section 2.2 combines Census of Manufactures employment and establishments, population-census classified employment and manufacturing occupations, and Census and County Business Patterns (CBP) HSPS-service employment for the long event study. The Quarterly Census of Employment and Wages (QCEW) independently supplies the 2000 service-composition outcome in Table 1. Unless stated otherwise, additive counts are aggregated to fixed geography before ratios are formed.

A.2.1 Fixed county geography

Historical county identifiers and boundary crosswalks come from IPUMS NHGIS (Schroeder et al., 2026). For historical censuses and Census of Manufactures tabulations, I assign each source-year county to the 1850 county with the greatest geographic overlap and sum additive counts. This accommodates microdata that locate individuals only by county of residence. Modern establishment counts are allocated across fixed 1850 counties using polygon-overlap shares. Numerators and denominators are mapped separately before forming ratios; employment shares are never interpolated directly.

A.2.2 Classified employment and the manufacturing peak

Manufacturing employment, establishments, wage bills, population, and urban population come from the historical county files of Haines et al. (2010), which compile population censuses and Census of Manufactures tabulations. Manufacturing employment is the average number of wage earners in the Census of Manufactures. Total classified employment comprises employed residents consistently assignable to primary industries, manufacturing, or services through harmonized IND1950:

$$\begin{aligned} \text{primary industries:} &\quad \{105, 116, 126\}, \\ \text{manufacturing:} &\quad 306\text{–}499, \\ \text{services:} &\quad 506\text{–}946. \end{aligned}$$

Mining, construction, unclassified employment, and industries outside these groups are excluded. This denominator provides the common economic base for peak manufacturing exposure and manufacturing HSPS.

For county c in year t, define

$$M_{ct} = \frac{L_{ct}^{M}}{L_{ct}^{C}}, \tag{38}$$

where L_{ct}^{M} is manufacturing employment and L_{ct}^{C} is classified employment. The manufacturing peak and its date are

$$M_{c}^{\text{peak}} = \max_{t \in \{1880,1900,1920,1930,1940\}} M_{ct}, \qquad P_{c} = \min\{t : M_{ct} = M_{c}^{\text{peak}}\}. \tag{39}$$

Ties are assigned to the earliest peak year. The largely destroyed 1890 population census cannot supply classified employment, and the required county manufacturing tabulations are unavailable for 1910, so

neither year enters the peak search. Establishment scale is log manufacturing workers per establishment in P_c; peak exposure, scale, and manufacturing HSPS therefore refer to the same county-year.

A.2.3 HSPS capability inside manufacturing

Occupation-by-industry counts use IPUMS Ancestry Full Count censuses through 1940 and harmonized IPUMS USA files thereafter (Ruggles et al., 2024, 2025). All employed individuals enter regardless of household role, as counts in full-count years and with person weights in sample years. Industries and occupations use harmonized IND1950 and OCC1950.

Manufacturing HSPS counts the accounting, coordination, personnel, technical, and scientific occupations in Table A.1 among workers in IND1950 industries 306–499:

$$MHSPS_{ct} = 100 \times \frac{\text{core HSPS-occupation workers in manufacturing}_{ct}}{L^C_{ct}}. \tag{40}$$

The event study uses this definition. The occupation set is fixed over time by function within manufacturing, independently of census-year educational attainment.

Table A.1: Occupation-based manufacturing HSPS classification

Function	Exact occupation content	OCC1950 codes
Accounting and record keeping	Accountants and auditors; bookkeepers; bill and account collectors; office-machine operators; shipping and receiving clerks	0, 310, 321, 341, 342
Purchasing, logistics, and coordination	Buyers and department heads in stores; buyers and shippers of farm products; purchasing agents and buyers not elsewhere classified; dispatchers and starters of vehicles	200, 201, 280, 322
Personnel administration	Personnel and labor-relations workers	72
Technical and scientific work	Draftsmen; engineers by specialty; statisticians and actuaries; surveyors; testing technicians; technicians not elsewhere classified; inspectors not elsewhere classified	35, 41–49, 83, 92, 95, 96, 533

Notes: Manufacturing is defined by IND1950 codes 306–499. The four groups jointly form the core manufacturing-HSPS measure used throughout the main analysis. Legal occupations and commercial intermediaries are not included.

A.2.4 HSPS service industries and the long event-study panel

Modern HSPS services are industry based: Information; Professional, Scientific, and Technical Services; and Management of Companies and Enterprises (NAICS 51, 54, and 55). These sectors supply information, professional and technical expertise, and organizational direction. NAICS 54 includes accounting, legal, engineering, scientific, technical, and consulting suppliers; NAICS 55 identifies enterprise management and coordination; and NAICS 51 covers information production and transmission. NAICS 56 is excluded because its two-digit aggregate combines business support with routine administration, cleaning, security, and waste management. NAICS 52–53 are excluded because they include substantial financial intermediation, insurance distribution, property operation, and rental activity outside this organizational and technical core.

The manufacturing and service measures are functional analogues: occupations identify capabilities within manufacturing firms, while industries identify specialized establishments supplying related information, technical, and organizational services. The latter is the narrow U.S. analogue of the business-and-professional component of the cross-country HSPS measure in Figure 1.

To measure when HSPS services become quantitatively important in the county economy, the event study divides their employment by classified employment:

$$SHSPS_{ct} = 100 \times \frac{\text{employment in HSPS service industries}_{ct}}{L^{C}_{ct}}. \tag{41}$$

The service numerator uses population-census industries through 1950, imputed early CBP in 1960 and 1970, and the harmonized, suppression-imputed CBP panel of Eckert et al. (2020) (EFSY) thereafter. EFSY supplies a common postwar industry classification and imputes confidential cells subject to CBP hierarchy adding-up restrictions. Table A.2 gives the concordance.

Table A.2: HSPS service-industry measurement by data vintage

Event-study year	Source and classification	HSPS service-industry numerator
1850–1950	Population-census microdata; harmonized IND1950	Communications, advertising, accounting, business, legal, engineering, and professional and technical services: codes 578, 579, 806, 807, 808, 856, 857, 879, 898, and 899.
1960	Imputed 1959 CBP; SIC	Communications, business services, legal services, and engineering, accounting, research, and related professional services: SIC groups 4800, 7300, 8100, and 8900.
1970	Imputed 1970 CBP; SIC	The same SIC analogue used for 1960.
1980–2000	EFSY harmonized and imputed CBP; common 2012 NAICS	All mutually exclusive EFSY employment cells beginning with NAICS 51, 54, or 55.

Notes: The early CBP files are from Eckert et al. (2022); the suppression imputations and common-NAICS panel are from Eckert et al. (2020). The EFSY long panel stores employment in mutually exclusive cells at the most detailed assignable industry level, so residual aggregate cells do not duplicate their descendants. Management companies are not separately identified in the historical IND1950 and SIC classifications, and those years use the closest functional analogue. IND1950 code 808 includes some activities later assigned to NAICS 56 that cannot be separated in the historical classification.

A.2.5 The QCEW long-run service-composition outcome

The outcome variable in Table 1 measures HSPS employment within total private services. The 2000 endpoint uses public BLS QCEW files (U.S. Bureau of Labor Statistics, 2004), retaining the modern HSPS numerator. QCEW directly publishes a county private service-providing aggregate; the EFSY denominator instead requires summing mutually exclusive imputed CBP components.

For county c in year t, define

$$Y^{QCEW}_{ct} = \frac{E^{\text{public}}_{c,51,t} + E^{\text{public}}_{c,54,t} + E^{\text{public}}_{c,55,t}}{E^{\text{public}}_{c,102,t} - E^{\text{public}}_{c,22,t}}, \tag{42}$$

where ownership code 5 restricts all components to private employment, code 102 is QCEW's service-providing aggregate, and code 22 is private utilities. The denominator includes wholesale and retail trade,

private transportation and warehousing, information, finance, real estate, professional and business services, private education and health, leisure, hospitality, and other private services. Private ownership excludes public administration and the U.S. Postal Service.

The main-text numerator sums public numeric records in NAICS 51, 54, and 55, without estimating latent confidential employment. In the reconstructed 2000 file, records failing disclosure are absent; later files identify suppressed records with an N flag and a zero numeric field. Unpublished contributions therefore enter this sum at zero even though the disclosed aggregate denominator includes them. Section C.2.5 uses the first explicit-flag year for QCEW-specific hierarchical imputation; Section C.2.6 compares QCEW with EFSY at county and commuting-zone levels where data suppression issues are much less common.

The unweighted outcome mean is 6.36 percent, reflecting equal county weights and greater disclosure losses in small cells. Private-service-employment weighting yields 13.97 percent. Summing available public county records before forming the ratio yields 14.02 percent, versus 14.58 percent from official national QCEW aggregates. Thus almost all national HSPS employment is observed and disclosure gaps concentrate in small counties. This aggregate comparison does not establish innocuous county measurement error; the flag-aware and commuting-zone checks address that concern.

A.2.6 Historical controls and estimation samples

The baseline county covariates are measured in 1850 and come from the Haines files: log population, the urban population share, the school-enrollment rate, the number of colleges, and log enslaved population plus one. The event study additionally includes predetermined 1870 employment shares in process, machinery-producing, and traditional manufacturing. Each baseline covariate is interacted with a linear calendar-time trend in the event-study specification.

The event-study sample contains 1,116 fixed-geography counties peaking in 1900, 1920, or 1940. These cohorts supply a common forty-year pre-period and have the treatment and baseline controls observed. Event time -40 is omitted. Outcome availability varies across county-event-time cells because the manufacturing and service series draw on different source years. Confidence intervals use a spatial covariance estimator allowing arbitrary covariance among observations whose county centroids lie within 100 kilometers.

Table 1 uses the same 1,487 counties. The common sample requires the QCEW 2000 outcome, the manufacturing peak, peak-year establishment scale and manufacturing HSPS, the 1850 covariates, county coordinates, all three shift-share measures, and the complete pre-1880 histories used in the IV specifications. Holding this sample fixed ensures that differences across columns do not reflect changes in county composition. Cross-sectional inference uses Conley standard errors with a 100-kilometer cutoff.

A.3 The Chandler throughput shift-share

The shift-share design begins with county-by-manufacturing-industry employment in the 1880 full-count Census of Population, harmonized to IND1950. Workers with unclassifiable industries do not enter the county-industry counts. Following Chandler (1990), I distinguish industries according to the production regime through which national expansion occurred. High-throughput industries are those in which continuous-process or large-batch production, or the fabrication and assembly of complex standardized machinery, required large fixed investments and extensive managerial coordination. Traditional low-throughput industries comprise wood products, textiles and apparel, printing, and leather, where

increasing output remained more readily divisible across smaller establishments. Table A.3 reports the concordance and source for every broad industry. The all-industry instrument uses every classifiable manufacturing industry in IND1950 codes 306–499.

Table A.3: Manufacturing industries in the Chandler throughput design

Group	Broad industry	IND1950 codes	Chandler source and classification rationale
High throughput	Glass and cement	316–317	Chandler (1990, pp. 20, 113–120). Chandler identifies plate and flat glass as scale intensive and describes continuous-process technologies and substantial scale economies in glass and cement.
High throughput	Primary iron, steel, and nonferrous metals	336–338	Chandler (1990, pp. 20, 23, 70–71, 121–140). Refining and smelting, including the Bessemer and open-hearth processes and electrolytic refining, required capital-intensive plants operating at high throughput.
High throughput	Nonelectrical machinery	356–358	Chandler (1990, pp. 23, 65–68, 194–205). Agricultural, office, and industrial machinery exemplify complex standardized machinery fabricated and assembled from interchangeable parts and supported by specialized distribution and service organizations.
High throughput	Electrical machinery	367	Chandler (1990, pp. 20, 68–69, 212–220). Electrical-equipment producers concentrated production in large plants and required technically trained organizations to design, manufacture, install, and service complex equipment.
High throughput	Transportation equipment	376–379	Chandler (1990, pp. 20, 205–212). These codes concord to automobiles, aircraft, ships, and railroad equipment. Production required large-scale fabrication and assembly and extensive managerial coordination.
High throughput	Photographic equipment	387	Chandler (1990, pp. 20, 67). Chandler identifies mass-produced cameras as scale intensive and discusses Eastman Kodak's mass production of cameras and photographic film.

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Table A.3 – continued from previous page

Group	Broad industry	IND1950 codes	Chandler source and classification rationale
High throughput	Processed foods	406–409	Chandler (1990, pp. 23, 62–65). Meatpacking, dairy processing, canning, and grain milling relied on mechanized processing and packaging, high-volume plants, and coordinated purchasing, distribution, and national marketing.
High throughput	Beverages	418	Chandler (1990, pp. 23, 64–65). Chandler places brewing and distilling among refining and processing industries and emphasizes the production and refrigerated-distribution networks constructed by national brewers.
High throughput	Tobacco	429	Chandler (1990, pp. 20, 23, 63–65). Mechanized processing and packaging permitted high-volume cigarette production for national markets.
High throughput	Pulp and paper mills	456	Chandler (1990, pp. 20, 112–113). Chandler identifies newsprint and describes pulp, newsprint, and paperboard as acquiring substantial scale economies after the transition to capital-intensive machinery.
High throughput	Chemicals, synthetic fibers, drugs, and paint	466–469	Chandler (1990, pp. 20, 23–26, 69–70, 170–193). Chemical production is a canonical continuous- and large-batch activity in which interdependent processes generated economies of scale and scope.
High throughput	Petroleum refining and coal products	476–477	Chandler (1990, pp. 20, 23–26, 94–104). Petroleum refining is Chandler's canonical illustration of throughput economies: large sunk investments were profitable only when managerial coordination maintained continuous flows from inputs through refining and distribution.
High throughput	Rubber products	478	Chandler (1990, pp. 20, 105–111). Chandler singles out tires and documents the large-batch processing, integrated distribution, and organizational investments of leading rubber producers.

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Table A.3 – continued from previous page

Group	Broad industry	IND1950 codes	Chandler source and classification rationale
Traditional low throughput	Logging, wood products, and furniture	306–309	Chandler (1990, pp. 20, 22–23). Chandler places lumber and furniture among traditional labor-intensive industries in which larger establishments obtained limited cost advantages over smaller ones.
Traditional low throughput	Textiles and apparel	436–439, 446, 448–449	Chandler (1990, pp. 20, 22–23). Textiles from natural fibers and apparel are labor intensive: increasing output primarily required adding machines and workers rather than exploiting large throughput economies.
Traditional low throughput	Printing and publishing	459	Chandler (1990, pp. 20, 22–23). Chandler includes printing and publishing among labor-intensive industries in which scale advantages were present but not substantial.
Traditional low throughput	Leather and footwear	487–489	Chandler (1990, pp. 20, 105–106). Chandler contrasts concentrated rubber footwear with leather footwear, where scale economies remained limited and production was dispersed across establishments.

Notes: Chandler states the general distinction on pp. 22–24: high-throughput industries used capital-intensive continuous or large-batch processes, or fabricated and assembled complex standardized machinery, whereas traditional industries remained comparatively labor intensive and obtained smaller cost advantages from greater plant size. Chandler's categories are expressed in the historical classifications used in *Scale and Scope*; the IND1950 cells are the paper's concordance to those categories. The baseline retains only cells that can be matched unambiguously. Ambiguous adjacent industries are excluded from both groups. Page references use the printed pagination of Chandler (1990).

For production regime $g \in \{A, H, L\}$, let $L_{cj,1880}$ denote county c's employment in manufacturing industry j in 1880 and define

$$\omega^{g}_{cj,1880} = \frac{L_{cj,1880}}{\sum_{\ell \in g} L_{c\ell,1880}}. \tag{43}$$

Here A denotes all manufacturing industries, H the high-throughput group, and L the traditional low-throughput group. The leave-county-out national shift is

$$G_{j,-c,1880\text{–}1900} = \frac{L_{-c,j,1900}}{L_{-c,j,1880}} - 1, \tag{44}$$

and the instrument is

$$Z^{g}_{c} = \sum_{j \in g} \omega^{g}_{cj,1880} G_{j,-c,1880\text{–}1900}. \tag{45}$$

Within-group normalization removes mechanical variation in the initial size of the high- and low-throughput regimes: the instruments compare initial industry composition rather than initial regime

size. All three use the same base year, horizon, leave-county-out rule, and normalization and are standardized within the estimation sample.

The 1880 base predates the observed manufacturing peak for 89 percent of counties and coincides with the earliest observed peak for the remainder. Removing county c from each national shift prevents its own later employment growth from entering its instrument mechanically. The preferred specification includes state and peak-cohort fixed effects; the five 1850 covariates defined above; and complete 1850, 1860, and 1870 histories of manufacturing employment and manufacturing HSPS relative to classified employment and of HSPS services relative to private-service employment. The identifying assumption is conditional: after accounting for location, initial development, and these observed pre-1880 trajectories, differential national growth across the industries represented in a county's 1880 manufacturing mix affects service composition through the local manufacturing expansion it induces.

A.4 Treatment of missing, confidential, and structurally zero observations

I distinguish a structural zero from an unpublished positive count. In full-count Census tabulations, the absence of workers in an otherwise observed county-industry cell contributes zero. In QCEW files with disclosure flags, an N-coded record is confidential and its numeric employment field is not treated as an observed zero in the flag-aware exercises below. An absent county-industry macro record is treated as a structural absence under the BLS publication convention. I do not replace an unavailable county outcome or denominator with zero. Additive counts are aggregated before ratios are calculated, and a ratio is missing whenever its denominator is unavailable or nonpositive.

The main-text 2000 QCEW numerator sums released numeric values without allocating employment to omitted detailed cells. EFSY instead uses constrained CBP imputations. The QCEW robustness exercises preserve disclosed counts, identify unknown cells through explicit flags, and allocate only the residual required by disclosed parents. The sole sample-preserving imputation for historical regressors replaces 48 unavailable peak-year manufacturing wages with peak-cohort medians and includes a replacement indicator.

B Additional Cross-Country Evidence

This appendix describes the two country samples used in Section 2 and reports the additional comparisons supporting its two facts: HSPS-employment growth conditional on income growth, peak timing, fixed effects, and initial service-sector size; the contribution of individual service industries; and GDP-per-capita growth in service economies, including those whose manufacturing peaks occurred before 2000.

B.1 Data, historical manufacturing peaks, and samples

The service-employment analysis uses all 51 economies in the GGDC/UNU–WIDER Economic Transformation Database (ETD), observed annually from 1990 through 2018. HSPS employment is the sum of business services, finance, and real estate. Generic market services comprise trade and other services; trade includes accommodation and food services. Both aggregates are expressed as shares of total employment. Transport and government services enter neither aggregate. For the service-sector-size control, I

additionally include transport, so that non-government services comprise trade, transport, business services, finance, real estate, and other services. Government employment remains in the total-employment denominator.

The growth analysis uses a separate manufacturing history. Manufacturing and service-employment data come from ILO series distributed by Gapminder, and real PPP GDP per capita comes from the World Bank via Gapminder. Where available, I use GGDC 10-sector observations to construct longer manufacturing employment share time series. A country's peak manufacturing employment share is the maximum of the longest data series available for that country. The Gapminder manufacturing series begins in 1991, and earlier observations come from GGDC where available.

The growth scatter contains 174 countries with complete plotted variables. Its median cutoff, 15.1 percent, is calculated among the 178 countries with recorded manufacturing peaks before the plotting restrictions. The regressions use 173 countries with complete outcomes and controls; South Sudan is excluded because its World Bank income classification based on 2009 GNI per capita (FY2011) is unavailable. Of these 173 countries, 104 have average service employment above 50 percent over 2010–2019, and 90 also have a recorded manufacturing peak before 2000.

Recorded maxima need not identify historical turning points. In the 174-country growth-scatter sample, 108 recorded peaks occur in the 1990s, including 79 at the first available observation. Source coverage and differences between historical series therefore limit the interpretation of peak dates.

B.2 Manufacturing peaks and HSPS-employment growth

Let H_{ct} denote HSPS employment as a percentage of total employment, $\Delta H_c = H_{c,2018} - H_{c,1990}$, and P_c the historical manufacturing peak in percentage points. Let T_c be the recorded peak year and $\tau_c = (T_c - 1990)/10$. The most extensive specification is

$$\Delta H_c = \alpha + \beta(P_c/10) + \gamma \Delta \ln y_c + \rho\tau_c + \theta S_{c,1990} + \delta_{r(c)} + \lambda_{g(c,1990)} + \varepsilon_c, \tag{46}$$

where $\Delta \ln y_c = \ln y_{c,2018} - \ln y_{c,1990}$ and $S_{c,1990}$ is non-government service employment as a percentage of total employment in 1990. GDP per capita is from the Maddison Project Database 2023, which supplies both endpoints for all 51 economies. Region effects distinguish Africa, the Americas, and Asia; initial income-group effects use the four World Bank categories based on 1990 GNI per capita (FY1992). The two sets of effects enter additively. Peak timing enters linearly; its coefficient is scaled per ten calendar years later.

Table B.1 adds income growth in column (2), peak timing in column (3), region and initial income-group effects in column (4), and the initial service share in column (5). All columns retain the same 51 countries, give them equal weights, and use HC3 standard errors with residual-degrees-of-freedom t tests.

A ten-percentage-point larger manufacturing peak is associated with a 2.87-percentage-point greater increase in the HSPS share between 1990 and 2018 without controls. With the full set of controls, including peak timing, the estimate is 2.26 percentage points (HC3 standard error 0.95; $p = 0.022$). The positive association thus remains statistically significant at the five-percent level.

Disaggregating service-employment growth.

For each non-government service industry j, I estimate $\Delta s_{cj} = \alpha_j + \beta_j(P_c/10) + \varepsilon_{cj}$, where Δs_{cj} is the 1990–2018 change in the industry's share of total employment, in percentage points. These separate

Table B.1: Manufacturing peaks and HSPS-employment growth

	Change in HSPS share of total employment, 1990–2018				
	(1)	(2)	(3)	(4)	(5)
Manufacturing peak (10 pp)	2.867***	2.937***	2.596***	2.117**	2.261**
	(0.507)	(0.501)	(0.616)	(0.801)	(0.953)
GDP-per-capita gain ($\Delta \ln y$)		−1.202	−0.995	−0.303	−0.309
		(0.732)	(0.758)	(1.004)	(1.053)
Recorded peak year (10 years)			−0.323	−0.134	−0.188
			(0.208)	(0.331)	(0.371)
Service share, 1990 (pp)					−0.032
					(0.080)
Region and initial income-group FE	No	No	No	Yes	Yes
Observations	51	51	51	51	51
R^2	0.530	0.550	0.573	0.619	0.622

Notes: All columns use the same 51 ETD economies. HSPS comprises business services, finance, and real estate. Manufacturing peaks use the full available GGDC 10-sector–ETD histories, with ETD preferred in overlapping years and exact ties assigned to the earliest year. Peak timing is $(T_c - 1990)/10$; its coefficient is per ten calendar years later. GDP-per-capita gain is the log change from 1990 to 2018 using Maddison 2023. The initial service share sums trade, transport, business services, finance, real estate, and other services, divided by total employment, in percent. Region and initial income-group effects enter additively; income groups use 1990 GNI (FY1992). All regressions include an intercept and give equal weight to each country. HC3 standard errors are in parentheses; tests use the residual-degrees-of-freedom t distribution. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

regressions use all 51 countries and no additional controls, decomposing the unadjusted HSPS association in column (1).

Business services have the largest coefficient, 1.59 percentage points, followed by finance (0.67) and real estate (0.61). Business services account for approximately 56 percent of the aggregate unadjusted HSPS coefficient. Their ETD category includes information and communication; professional, scientific, and technical activities; and administrative and support services.

B.3 GDP-per-capita growth in service economies after manufacturing peaks

The growth regressions relate historical manufacturing peaks to average annual GDP-per-capita growth over 2010–2019. The most extensive specification is

$$g_c = \alpha + \alpha_{r(c)} + \alpha_{i(c)} + \beta^m (M_c^{\max}/0.10) + \beta^s (\overline{S}_{c,2010:2019}/0.10) + \delta (\overline{M}_{c,2010:2019}/0.10) + \gamma I_c + \varepsilon_c. \quad (47)$$

Here g_c is 100 times the mean annual log GDP-per-capita change for years ending in 2010–2019, including the change from 2009 to 2010. I also control for the year of the peak. Employment shares are fractions of total employment; bars denote averages over 2010–2019. All 173 regression countries have manufacturing observations in each of these ten years. Region effects use five broad geographic regions,

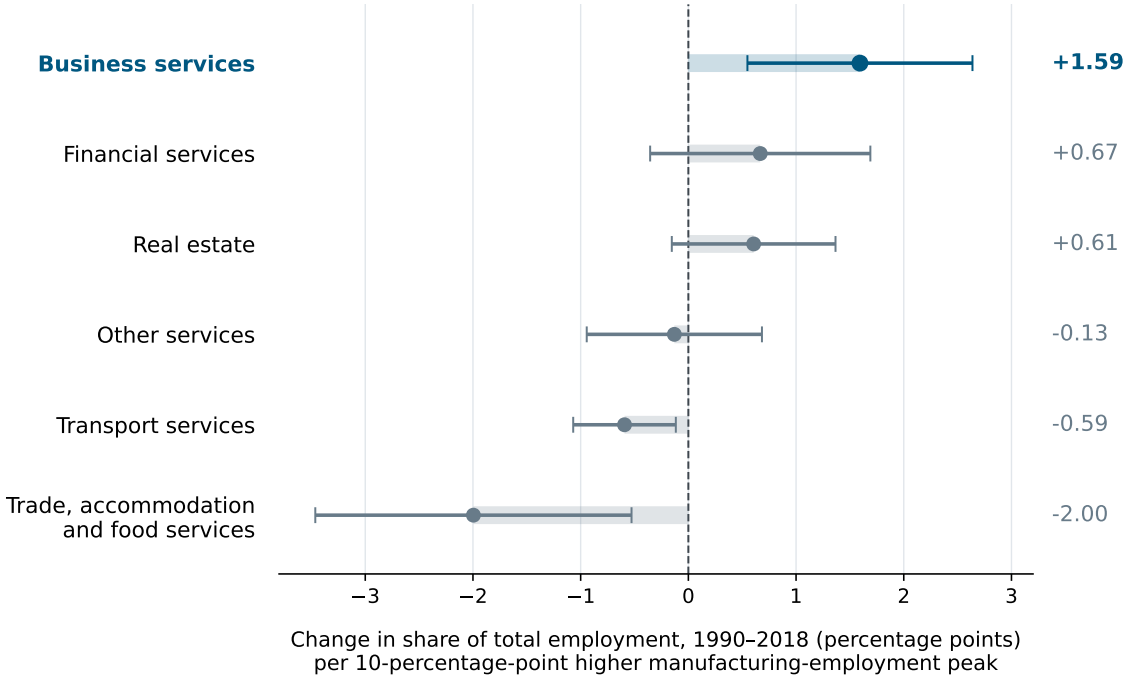


Figure B.1: Manufacturing peaks and service-industry employment growth. Points report coefficients from separate regressions of each industry's 1990–2018 change in its share of total employment on the historical manufacturing peak. Coefficients are percentage points per ten-percentage-point larger peak. All regressions use the same 51 ETD economies, the full GGDC–ETD peak histories, equal country weights, an intercept, and no additional controls. Bars are pointwise 95 percent confidence intervals using HC3 standard errors and residual-degrees-of-freedom t critical values. Trade includes accommodation and food services; government services are excluded.

and income-group effects use the four World Bank categories based on 2009 GNI per capita (FY2011). Institutional quality I_c is the World Bank control-of-corruption percentile rank.

Table B.2 reports two specifications for the full sample and for economies where services employ more than half of workers, first without and then with contemporary manufacturing employment. Column (5) repeats the latter specification for the 90 service-majority economies whose recorded manufacturing peak occurred strictly before 2000. It therefore applies the two restrictions jointly, using the same covariates as columns (2) and (4). All columns use HC3 standard errors and residual-degrees-of-freedom t tests, as in the Section 2 replication.

In the full sample, adding contemporary manufacturing employment reduces the peak coefficient from 0.66 to 0.41; the latter is positive but statistically imprecise ($p = 0.387$). Among service-majority economies, the coefficient with both contemporary sectoral shares is 1.37 ($p = 0.033$). Restricting this group further to manufacturing peaks before 2000 yields a coefficient of 1.636 ($p = 0.058$).

The earlier-peak comparison shows that a positive association is present when recorded manufacturing peaks precede the start of the growth window by at least a decade and services subsequently employ a majority of workers. It remains descriptive, subject to the historical-coverage limitations above.

Table B.2: Manufacturing peaks and growth in service economies

	All countries		Services $> 50\%$		Services $> 50\%$ Peak before 2000
	(1)	(2)	(3)	(4)	(5)
Manufacturing peak (10 pp)	0.655^{**}	0.414	0.664^{*}	1.363^{**}	1.636^{**}
	(0.272)	(0.551)	(0.390)	(0.649)	(0.629)
Service share, 2010–2019 (10 pp)	-0.559^{**}	-0.539^{**}	-1.036^{**}	-1.320^{**}	-1.575^{***}
	(0.215)	(0.218)	(0.476)	(0.616)	(0.563)
Manufacturing share, 2010–2019 (10 pp)		0.337		-1.156	-1.992^{**}
		(0.624)		(1.021)	(0.854)
Recorded peak year (10 years)	0.045	0.008	0.020	0.134	0.446^{**}
	(0.122)	(0.141)	(0.197)	(0.226)	(0.220)
Region and income-group FE	Yes	Yes	Yes	Yes	Yes
Institutional quality control	Yes	Yes	Yes	Yes	Yes
Observations	173	173	104	104	90
R^2	0.361	0.362	0.278	0.291	0.402

Notes: The dependent variable is average annual log GDP-per-capita growth over 2010–2019, multiplied by 100. Columns (1)–(2) use 173 countries; columns (3)–(4) use the 104 whose average service-employment share exceeds 50 percent. Column (5) additionally restricts the recorded manufacturing peak year to before 2000, leaving 90 countries. All five columns control linearly for peak timing using $(T_c - 1990)/10$, where T_c is the recorded manufacturing peak year; its coefficient is per ten calendar years later. The service and contemporary manufacturing shares are averages over 2010–2019. The manufacturing peak is the maximum over the available GGDC–Gapminder history through 2019, with exact ties assigned to the earliest year. All employment shares use total employment as the denominator; their coefficients are per ten-percentage-point increase. Region and income-group effects are additive; income groups use 2009 GNI (FY2011). All columns control for the World Bank control-of-corruption percentile rank. South Sudan is omitted because its historical income classification is unavailable. All regressions include an intercept and give equal weight to each country. HC3 standard errors are in parentheses; tests use the residual-degrees-of-freedom t distribution. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C Additional U.S. County Evidence

C.1 Timing of county manufacturing peaks

County manufacturing peaks follow equation (1): the maximum manufacturing-employment share of total classified employment over 1880, 1900, 1920, 1930, and 1940. The years 1890 and 1910 are excluded because the required population and manufacturing tabulations, respectively, are unavailable.

Peak dates vary from the turn of the twentieth century to the interwar period, providing the staggered timing used in the event-study analysis. Figure C.1 plots the distribution of peak dates across counties and peak sizes.

C.2 Robustness of the U.S. county evidence

The following exercises assess outcome measurement and identification: endpoint and peak-window choices; separate high- and low-throughput event studies; industry-share diagnostics; QCEW confidentiality and EFSY comparisons; national-shock inference; historical controls and spatial cutoffs; and the role of corporate headquarters.

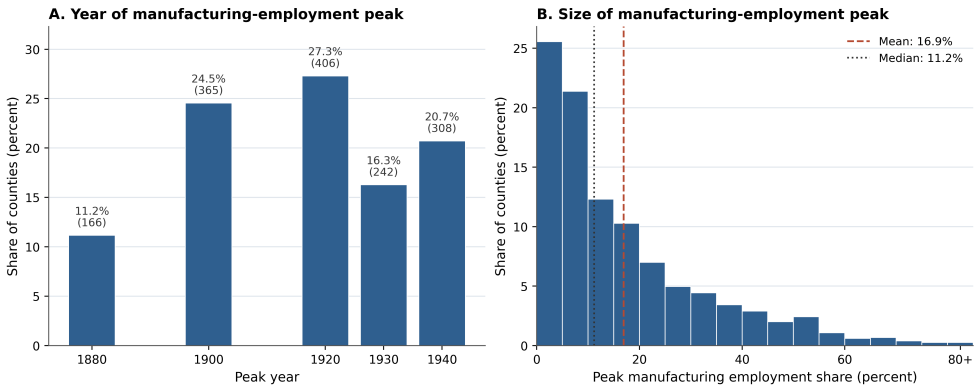


Figure C.1: Distribution of county manufacturing peaks. Panel A reports the year in which each county reached its peak manufacturing-employment share, and Panel B reports the magnitude of that peak. The sample consists of the 1,487 counties used in Section 2.

C.2.1 QCEW endpoint and peak-window robustness

Table C.1 moves the public-record QCEW endpoint from 2000 to 2001 and 2010. The baseline peak coefficient remains positive. Adding manufacturing scale and HSPS attenuates it by 68 percent in 2000, 68 percent in 2001, and 57 percent in 2010; manufacturing HSPS remains large and precisely estimated.

Table C.2 reports the IV results. The high-throughput estimate is 0.241 in 2010, with a conventional p-value of 0.045 and an Anderson–Rubin p-value of 0.051. Excluding counties whose observed peaks occur at the 1880 or 1940 boundaries raises the high-throughput estimate to 0.318 in 2000 and 0.273 in 2010. The result is therefore not generated by counties whose true peak may lie outside the observed window.

The disclosed service aggregate retains confidential non-HSPS employment in the denominator; summing public two-digit records instead removes confidential employment from both sides of the ratio. Table C.3 shows that including private utilities is immaterial and reproduces the public-record result in 2001, when explicit suppression flags become available. Section C.2.5 uses those flags for imputation.

Table C.1: Alternative QCEW endpoints

	2000	2001	2010
Panel A. Baseline			
Peak manufacturing employment share	0.048***	0.043***	0.042***
	(0.010)	(0.009)	(0.009)
Panel B. Add manufacturing scale and HSPS			
Peak manufacturing employment share	0.015	0.014	0.018*
	(0.012)	(0.011)	(0.010)
Log manufacturing establishment scale at peak	0.0041*	0.0033	0.0024
	(0.0023)	(0.0021)	(0.0025)
Manufacturing HSPS / classified employment at peak	1.927***	1.884***	1.671***
	(0.535)	(0.508)	(0.506)
Attenuation of peak coefficient (%)	68.3	67.8	56.9
Observations	1,487	1,511	1,509

Notes: Every outcome uses the QCEW public-record numerator and disclosed service-aggregate denominator. Each column uses the maximum available common sample for that endpoint. Standard errors are Conley with a 100-kilometer cutoff. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table C.2: Outcome endpoint and peak-window robustness

	Full sample 2000	Interior peaks 2000	Full sample 2010	Interior peaks 2010
All-industries IV	0.133	0.198*	0.092	0.188*
	(0.105)	(0.111)	(0.090)	(0.105)
First-stage Conley F	14.67	16.14	13.77	14.36
Anderson–Rubin p-value	0.203	0.093	0.330	0.100
High-throughput IV	0.283**	0.318***	0.241**	0.273**
	(0.118)	(0.121)	(0.120)	(0.117)
First-stage Conley F	12.87	12.57	11.00	10.29
Anderson–Rubin p-value	0.009	0.008	0.051	0.028
Observations	1,487	1,013	1,509	1,031

Notes: The 2000 and 2010 outcomes use the same QCEW public-record construction. Interior-peak samples exclude counties whose observed manufacturing peak occurs at either boundary of the 1880–1940 window. All other variables and specifications match Table 1. Standard errors are Conley with a 100-kilometer cutoff. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table C.3: QCEW denominator and public-record bridge

Year	Construction	Estimate	SE	p-value	AR p-value
2000	Public numerator; disclosed service aggregate	0.283	0.118	0.016	0.009
2000	Public numerator; sum of public components	0.188	0.123	0.126	0.117
2000	Public numerator; aggregate including utilities	0.281	0.117	0.017	0.009
2001	Public numerator; disclosed service aggregate	0.279	0.116	0.016	0.013
2010	Public numerator; disclosed service aggregate	0.241	0.120	0.045	0.051
2010	Public numerator; sum of public components	0.138	0.127	0.275	0.289
2010	Public numerator; aggregate including utilities	0.239	0.119	0.045	0.052

Notes: Each row reports the high-throughput IV with otherwise identical treatment, controls, sample rule, and 100-kilometer Conley inference. The component-sum construction adds only public two-digit values. The aggregate-denominator construction retains all employment incorporated by BLS in the disclosed private service-providing total. The final construction in each endpoint shows that subtracting published private utilities is immaterial.

C.2.2 High- and non-High-throughput manufacturing peaks

To distinguish general local development from HSPS formation, I partition manufacturing into Chandler high-throughput industries and all remaining industries, labeled non-high-throughput. Each group's peak is its maximum employment share over 1880, 1900, 1910, 1920, 1930, and 1940. Separate event studies align counties on each type-specific peak and interact its magnitude with event time, conditioning on the other type's magnitude interacted with the same windows. This compares the population, urbanization, manufacturing-HSPS, and service-HSPS paths associated with each type's own peak, rather than its share at the county-wide peak:

$$Y_{ct} = \alpha_c^g + \lambda_{s(c),t}^g + \mu_{k_{ct}^g}^g + \sum_{b \in \mathcal{B}} \mathbf{1}\{k_{ct}^g \in b\} \left[\beta_b^g \frac{P_c^g}{0.10} + \rho_b^g \frac{P_c^{\bar{g}}}{0.10} \right] + X_c' \Gamma^g \frac{t - 1900}{10} + \varepsilon_{ct}^g, \tag{48}$$

Here α_c^g, $\lambda_{s(c),t}^g$, and $\mu_{k_{ct}^g}^g$ are county, state-by-calendar-year, and exact decennial event-time fixed effects. The predetermined controls X_c are 1850 population, urbanization, schooling, colleges, and enslaved population, plus 1870 process-, machinery-, and traditional-industry shares; Γ^g gives each a linear calendar-decade trend, and ε_{ct}^g is the error. Each type and outcome is estimated separately. The plotted β_b^g associates a 10-percentage-point larger type-g peak with outcome Y_{ct} in window b, conditional on the other type's magnitude through ρ_b^g; the other type's peak year does not set the clock. Standard errors use the 100-kilometer spatial estimator. Figure C.2 uses uniform 20-year windows and ends at $+60$, the last horizon supported by every eligible peak cohort.[28] Service numerators and denominators share an employment universe within each vintage: harmonized Census records through 1950, imputed early CBP in 1960 and 1970, and EFSY CBP thereafter, as in Figure 3.

Both manufacturing types predict rising population and urbanization before and after their peaks. Over $+10$ to $+40$, a 10-percentage-point larger high-throughput peak is associated with 0.375 log points more population and a 6.33-percentage-point larger urban share; low-throughput associations are 0.143 log points and 2.32 percentage points, both distinguishable from zero. HSPS outcomes differ: the manufacturing-HSPS share is 0.831 percentage points higher after a larger high-throughput peak and 0.396 points lower after a larger low-throughput peak. At $+50/+60$, the service-HSPS coefficients are 0.429 points ($p = 0.006$) and -0.095 points ($p = 0.362$), a difference of 0.525 points ($p = 0.003$). This sequence is consistent with high-throughput industrialization developing HSPS work first within manufacturing and later in services. Population growth and urbanization alone cannot explain the pattern: low-throughput peaks predict both without a positive HSPS response.

28 For the high-throughput service regression, support falls from 1,417 county observations at $+60$ to 226 at $+70$ and six at $+100$.

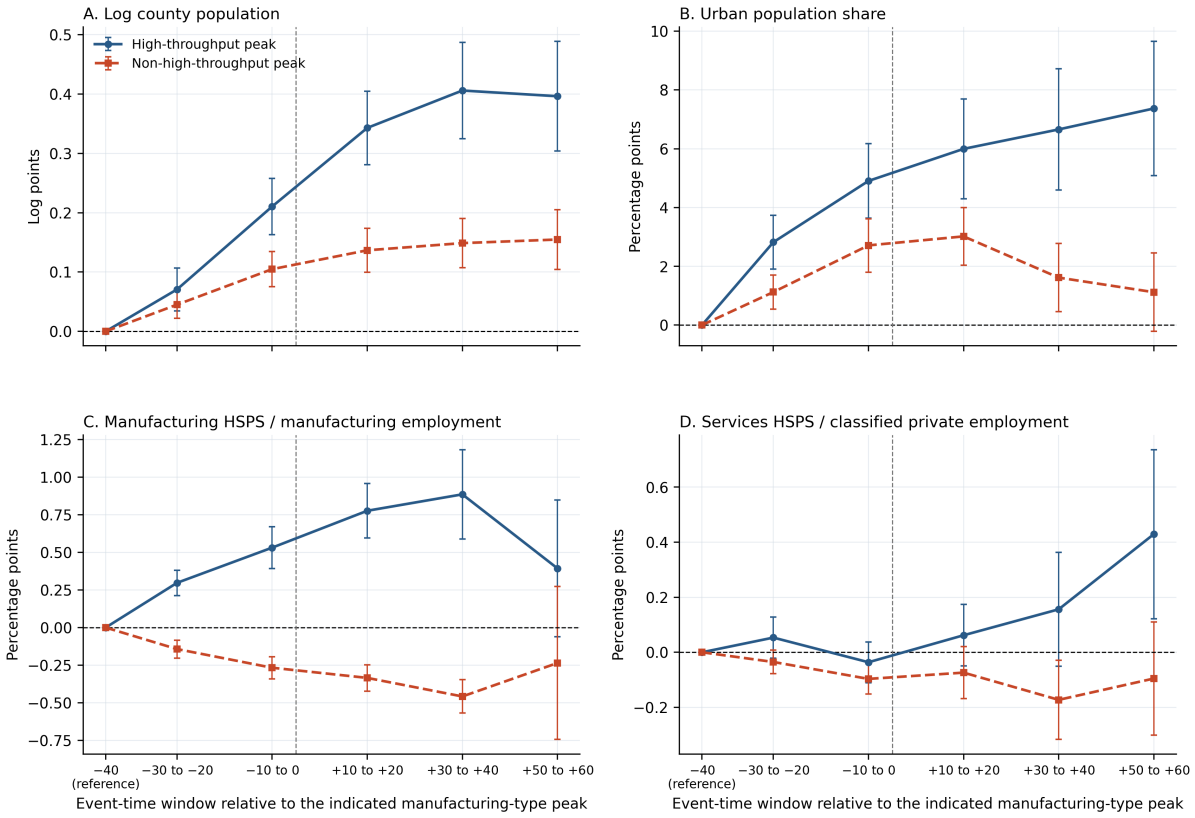


Figure C.2: Local development and HSPS employment around separate manufacturing-type peaks. The blue path is aligned on the county's high-throughput manufacturing-employment peak and the red path on its separately calculated non-high-throughput peak. Each point is the coefficient associated with a 10-percentage-point larger peak share in the indicated manufacturing type, conditional on the magnitude of the other type's peak interacted with every event-time window. The windows are $-30/-20$, $-10/0$, $+10/+20$, $+30/+40$, and $+50/+60$; event time -40 is omitted. Panels A and B report log county population and the urban population share. Panel C reports manufacturing HSPS employment divided by manufacturing employment; Panel D reports HSPS-service employment divided by classified private employment. The high- and low-peak regressions use different eligible county sets because each requires an observation at -40 relative to its own peak: 1,419 and 1,158 counties, respectively, for the service outcome. Both specifications include county, state-by-calendar-year, and exact decennial event-time fixed effects and predetermined baseline-covariate linear trends. Bars are 95-percent confidence intervals based on a 100-kilometer spatial covariance estimator.

C.2.3 HSPS Placebos: Skill Upgrading vs. Organizational Capabilities

The increase in HSPS services could reflect a general tendency for historically industrial counties to attract skill-intensive services rather than the organizational-capability channel emphasized in the paper. Figure C.3 therefore applies the event-study specification in equation (4) to three skill-intensive service groups outside NAICS 51, 54, and 55: health care, education, and welfare, religious, and nonprofit services. The treatment, denominator, sample, fixed effects, predetermined-covariate trends, and spatial covariance estimator are unchanged.

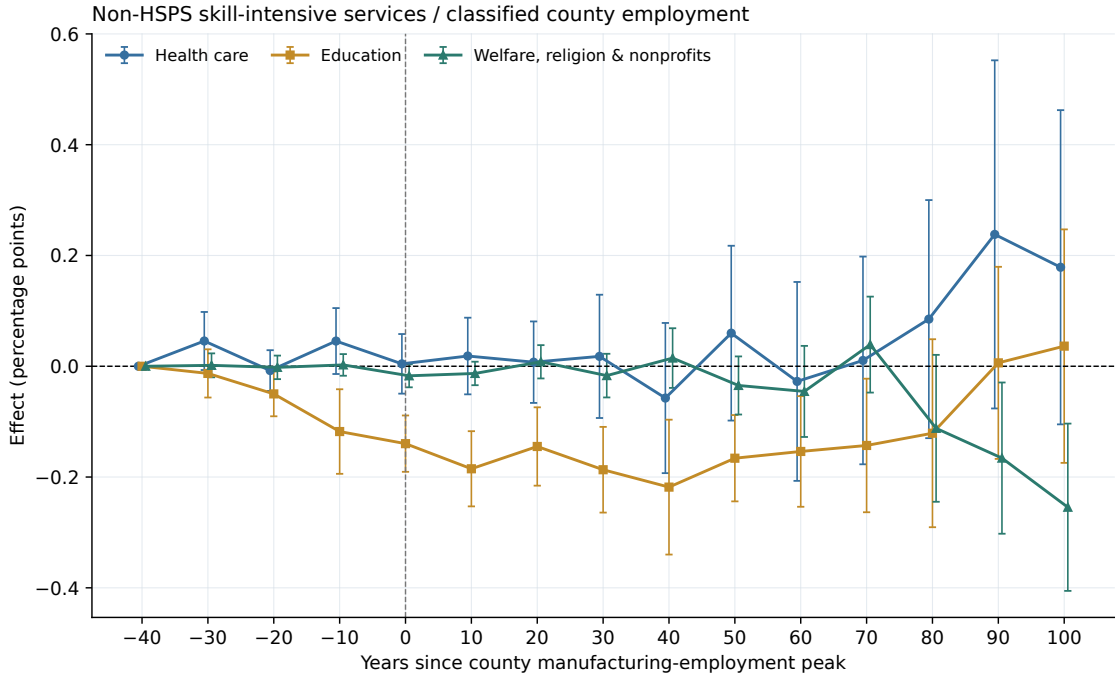


Figure C.3: Placebo outcomes: non-HSPS skill-intensive services. Each path reports the effect of a 10-percentage-point larger peak manufacturing-employment share on employment in the indicated service group divided by classified county employment. Health care corresponds to IND1950 868–869, SIC 80, and NAICS 621–623; education to IND1950 888, SIC 82, and NAICS 61; and welfare, religious, and nonprofit services to the closest available analogues in IND1950 896–897, SIC 86, and NAICS 624 and 813. The regressions use the same 1,116 counties and specification as Figure 3. Event time -40 is omitted. Bars report 95-percent confidence intervals based on a 100-kilometer spatial covariance estimator.

The placebo outcomes do not reproduce the late increase in HSPS services. The pattern in Figure 3 is therefore not a general consequence of a broad rise in industries which are intensive in skilled labor; it instead reflects a rise in specific industries which are tied to the organizational demands of large firms.

C.2.4 Identification from initial industry shares

The shift-share design admits shares and shocks interpretations. Under Goldsmith-Pinkham et al. (2020), national growth rates combine predetermined 1880 county industry shares into one instrument, and 2SLS is a weighted combination of share-specific just-identified estimates. Figure C.4 reports these Rotemberg weights after residualizing treatment, shares, and outcome on the complete Table 1 conditioning set.

Twenty-six high-throughput shocks are usable. Blast furnaces and steel works, grain milling, tobacco, primary nonferrous metals, and other primary iron and steel have the largest absolute weights.

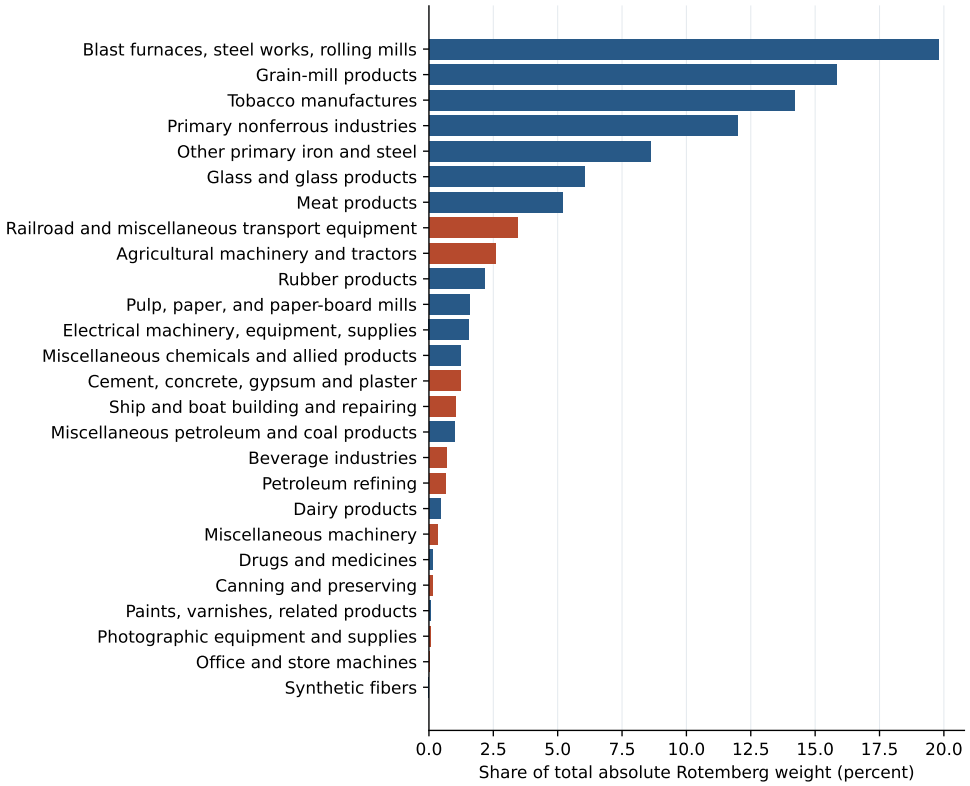


Figure C.4: Rotemberg weights in the high-throughput design. Absolute Rotemberg-weight shares are calculated on the QCEW-2000 estimation sample after partialling out the state and peak-cohort fixed effects, five 1850 controls, and complete 1850–1870 histories used in Table 1. Colors distinguish positive from negative weights. The underlying machine-readable table reports signed weights, first stages, and just-identified estimates for every usable industry.

The largest accounts for 19.8 percent, the top three for 49.8 percent, and the top five for 70.4 percent; negative weights account for 10.2 percent of absolute weight. The inverse-Herfindahl effective number of Rotemberg weights is 8.6, and the exposure-weighted effective number of national shocks is 4.7. Identification is concentrated but does not rest on one industry.

Figure C.5 removes each high-throughput industry and rebuilds both stages. All second-stage coefficients remain positive, and 27 of the 28 reported concordance estimates remain significant at 5 percent. Removing grain milling reduces first-stage precision but leaves a positive estimate of 0.287. Removing primary metals, steel, or transportation equipment does not eliminate the result.

The within-group weights in equation (43) sum to one wherever the instrument is defined. The benchmark shifter therefore asks which high-throughput industries a county initially specialized in, conditional on having a classifiable high-throughput base; it does not use the initial size of that base as an omitted exposure share. Table C.4 nevertheless reports economy-scaled alternatives in which the incomplete high-throughput share is included separately, as required under the quasi-experimental exposure framework of Borusyak et al. (2022). It also alters the mapping of debatable sectors. The estimate remains positive and statistically significant when processed foods, machinery, or transportation equipment are removed separately, under a narrower continuous-process classification, and under a broader classification. Removing all processed foods weakens the first stage and makes the estimate imprecise; that specification discards several industries that Chandler explicitly associates with high-throughput production.

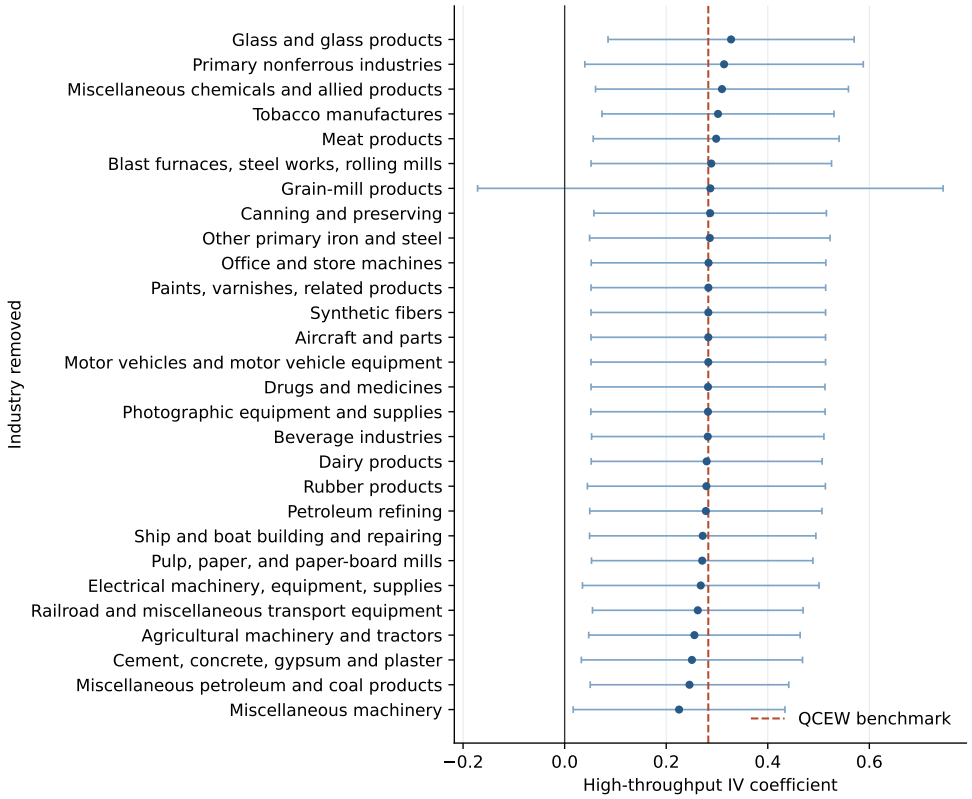


Figure C.5: Leave-one-industry-out high-throughput estimates. Each row removes the labeled IND1950 industry from the high-throughput set, reconstructs the leave-county-out shifter and its within-group weights, and re-estimates the first and second stages on the Table 1 sample. Points are 2SLS estimates; bars are 95-percent confidence intervals based on 100-kilometer Conley standard errors. The vertical line marks the benchmark estimate.

The comparison that most directly tests the mechanism holds the instrument construction fixed and changes the production regime supplying the national shifts. Table C.5 enters the high-throughput and comparison-group shifters jointly. High-throughput national growth predicts both a larger manufacturing peak and a larger long-run HSPS-service share. Growth in all remaining manufacturing industries or in the narrower traditional low-throughput set does not predict the service outcome. Equality of the second-stage coefficients is rejected at 10 percent against all non-high-throughput industries and at 5 percent against the traditional group. A generic local-demand or agglomeration account would need to explain why national growth outside high-throughput manufacturing generates industrial exposure without the same long-run service-composition response.

Table C.4: Alternative throughput classifications and exposure normalizations

Specification	IV estimate	SE	p-value	First-stage F	AR p-value
Panel A. Throughput classification					
Benchmark Chandler set	0.283	0.118	0.016	12.87	0.009
Exclude processed foods	0.325	0.257	0.207	3.63	0.159
Exclude machinery	0.189	0.098	0.055	28.98	0.041
Exclude processed foods and machinery	0.366	0.175	0.036	9.19	0.011
Exclude transportation equipment	0.251	0.102	0.014	16.38	0.008
Narrow continuous/process set	0.183	0.078	0.018	29.69	0.011
Broader Chandler set	0.267	0.120	0.026	12.45	0.014
Panel B. Exposure normalization					
Within-high shares, benchmark	0.283	0.118	0.016	12.87	0.009
Within-high shares; control high-throughput share of manufacturing	0.322	0.135	0.017	11.36	0.009
Economy-scaled exposure; control included-share sum	0.206	0.080	0.010	40.29	0.015
Within-high shares; control high-throughput share of county employment	0.822	0.674	0.223	1.59	0.016
Economy-scaled exposure; county-employment denominator	−0.419	1.313	0.749	0.38	0.645

Notes: Every row instruments the same peak-manufacturing treatment and uses the QCEW-2000 outcome, sample rule, fixed effects, and predetermined controls in Table 1. The within-high specifications normalize 1880 industry employment by the county's total employment in the relevant high-throughput set. The economy-scaled specification normalizes by the stated broader base and includes the sum of included exposure shares as a separate control. First-stage F statistics and conventional inference use a 100-kilometer Conley covariance estimator; AR entries are the corresponding reduced-form tests.

Table C.5: High-throughput and comparison-industry contrasts

Joint instrument set	High throughput	Comparison group	p-value, equality
Panel A. First stage: peak manufacturing employment share			
High throughput versus all non-high-throughput industries	1.928	0.193	0.014
High throughput versus traditional low-throughput industries	1.928	0.041	0.011
Panel B. Reduced form: 2000 HSPS share of private services			
High throughput versus all non-high-throughput industries	0.546	−0.049	0.053
High throughput versus traditional low-throughput industries	0.547	−0.074	0.021

Notes: Entries are percentage-point effects of a one-standard-deviation shifter from regressions that include both standardized group shifters simultaneously. Every regression uses the QCEW-2000 sample and the complete Table 1 conditioning set. The final column tests equality of the two coefficients using a 100-kilometer Conley covariance estimator. Panel B reports reduced forms because each group shifter predicts the common manufacturing-peak treatment with a different strength.

C.2.5 Flag-aware QCEW imputation

The public-record outcome is well-known to suppress data on employment from smaller counties to protect confidentiality of businesses. To address whether this may systematically bias the estimates, I perform two additional robustness checks: imputing suppressed data and aggregating to commuting zones. First I use the flags available beginning in 2001 which provide employment bins for suppressed data in QCEW to estimate the outcome variable in data suppressed counties. QCEW is preferable to EFSY for this specific purpose because it reports, in addition to a denominator capturing all private service employment, an aggregate employment count of NAICS industries 54-55 and 56 (parent code 1024). This permits a more precise estimate for the dependent variable than would be possible from EFSY. The disclosed county private service-providing total, code 102, supplies the top adding-up constraint, and disclosed service supersectors 1021–1029 supply the next set of margins. Two of these supersectors are especially informative for the outcome. Code 1022 equals Information (NAICS 51), while code 1024 is the higher-level Professional and Business Services aggregate that equals NAICS 54, 55, and 56. Consequently, whenever NAICS 56 is public, NAICS 54 plus 55 is observed exactly as code 1024 less NAICS 56 even if the separate 54 and 55 records are confidential. When NAICS 56 is also confidential, the disclosed 1024 total still tightly constrains the joint residual that can be assigned to 54, 55, and 56. Public three-digit descendants provide additional lower bounds whenever a two-digit parent is confidential.

Within each disclosed parent, I allocate only the residual using state-industry shares, with national-industry and equal-share sensitivities. Allocations never use the instrument, manufacturing peak, long-run outcome, or regression residuals. Zero- and one-worker lower bounds for confidential macro cells give virtually identical results because parent margins determine nearly all allocated employment. QCEW and CBP differ in administrative systems, reference periods, and treatment of auxiliary establishments. EFSY estimates CBP employment, rather than missing QCEW records, and therefore provides an independent-source check; the QCEW hierarchy reconstructs the latent main-outcome numerator.

Across 3,194 located counties, the preferred reconstruction implies an employment-weighted HSPS share of 14.38 percent, compared with 14.53 percent in the official national QCEW aggregates. The state- and national-share allocations differ by less than 0.01 percentage point in the aggregate, and changing the lower bound from zero to one worker changes the high-throughput estimate by less than 0.001. Unallocated hierarchy margins and published-child overages total 1,504 and 2,680 workers, respectively, and are negligible relative to national service employment.

The high-throughput coefficient is 0.187 (SE 0.100, $p = 0.061$, AR $p = 0.066$) under state-industry shares and 0.185 (SE 0.099, $p = 0.063$, AR $p = 0.066$) under national-industry shares. Equal splitting ignores observed state and national composition and gives 0.159 (SE 0.100, $p = 0.113$, AR $p = 0.121$). State shares are preferred because they reflect regional composition while remaining external to county treatment and outcome.

Table C.7 starts with the main Table 1 result. EFSY's fully imputed native-2000 CBP outcome gives a positive but imprecise high-throughput estimate of 0.103 (SE 0.080, AR $p = 0.202$). The QCEW flag-aware estimate is 0.187 (SE 0.100, conventional $p = 0.061$, AR $p = 0.066$). QCEW's separate code-1024 parent for NAICS 54–55–56 supplies an additional constraint on confidential employment, helping explain the source difference. Its all-industry first stage remains strong (Conley $F = 13.55$), but the second-stage estimate is 0.062 (AR $p = 0.519$). Averaging flag-aware outcomes over 2001–2003 preserves this contrast: the all-industry estimate is an insignificant 0.065, while the high-throughput estimate is 0.164 and significant at 10 percent under conventional and Anderson–Rubin tests.

Table C.6: Disclosure flags and QCEW-imputation audit, 2001

Sector	County records	Confidential records	Confidential with public local parent	Confidential with public three-digit descendant
NAICS 51	3,129	583	0	36
NAICS 54	3,197	948	826	0
NAICS 55	2,160	1,153	1,124	0
Aggregate reconciliation				
Preferred state-share allocation	14.3825%		employment-weighted HSPS share	
National-share allocation	14.3788%		employment-weighted HSPS share	
Equal-share allocation	14.4228%		employment-weighted HSPS share	
Official national QCEW aggregates	14.5272%		employment-weighted HSPS share	

Notes: A public local parent is QCEW supersector 1022 for Information or 1024 for Professional and Business Services. The latter equals NAICS 54, 55, and 56. Published cells remain fixed. State, national, and equal shares are alternative priors used only to split an otherwise unidentified residual among confidential children of a disclosed parent. Official national employment is 12,179,566 in NAICS 51, 54, and 55 and 83,839,631 in private services net of utilities.

The last two rows hold the year, denominator, controls, instruments, and 1,501-county sample fixed and change only the numerator source. The QCEW reconstruction gives 0.180, while the EFSY numerator gives 0.138 and is not statistically significant. The EFSY-minus-QCEW difference is -0.043 with a standard error of 0.098 and a p-value of 0.662. The data do not reject equality of the two estimates.

The flag-aware outcome also preserves the channel result. At the county level, the baseline peak coefficient is 0.029. Adding manufacturing scale and manufacturing HSPS reduces it to 0.012, an attenuation of 59 percent; manufacturing HSPS remains positive and precisely estimated at 1.260. At the commuting-zone level, the point estimate on the peak is already less precise, but the joint specification again leaves a small residual peak coefficient while both scale and manufacturing HSPS remain positive. EFSY gives a 75-percent attenuation at the commuting-zone level in both 2000 and 2001.

Table C.7: County IV results under alternative disclosure treatments

Outcome construction	OLS	All-industry IV	F_{all}	AR p_{all}	High-throughput IV	F_{high}	AR p_{high}
QCEW public records, 2000	0.029**	0.133	14.67	0.203	0.283**	12.87	0.009
	(0.012)	(0.105)			(0.118)		
EFSY-imputed CBP, native 2000	0.028***	0.031	14.67	0.747	0.103	12.87	0.202
	(0.010)	(0.094)			(0.080)		
Flag-aware QCEW, state shares, 2001	0.018*	0.062	13.55	0.519	0.187*	11.25	0.066
	(0.009)	(0.093)			(0.100)		
Flag-aware QCEW, 2001–2003 average	0.018**	0.065	14.30	0.458	0.164*	13.00	0.075
	(0.009)	(0.085)			(0.091)		
Flag-aware QCEW, matched 2001 sample	0.017*	0.065	13.17	0.511	0.180*	11.07	0.076
	(0.009)	(0.095)			(0.100)		
EFSY numerator / QCEW denominator, matched 2001 sample	0.034***	−0.013	13.17	0.910	0.138	11.07	0.186
	(0.011)	(0.113)			(0.102)		
Observations	1,487; 1,487; 1,511; 1,497; 1,501; 1,501, respectively						

Notes: Every specification matches Table 1. The matched 2001 rows use the same QCEW service denominator and differ only in the NAICS 51, 54, and 55 numerator. F statistics use 100-kilometer Conley first-stage standard errors; AR entries are the corresponding Conley reduced-form tests. The EFSY-minus-QCEW high-throughput coefficient in the matched sample is -0.043 (SE 0.098, $p = 0.662$). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table C.8: Imputation and commuting-zone aggregation

	QCEW public county, 2000	QCEW flag-aware county, 2001	QCEW flag-aware CZ, 2001	EFSY native CZ, 2000	EFSY harmonized CZ, 2001
Panel A. Baseline					
Peak manufacturing employment share	0.048***	0.029***	0.022	0.038**	0.037**
	(0.010)	(0.008)	(0.017)	(0.016)	(0.017)
Panel B. Add manufacturing scale and HSPS					
Peak manufacturing employment share	0.015	0.012	−0.012	0.010	0.009
	(0.012)	(0.011)	(0.020)	(0.022)	(0.023)
Log manufacturing establishment scale at peak	0.0041*	0.0016	0.0050*	0.0028	0.0028
	(0.0023)	(0.0021)	(0.0030)	(0.0033)	(0.0034)
Manufacturing HSPS / classified employment at peak	1.927***	1.260***	1.382*	1.729**	1.693**
	(0.535)	(0.433)	(0.814)	(0.834)	(0.837)
Attenuation of peak coefficient (%)	68.3	58.7	153.3	74.8	74.6
Observations	1,487	1,511	391	391	391

Notes: The first two columns use fixed 1850 counties. The last three aggregate additive historical and modern employment counts to 1990 commuting zones before forming shares. CZ fixed effects use the dominant historical state and the CZ peak cohort. The QCEW flag-aware columns use the preferred state-share allocation. The attenuation percentage exceeds 100 when the residual peak coefficient changes sign. Standard errors are Conley with a 100-kilometer cutoff. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

C.2.6 EFSY imputation and commuting-zone aggregation

Commuting-zone aggregation reduces the influence of confidentiality in small county-industry cells. I map 2000 and 1850 counties to commuting zones using weighted area overlap, then recompute outcomes, controls, and instruments with leave-commuting-zone-out national growth.

Table C.9 compares the 2001 outcomes on the same 391 commuting zones. Flag-aware QCEW gives 0.324 (SE 0.141, $p = 0.022$, AR $p = 0.005$), versus EFSY's 0.204 (SE 0.125, conventional $p = 0.102$, AR $p = 0.094$). Holding the QCEW denominator fixed and changing only the numerator raises EFSY's estimate to 0.248. The EFSY-minus-QCEW difference is -0.076 (SE 0.080, $p = 0.343$), so equality is not rejected. The source discrepancy is smaller at this geography, where confidentiality is less consequential.

The EFSY endpoint checks give 0.279 in native-2000 data (SE 0.127, $p = 0.028$, AR $p = 0.012$), 0.248 in harmonized-2000 data (SE 0.130, $p = 0.056$, AR $p = 0.035$), and 0.279 in harmonized-2010 data (SE 0.135, $p = 0.039$, AR $p = 0.009$). Thus EFSY recovers the high-throughput result after commuting-zone aggregation despite its imprecise county estimate, consistent with concentrated small-cell confidentiality and HSPS activity serving wider labor markets.

The difference between QCEW and EFSY is concentrated in the two sectors for which institutional coding and local margins differ most. In the 2000 county bridge, QCEW and EFSY give similar high-throughput coefficients for NAICS 51, 0.072 and 0.077. The QCEW coefficients for NAICS 54 and 55 are 0.141 and 0.070, whereas the corresponding EFSY coefficients are 0.034 and 0.003. EFSY's harmonization is built to recover CBP employment consistently across industry vintages; the QCEW allocation is built to recover the QCEW concept used by the main outcome. Because the two systems differ in their reference period, establishment frame, treatment of auxiliary establishments, and available hierarchy margins, neither is a mechanical correction of the other.

The QCEW reconstruction directly audits the main result by retaining its employment concept and annual-average reference period, identifying confidentiality through explicit flags, and exploiting the disclosed code-1024 parent for NAICS 54, 55, and 56 as a local constraint on the numerator.

Table C.9: QCEW and EFSY IV estimates at the commuting-zone level

Outcome construction	OLS	All-industry IV	F_{all}	AR p_{all}	High-throughput IV	F_{high}	AR p_{high}
Flag-aware QCEW, 2001	0.004	−0.274	0.17	0.736	0.324**	9.03	0.005
	(0.020)	(1.176)			(0.141)		
EFSY harmonized, 2001	0.021	−0.537	0.17	0.541	0.204	9.03	0.094
	(0.020)	(1.632)			(0.125)		
EFSY numerator / QCEW denominator, 2001	0.044*	−1.015	0.17	0.314	0.248*	9.03	0.070
	(0.026)	(2.804)			(0.141)		
EFSY native classification, 2000	0.028	−0.640	0.17	0.428	0.279**	9.03	0.012
	(0.018)	(1.877)			(0.127)		
EFSY harmonized, 2000	0.023	−0.362	0.17	0.669	0.248*	9.03	0.035
	(0.020)	(1.309)			(0.130)		
EFSY harmonized, 2010	0.018	0.020	0.17	0.981	0.279**	9.03	0.009
	(0.018)	(0.839)			(0.135)		
Commuting zones			391				

Notes: All rows use the same standard 1990 commuting zones, historical treatment, controls, fixed effects, and commuting-zone shift-share instruments. The QCEW row aggregates the preferred county-level state-share reconstruction. The EFSY 2001 row uses an EFSY numerator and EFSY private-service denominator; the following row changes only the numerator while retaining the QCEW denominator. The all-industry instrument is weak after aggregation and its CZ second-stage estimates are not interpreted. Coefficient and first-stage standard errors are Conley with a 100-kilometer cutoff. AR entries report the corresponding reduced-form test. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

C.2.7 Identification and inference from national industry shocks

Under Borusyak et al. (2022), identification instead rests on national industry-growth shocks, so sampling variation is organized by industry. The exact implementation retains county–industry leave-county-out shocks and clusters equivalent shock-level scores by industry. A common-shock approximation replaces the small leave-out differences with one national shock per industry and estimates the collapsed regression. Both account for the shared disturbances facing counties with similar industry shares (Adão et al., 2019).

In Table C.10, the exact transformation preserves the county 2SLS coefficient but changes uncertainty. The public-record QCEW estimate is 0.283 with an industry-clustered SE of 0.159; conventional and Anderson–Rubin tests reject at 10 percent ($p = 0.088$ and $p = 0.062$). The common-shock approximation yields 0.284 (SE 0.132, AR $p = 0.047$). Flag-aware 2001 and averaged 2001–2003 outcomes give smaller positive estimates, with exact-shock AR p-values of 0.065 and 0.082.

Precision is limited by 26 usable high-throughput shocks and an exposure-weighted effective count of approximately 4.7. Exact shock-clustered first-stage F statistics are 5.26 for the main outcome and 5.02 for the flag-aware outcome, below county-level Conley values. The all-industry shock-level first stage is also weak, and neither its reduced form nor second stage is distinguishable from zero. The shocks interpretation supports a positive high-throughput effect at 10 percent but is less precise than inference organized around 1,487 counties.

Table C.10: Industry-shock inference for the shift-share estimates

Outcome and inference	IV estimate	SE	p-value	First-stage F	AR p-value	Industries
Panel A. High-throughput instrument						
QCEW public, 2000: county-level Conley	0.283	0.118	0.016	12.87	0.009	26
QCEW public, 2000: exact leave-out shocks	0.283	0.159	0.088	5.26	0.062	26
QCEW public, 2000: common national shocks	0.284	0.132	0.041	3.94	0.047	26
Flag-aware QCEW, 2001: county-level Conley	0.187	0.100	0.061	11.25	0.066	26
Flag-aware QCEW, 2001: exact leave-out shocks	0.187	0.121	0.135	5.02	0.065	26
Flag-aware QCEW, 2001: common national shocks	0.188	0.106	0.089	3.83	0.034	26
Flag-aware QCEW, 2001–2003: county-level Conley	0.164	0.091	0.069	13.00	0.075	26
Flag-aware QCEW, 2001–2003: exact leave-out shocks	0.164	0.108	0.142	5.39	0.082	26
Panel B. All-industry instrument, QCEW public outcome						
County-level Conley	0.133	0.105	0.202	14.67	0.203	57
Exact leave-out shocks	0.133	0.223	0.551	1.59	0.623	57
Common national shocks	0.133	0.141	0.349	3.29	0.422	57
Panel C. Concentration diagnostics for high-throughput shocks						
Exposure-weighted effective number of shocks				4.71		
Inverse-Herfindahl effective number of Rotemberg weights				8.62		
Largest / three largest / five largest absolute weights				19.8% / 49.8% / 70.4%		
Negative share of total absolute weight				10.2%		

Notes: County-level rows use the 100-kilometer Conley estimator reported elsewhere. Exact leave-out rows retain the county-specific leave-out shock and cluster the equivalent shock-level score by IND1950 industry. Common-shock rows use one leave-sample-out national growth shock per industry. Their first-stage F is the shock-level statistic. p-values use finite-industry degrees of freedom; AR entries are the corresponding reduced-form tests. The bounded 95-percent exact-shock AR confidence set for the main high-throughput estimate is $[-0.023, 2.108]$; for the flag-aware outcome it is $[-0.014, 2.181]$.

C.2.8 Predetermined balance, controls, and spatial inference

Table C.11 tests pre-1880 outcomes against the standardized high-throughput shifter, five 1850 covariates, state fixed effects, and the other two outcome families' histories. Exposure predicts the 1860–1870 change in manufacturing and, at 10 percent, HSPS services; it does not predict 1850–1860 changes in manufacturing, manufacturing HSPS, or services, or the 1860–1870 manufacturing-HSPS change. Nonzero level correlations further motivate conditioning on complete observed 1850–1870 paths rather than asserting unconditional balance of the 1880 mix.

Table C.12 then varies the conditioning set and the spatial cutoff. Omitting the pre-1880 histories reduces the high-throughput coefficient from 0.283 to 0.184 rather than creating it. Adding proximity to coal, waterpower, ports, the Ohio and Mississippi rivers, and the Great Lakes raises the estimate to 0.320. Adding peak-year wages, population, and urbanization lowers first-stage precision and produces a positive but imprecise estimate of 0.242. Adding the manufacturing employment share in 2000 leaves a positive estimate of 0.196 that is significant at 5 percent. This last result rules out the mechanical explanation that high-throughput exposure changes service composition simply because it leaves more manufacturing employment in place at the endpoint. These variables are post-1880 and can be caused by industrialization, so both rows are channel-blocking exercises rather than preferred identification specifications. The point estimate is invariant to the Conley radius by construction, while the 95-percent uncertainty widens at long cutoffs. It remains significant at 5 percent through 300 kilometers and at 10 percent at 500 kilometers; the AR test likewise rejects at 10 percent throughout.

Table C.11: High-throughput exposure and pre-1880 changes

Pre-period outcome	Coefficient	SE	p-value
Manufacturing share, 1850–1860 change	-0.0027	0.0018	0.126
Manufacturing share, 1860–1870 change	0.0047	0.0017	0.007
Manufacturing HSPS, 1850–1860 change	0.000012	0.000007	0.100
Manufacturing HSPS, 1860–1870 change	0.000006	0.000007	0.399
HSPS services, 1850–1860 change	0.151	0.171	0.378
HSPS services, 1860–1870 change	-0.225	0.122	0.064

Notes: Each row regresses the indicated change on the standardized high-throughput shifter, state fixed effects, the five 1850 controls, and the complete histories of the other two outcome families, using the 1,487-county QCEW-2000 sample. The tested outcome's own history is omitted from the conditioning set. Standard errors are Conley with a 100-kilometer cutoff. Peak-cohort fixed effects are excluded because the peak is realized after the 1880 exposure shares.

Table C.12: Conditioning-set and spatial-inference robustness

Specification	IV estimate	SE	p-value	First-stage F	AR p-value
Preferred controls, 100 km	0.283	0.118	0.016	12.87	0.009
No peak-cohort fixed effects	0.298	0.115	0.010	13.46	0.004
Omit 1850–1870 histories	0.184	0.047	< 0.001	45.26	< 0.001
Add historical geography controls	0.320	0.120	0.008	13.43	0.002
Add peak-year wages, urbanization, and population	0.242	0.197	0.218	5.45	0.188
Add 2000 manufacturing employment share	0.196	0.096	0.042	14.81	0.042
Preferred controls, 200 km	0.283	0.114	0.013	8.37	0.018
Preferred controls, 300 km	0.283	0.134	0.034	5.20	0.038
Preferred controls, 500 km	0.283	0.170	0.096	2.83	0.067

Notes: Every row uses the QCEW-2000 outcome and high-throughput instrument. Historical-geography controls are log distance to coal, waterpower, the nearest port, the Ohio River, and the Mississippi River and an indicator for Great Lakes access. The peak-year row includes average manufacturing wages, the urban population share, log population, and the missing-wage indicator. The endpoint-manufacturing row adds manufacturing employment divided by total classified employment in 2000. Rows 1–6 use a 100-kilometer cutoff; rows 7–9 vary the spatial radius. AR entries are the reduced-form tests computed with the same covariance estimator.

These exercises preserve the main empirical pattern. Manufacturing scale and HSPS account for most of the peak-industrialization–service-composition relationship. Both manufacturing types predict population growth and urbanization, but only high throughput predicts later HSPS services; higher manufacturing peaks predict HSPS employment rather than a larger overall service sector. Endpoint, peak-window, classification, leave-one-industry-out, exposure-normalization, geography, and spatial checks support this pattern. Flag-aware QCEW preserves the high-versus-all contrast at 10 percent; EFSY is imprecise at county level but positive and significant at commuting-zone level in 2000 and 2010, with estimates statistically indistinguishable from QCEW. Exact industry-shock Anderson–Rubin inference also rejects at 10 percent despite only 26 high-throughput shocks. The evidence therefore links national growth in production requiring large-scale coordination to deeper local industrialization and persistent service reallocation toward organizational functions; broad manufacturing growth alone does not show the same pattern.

C.2.9 Corporate headquarters and NAICS 55

NAICS 55 includes headquarters and management establishments of firms operating in manufacturing. Such establishments supply the managerial services central to the mechanism, but could nevertheless account for the measured service response. I therefore decompose the flag-aware numerator into NAICS 51, 54, and 55 and re-estimate the high-throughput IV for each.

Table C.13: Excluding management of companies and enterprises from HSPS services

	Counties	Commuting zones
Full HSPS numerator: NAICS 51, 54, and 55	0.187^{*}	0.324^{**}
	(0.100)	(0.141)
Excluding NAICS 55: NAICS 51 and 54	0.147	0.254^{**}
	(0.090)	(0.121)
NAICS 51: Information	0.024	0.079^{*}
	(0.034)	(0.046)
NAICS 54: Professional, scientific, and technical services	0.122	0.175^{**}
	(0.086)	(0.089)
NAICS 55: Management of companies and enterprises	0.040^{*}	0.069^{*}
	(0.021)	(0.036)
First-stage Conley F statistic	11.25	9.03
Anderson–Rubin p-value, excluding NAICS 55	0.111	0.012
Observations	1,511	391

Notes: Each coefficient row reports the high-throughput IV estimate from the specification in Table 1. The dependent variable is flag-aware 2001 QCEW private employment in the indicated sector or sectors divided by the same private-service denominator. County counts are mapped to fixed 1850 boundaries before shares are formed; commuting-zone counts are summed before shares are formed. All specifications include state and manufacturing-peak-cohort fixed effects, the five 1850 controls, and complete 1850–1870 histories. Standard errors in parentheses use a 100-kilometer Conley cutoff. The three component coefficients add exactly to the full-numerator coefficient. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Excluding NAICS 55 reduces the county estimate from 0.187 to 0.147 and the commuting-zone estimate from 0.324 to 0.254: NAICS 51 and 54 retain approximately 79 percent of the effect at both geographies. The commuting-zone estimate remains significant conventionally ($p = 0.035$) and under Anderson–Rubin inference ($p = 0.012$). NAICS 54 is the largest component at both geographies and independently significant at commuting-zone level. Headquarters contribute about one-fifth of the estimate rather than driving the persistent HSPS-service response.

D Model Details and Policy Extension

This appendix documents historical examples of producer-service codification, the equilibrium and proposition proofs, calibration and sensitivity exercises, and the policy extension.

D.1 Historical examples of producer-service codification

The model distinguishes producer-service functions from how they are performed. Work organization, quality control, inventory management, accounting, procurement, and personnel management can rely on local experience, tacit knowledge, and ad-hoc supervision, the model's *LSPS*, or on codified records,

procedures, metrics, job categories, and decision rules, the model's *HSPS*. Repeated, measurable tasks make manufacturing and industrial procurement natural settings for codification. The resulting routines can then diffuse to department stores, hospitals, banks, retailers, offices, and universities.

Table D.1 summarizes the mapping from the model to several historical examples.

Table D.1: Examples of producer-service tasks moving from low-skill to high-skill mode

Producer-service task	Low-skill mode	Industrial codification	Later service sector use
Work organization	Foreman judgment, local rules of thumb, ad-hoc supervision	Motion study, workflow design, timing of tasks, written procedures, training systems	Department stores, offices, hospitals, and other service organizations use workflow design, scheduling, clerical organization, and personnel practices
Quality control	Inspection by eye, ad-hoc rejection of defective output	Testing protocols, defect records, statistical process control, control charts	Clinical laboratories and hospitals use control charts and statistical quality-control routines to monitor measurement processes
Accounting and managerial control	Bookkeeping and transaction recording	Cost accounting, budgeting, return-on-investment calculations, divisional reporting, performance metrics	Retail chains, department stores, banks, hospitals, universities, and other multi-unit service organizations use budgeting and management-control systems
Inventory, procurement, and logistics	Visual stock checks, ad hoc ordering, relationship-based purchasing	Stock-control systems, procurement planning, supplier coordination, delivery scheduling	Retail chains, wholesalers, hospitals, restaurants, and distribution firms use inventory and logistics routines
Personnel and training	ad-hoc apprenticeship, personal hiring, on-the-job learning	Job analysis, hiring rules, training manuals, personnel departments, performance evaluation	Department stores, hotels, hospitals, offices, and other service organizations use formal hiring and training systems

Work organization: from factory motion study to department stores. In low-skill work organization, foremen and supervisors rely on experience and local knowledge. Repeated factory operations

made worker motions, delays, output, and defects observable and measurable, enabling scientific management to codify work organization through motion study, workflow design, written procedures, and training.

Frank and Lillian Gilbreth developed motion-study methods initially oriented toward industrial work. Lillian Gilbreth later adapted motion study and industrial psychology to service and household settings, including department-store employees and consumers (Graham, 1992, 1999). Her work at Macy's in 1925–1928 applied scientific-management and personnel-management ideas to hiring, training, employee handling, clerical organization, and work organization (Graham, 2000). This illustrates the model's diffusion mechanism: routines codified in industrial work subsequently serve other sectors.

Quality control: from industrial process control to clinical laboratories. ad-hoc quality control relies on direct inspection and experienced workers' judgment. Recording manufacturing defects, tolerances, downtime, and output variation enabled statistical process control. Walter Shewhart introduced the control chart at Bell Laboratories in the 1920s and formalized statistical monitoring of production variation in his later book (Shewhart, 1931; Best and Neuhauser, 2006).

Clinical laboratories subsequently adopted statistical quality control to monitor laboratory measurements. Levey and Jennings introduced control charts for clinical laboratories in 1950, applying process-control logic to repeated medical measurements (Levey and Jennings, 1950; Westgard et al., 1981). ad-hoc inspection became a routine based on records, thresholds, and statistical decision rules. In the model, public organizational knowledge K_t similarly facilitates service-sector adoption of tasks first formalized in industry.

Accounting and managerial control: from industrial cost control to retail management. Accounting moves from transaction recording and bookkeeping to high-skill managerial control when managers must compare plants, products, inventories, costs, divisions, and capital allocations. The historical literature traces cost accounting and managerial control to large industrial and transportation enterprises, scientific management, and formal budgeting and performance measures at firms such as DuPont and General Motors (Johnson, 1975; Kaplan, 1984; Chandler, 1977).

These routines also became tools for service firms. James O. McKinsey professionalized budgeting and managerial accounting as consulting services through McKinsey (1922) and McKinsey (1924). After a 1935 consulting engagement for the Chicago department-store chain Marshall Field's, he became its chairman and chief executive (Flesher, 1985). The transferable object was a producer-service routine for measuring, budgeting, and managing a complex organization: the model's transition from low-skill bookkeeping to codified high-skill managerial control.

D.2 Equilibrium details

Producer-service supply. Formal and ad-hoc producer services are supplied competitively using skilled and unskilled labor, respectively. If $\mathcal{L}_t^H$ and $\mathcal{L}_t^L$ denote the corresponding employment levels, output is

$$Y_t^H = A_t^H h_s \mathcal{L}_t^H, \qquad Y_t^L = A_t^L h_u \mathcal{L}_t^L. \tag{49}$$

Perfect competition gives $p_t^H = w_t^s / A_t^H$ and $p_t^L = w_t^u / A_t^L$, since wages are prices per efficiency unit. Formal services enter gross production, recurring coordination, and procedure creation and adaptation;

ad-hoc services enter gross production and recurring coordination. Both producer-service productivities follow (28) with exponent ϕ_c.

Aggregate state and prices. The beginning-of-period aggregate state is

$$\mathcal{X}_t \equiv \left(L_t, S_t, K_t, \{A_{jt}, Z_{jt}, \mu_{jt}\}_{j\in\mathcal{J}}, A_t^H, A_t^L, \{e_{i,t}\}_{i\in\{a,m,s\}}\right). \tag{50}$$

Here $\mu_{jt}(a,s)$ is the measure of active sector-j establishments over permanent ability and inherited formalization frontiers. Establishment mass, employment, and the employment-weighted formal share are

$$M_{jt} = \int d\mu_{jt}(a,s), \qquad N_{jt} = \int n_{jt}^*(a,s)\, d\mu_{jt}(a,s), \qquad x_{jt} = \frac{1}{N_{jt}} \int n_{jt}^*(a,s) q(s)\, d\mu_{jt}(a,s). \tag{51}$$

The aggregate x_{jt} summarizes the establishment distribution; each establishment's productive and recurring producer-service demands depend on its own inherited share $q(s)$.

Let

$$\mathcal{P}_t \equiv \left(\{p_{jt}, p_{jt}^c\}_{j\in\mathcal{J}}, p_t^H, p_t^L, w_t^s, w_t^u\right), \qquad p_{mt} \equiv 1, \tag{52}$$

where p_{jt} and p_{jt}^c are producer and consumer prices and manufacturing output is the numeraire. Let $\mathcal{Q}_t$ collect household allocations, establishment input and organizational policies, entry, and producer-service production.

Perfect-foresight competitive equilibrium. Given an initial state $\mathcal{X}_0$ and deterministic paths for exogenous fundamentals and government policies, a perfect-foresight competitive equilibrium is a sequence

$$\{\mathcal{P}_t, \mathcal{Q}_t, \mathcal{X}_{t+1}\}_{t=0}^{\infty} \tag{53}$$

such that, at every date t, the following conditions hold.

1. *Households.* Households choose sectoral consumption, fertility, and education optimally subject to their budget and time constraints. Children choose whether to acquire skill according to the random-utility rule in (24). Households take current prices and dividends as given and correctly anticipate future wages and skill premia.

2. *Final-good establishments.* For every sector j and state (a,s), an incumbent chooses current labor, materials, and output-linked producer services to solve (13). It chooses its next-period frontier $u \geq s$ according to (20). The inherited share $q(s)$ governs current production and recurring coordination; the chosen frontier affects production from $t+1$ onward.

3. *Producer-service suppliers.* Formal and ad-hoc suppliers maximize profits under (49). Their outputs equal the sums of output-linked demand, recurring organizational demand, and, for formal services, the inputs used to create and adapt procedures.

4. *Entry.* Potential entrants pay the setup cost before drawing $a \sim G$ and enter with $s = 0$. Entry and establishment counts satisfy (22).

5. *Market clearing and resource feasibility.* Agricultural, manufacturing, and consumer-service output is allocated to household consumption, intermediate use, and entry setup. Formal and ad-hoc

producer-service markets clear separately. Demand for skilled and unskilled efficiency units equals the corresponding household supply. Establishment profits net of organizational investment, entry costs, and taxes are distributed to households, and the government budget balances.

6. *Aggregation and laws of motion.* Individual allocations aggregate according to (51). Let $u^*_{jt}(a,s)$ be the organizational policy implied by the Bellman equation. For every measurable set B of establishment states,

$$\mu_{j,t+1}(B) = (1-\delta_f)\int \mathbb{1}\{(a, u^*_{jt}(a,s)) \in B\}\, d\mu_{jt}(a,s) + I_{j,t+1}\int \mathbb{1}\{(a,0)\in B\}\, dG(a). \tag{54}$$

   Public knowledge follows (21); population, skills, and productivities follow (27)– (28).

7. *Perfect foresight.* Forecasts of future aggregate prices and states coincide with their equilibrium realizations: for every $k \geq 1$,

$$\widehat{\mathcal{P}}_{t+k|t} = \mathcal{P}_{t+k}, \qquad \widehat{\mathcal{X}}_{t+k|t} = \mathcal{X}_{t+k}. \tag{55}$$

Perfect foresight concerns aggregate variables. Establishments retain idiosyncratic exit risk and entrants draw heterogeneous ability; with a continuum of establishments, these risks aggregate according to their known distributions without aggregate uncertainty.

D.3 Establishment and household decisions and proposition proofs

D.3.1 Proof of Proposition 1

I first derive the reduced static establishment problem used in the proof. Conditional cost minimization in the CES labor nest gives

$$w^v_{jt} = \left[\alpha_j^{\sigma_L}(w^s_t)^{1-\sigma_L} + (1-\alpha_j)^{\sigma_L}(w^u_t)^{1-\sigma_L}\right]^{\frac{1}{1-\sigma_L}}. \tag{56}$$

Because the labor aggregator is homogeneous of degree one, headcount is $n = \nu_{jt}\ell$, where

$$\nu_{jt} = \frac{\alpha_j^{\sigma_L}}{h_s}\left(\frac{w^v_{jt}}{w^s_t}\right)^{\sigma_L} + \frac{(1-\alpha_j)^{\sigma_L}}{h_u}\left(\frac{w^v_{jt}}{w^u_t}\right)^{\sigma_L}. \tag{57}$$

Recurring organizational cost per unit of the labor composite is therefore

$$c^o_{jt}(q) = \frac{\nu_{jt}}{\bar{N}}\left[q p^H_t + r(1-q)p^L_t\right]. \tag{58}$$

Fix (j,t,a,s) and write $q = q(s)$. Before conditional inputs are eliminated, the establishment solves

$$g^*_{jt}(a,s) = \max_{\ell, m^a, m^m, H^p, L^p \geq 0}\Big\{p_{jt}Z_{jt}(A_{jt}a\ell^\eta)^{\theta_{jt}}(m^a)^{\gamma^a_{jt}}(m^m)^{\gamma^m_{jt}}[P^p(H^p,L^p;q)]^{\gamma^P_{jt}} - w^v_{jt}\ell - p_{at}m^a - p_{mt}m^m - p^H_t H^p - p^L_t L^p - c^o_{jt}(q)\ell\Big\}. \tag{59}$$

For fixed P^p, CES duality implies expenditure $\omega_t^P(q)P^p$, where

$$\omega_t^P(q) = \left[q(p_t^H)^{1-\sigma_P} + (1-q)(p_t^L)^{1-\sigma_P}\right]^{\frac{1}{1-\sigma_P}}.$$

Next fix value added $v = A_{jt}a\ell^\eta$ and optimize over (m^a, m^m, P^p). If $\mathcal{R}$ denotes gross revenue at the conditional optimum, the first-order conditions imply

$$p_{at}m^a = \gamma_{jt}^a\mathcal{R}, \qquad p_{mt}m^m = \gamma_{jt}^m\mathcal{R}, \qquad \omega_t^P(q)P^p = \gamma_{jt}^P\mathcal{R}. \tag{60}$$

Substitution into the gross-output technology gives

$$\mathcal{R} = \left[p_{jt}Z_{jt}\left(\frac{\gamma_{jt}^a}{p_{at}}\right)^{\gamma_{jt}^a}\left(\frac{\gamma_{jt}^m}{p_{mt}}\right)^{\gamma_{jt}^m}\left(\frac{\gamma_{jt}^P}{\omega_t^P(q)}\right)^{\gamma_{jt}^P}\right]^{\frac{1}{\theta_{jt}}} v. \tag{61}$$

Conditional expenditure on materials and productive producer services is $(1-\theta_{jt})\mathcal{R}$, so revenue net of these inputs is $\theta_{jt}\mathcal{R} = p_{jt}^v(q)v$. Hence the static problem reduces to

$$g_{jt}^*(a,q) = \max_{\ell\geq 0}\left\{p_{jt}^v(q)A_{jt}a\ell^\eta - [w_{jt}^v + c_{jt}^o(q)]\ell\right\}. \tag{62}$$

The unique interior solution satisfies

$$\ell_{jt}^*(a,q) = \left[\frac{\eta p_{jt}^v(q)A_{jt}a}{w_{jt}^v + c_{jt}^o(q)}\right]^{\frac{1}{1-\eta}}, \qquad n_{jt}^*(a,q) = \nu_{jt}\ell_{jt}^*(a,q). \tag{63}$$

Proof of Proposition 1. By the envelope theorem,

$$\begin{aligned}\frac{\partial g_{jt}^*(a,q)}{\partial q} &= v_{jt}^*(a,q)\frac{\partial p_{jt}^v(q)}{\partial q} - \ell_{jt}^*(a,q)\frac{\partial c_{jt}^o(q)}{\partial q}\\ &= v_{jt}^*(a,q)\frac{\partial p_{jt}^v(q)}{\partial q} + \frac{n_{jt}^*(a,q)}{\bar{N}}(rp_t^L - p_t^H),\end{aligned} \tag{64}$$

where the second equality uses

$$\frac{\partial c_{jt}^o(q)}{\partial q} = \frac{\nu_{jt}}{\bar{N}}(p_t^H - rp_t^L).$$

The first-order condition for ℓ implies

$$\eta p_{jt}^v(q)v_{jt}^*(a,q) = [w_{jt}^v + c_{jt}^o(q)]\frac{n_{jt}^*(a,q)}{\nu_{jt}}.$$

Substituting into (64) gives

$$\frac{\partial g_{jt}^*(a,q)}{\partial q} = n_{jt}^*(a,q)\left[\frac{w_{jt}^v + c_{jt}^o(q)}{\eta\nu_{jt}}\frac{\partial \log p_{jt}^v(q)}{\partial q} + \frac{rp_t^L - p_t^H}{\bar{N}}\right] = n_{jt}^*(a,q)\Psi_{jt}(q).$$

Thus, whenever $\Psi_{jt}(q) > 0$, the marginal operating gain from formalization is positive and proportional to employment.

It remains to establish the converse effect from formalization to desired scale. Differentiating (63) gives

$$\frac{\partial \log n^*_{jt}(a,q)}{\partial q} = \frac{1}{1-\eta}\left[\frac{\partial \log p^v_{jt}(q)}{\partial q} + \frac{\nu_{jt}}{\bar{N}}\frac{rp^L_t - p^H_t}{w^v_{jt} + c^o_{jt}(q)}\right]. \tag{65}$$

If $\Psi_{jt}(q) > 0$, then

$$\frac{\partial \log p^v_{jt}(q)}{\partial q} > -\frac{\eta\nu_{jt}}{\bar{N}}\frac{rp^L_t - p^H_t}{w^v_{jt} + c^o_{jt}(q)}.$$

Hence, when $rp^L_t > p^H_t$, the bracket in (65) is strictly positive. Therefore

$$\frac{\partial \log n^*_{jt}(a,q)}{\partial q} > 0.$$

This proves both statements of the proposition. □

D.3.2 Proof of Proposition 2

I first derive the household decision rules used in the proof. Define

$$\zeta_t \equiv 1 + \frac{d_t}{\bar{w}_t}, \qquad \chi_t \equiv \tau_e(S_t).$$

For a household of employment origin i, the time cost of one child receiving education $e_{i,t}$ is

$$T_{i,t}(e_{i,t}) = \tau_i + \chi_t e_{i,t}.$$

Conditional on the aggregate growth and skill environment, the first-order condition for fertility gives

$$n_{i,t} = \frac{\gamma\zeta_t}{\tau_i + \tau_e(S_t)e_{i,t}}. \tag{66}$$

For an interior education choice, the first-order condition is

$$\frac{g_t}{(e_{i,t}+\tau_h)(e_{i,t}+\tau_h+g_t)} = \frac{\tau_e(S_t)}{\tau_i + \tau_e(S_t)e_{i,t}},$$

which rearranges to

$$\tau_e(S_t)(e_{i,t}+\tau_h)^2 = g_t[\tau_i - \tau_e(S_t)\tau_h].$$

Hence

$$e^*_{i,t} = \left[\sqrt{g_t\left(\frac{\tau_i}{\tau_e(S_t)} - \tau_h\right)} - \tau_h\right]_+, \tag{67}$$

where $[x]_+ \equiv \max\{x, 0\}$. Aggregate fertility is

$$\bar{n}_t = \sum_{i\in\{a,m,s\}} \lambda_{i,t} n_{i,t}. \tag{68}$$

Proof of Proposition 2. Fix $S \in (0,1)$ and define

$$t \equiv \tau_e(S), \qquad a \equiv \gamma\zeta(S) - \tau_s, \qquad b \equiv \tau_s - t\tau_h.$$

The maintained interiority conditions imply $a > 0$ and $b > 0$. Equations (67), (66), and (28) reduce in the terminal service economy to

$$g(L) = \left[1 + \lambda_A (SL)^{\phi_c}\right]^{1/\Delta} - 1, \tag{69}$$

$$e(L) = \left[\sqrt{\frac{b}{t} g(L)} - \tau_h\right]_+, \tag{70}$$

$$n(L) = \frac{\gamma\zeta(S)}{\tau_s + te(L)}. \tag{71}$$

Population therefore follows $L_{t+1} = T(L_t)$, where $T(L) = Ln(L)^{1/\Delta}$.

The function $g(L)$ is continuous and strictly increasing from zero to infinity. Hence $e(L)$ is continuous and nondecreasing, while $n(L)$ is continuous and nonincreasing. Moreover,

$$n(0) = \frac{\gamma\zeta(S)}{\tau_s} > 1, \qquad \lim_{L\to\infty} n(L) = 0.$$

There is therefore a unique $L^\infty(S) > 0$ satisfying $n(L^\infty(S)) = 1$.

To establish global stability, first consider an interval in which education is interior. Let $\epsilon_g(L) = d\log g(L)/d\log L$. Concavity of $[1+x]^{1/\Delta} - 1$ implies $0 < \epsilon_g(L) \le \phi_c$. From (70),

$$\frac{de(L)}{d\log L} = \frac{1}{2}[e(L) + \tau_h]\epsilon_g(L).$$

Since $b > 0$ implies $t[e(L) + \tau_h] < \tau_s + te(L)$,

$$-\frac{d\log n(L)}{d\log L} = \frac{t}{\tau_s + te(L)} \frac{de(L)}{d\log L} < \frac{\epsilon_g(L)}{2} \le \frac{\phi_c}{2}.$$

Therefore

$$\frac{d\log T(L)}{d\log L} = 1 + \frac{1}{\Delta}\frac{d\log n(L)}{d\log L} > 1 - \frac{\phi_c}{2\Delta} > 0.$$

Where education is at its corner, $n(L)$ is locally constant and $T(L)$ remains strictly increasing. Thus T is globally increasing. If $L < L^\infty(S)$, then $L < T(L) < L^\infty(S)$; if $L > L^\infty(S)$, then $L > T(L) > L^\infty(S)$. Hence the population sequence is monotone and bounded and converges to $L^\infty(S)$ from every $L_0 > 0$, while fertility converges to replacement.

At the fixed point, (71) implies

$$e^\infty(S) = \frac{\gamma\zeta(S) - \tau_s}{\tau_e(S)} = \frac{a}{t}.$$

Using the interior education rule then gives

$$g^{\infty}(S) = \frac{[a + t\tau_h]^2}{tb} = \frac{[\gamma\zeta(S) - \tau_s + \tau_e(S)\tau_h]^2}{\tau_e(S)[\tau_s - \tau_e(S)\tau_h]} = \mathcal{G}(S),$$

which proves (29). Inverting (69) at $g(L^{\infty}) = \mathcal{G}(S)$ gives (30).

Finally, since $t'(S) = -(1 - \tau_e^s) < 0$, logarithmic differentiation yields

$$\frac{\mathcal{G}'(S)}{\mathcal{G}(S)} = \frac{2\gamma\zeta'(S)}{a + t\tau_h} + t'(S)\left[\frac{2\tau_h}{a + t\tau_h} - \frac{1}{t} + \frac{\tau_h}{\tau_s - t\tau_h}\right]. \tag{72}$$

The first term is nonnegative when $\zeta'(S) \geq 0$. Since $t'(S) < 0$, the second term is strictly positive whenever the bracketed expression is negative. Rearranging this condition gives

$$\tau_h = 0 < \frac{\tau_s[\gamma\zeta(S) - \tau_s]}{\tau_e(S)[2\gamma\zeta(S) - \tau_s]},$$

Hence $\mathcal{G}'(S) > 0$ under the conditions stated in the proposition. □

D.3.3 Proof of Proposition 3

Proof of Proposition 3. Consider first the terminal service economy. For a stationary service establishment, the adaptation cost in (18) and the discounted recurring-coordination gain imply

$$q_f(K) = \min\left\{K, \left[1 - \frac{1}{\mathrm{e}\mathcal{A}_f}\right]_+\right\}.$$

The employment-weighted formalized-task share,

$$x(K) = \frac{1}{N^s}\int n_f q_f(K)\, d\mu^s(f),$$

is therefore weakly increasing in public organizational knowledge. Together with the producer-service demands in (10), the skill-supply rule (24), and labor-market clearing, this implies

$$K_{A,3} > K_{B,3} \quad \Longleftrightarrow \quad S_A^{\infty} > S_B^{\infty}.$$

Proposition 2 then gives

$$g_A^{\infty} > g_B^{\infty} \quad \Longleftrightarrow \quad K_{A,3} > K_{B,3}. \tag{73}$$

It remains to determine when the manufacturing path generates the larger public stock. Under the period-2 restrictions used in the proposition, the creation cutoff and diffusion law imply

$$K_{A,3} > K_{B,3} \quad \Longleftrightarrow \quad \bar{e}_m^o n_m > \bar{e}_c^o n_c.$$

Thus the comparison depends on education embodied in organizational labor and on the firm scale over which newly created procedures are used.

Taking $\tau_h \to 0$ in (67) gives

$$\frac{\bar{e}_m^o}{\bar{e}_c^o} = \left(\frac{g_m}{g_c}\frac{\tau_m}{\tau_s}\right)^{1/2}.$$

With $\Delta = 1$, (28) implies

$$\frac{g_m}{g_c} = (S_1 L_1)^{\phi_m - \phi_c}.$$

Under specialization and symmetry, establishment value is proportional to employment and entry is inversely related to average establishment size. Combining these relationships with (22), under the assumed equality of price and operating-surplus terms across the two paths, gives

$$\frac{n_m}{n_c} = \left(\frac{F_m}{F_c}\right)^{\frac{1}{1+\phi_E}}.$$

Consequently,

$$\frac{\bar{e}_m^o n_m}{\bar{e}_c^o n_c} = \left[\frac{\tau_m}{\tau_s}(S_1 L_1)^{\phi_m - \phi_c}\right]^{1/2} \left(\frac{F_m}{F_c}\right)^{\frac{1}{1+\phi_E}}.$$

The manufacturing path therefore generates more public organizational knowledge if and only if

$$\frac{\tau_m}{\tau_s}(S_1 L_1)^{\phi_m - \phi_c} \left(\frac{F_m}{F_c}\right)^{\frac{2}{1+\phi_E}} > 1,$$

which is exactly condition (31). Combining this result with (73) proves the proposition. □

D.4 Calibration details

This subsection documents target construction, assigned parameters, numerical implementation, fit, local identification, and robustness. The main text reports the parameter estimates and summarizes the fit.

D.4.1 Calibration moments and measurement

The calibration uses 100 moment rows collected in 13 blocks. For moment k in block b, the normalized discrepancy is

$$r_{bk}(\Theta) = \frac{m_{bk}(\Theta) - \widehat{m}_{bk}}{\max\{|\widehat{m}_{bk}|, 0.01\}}. \tag{74}$$

The objective averages squared discrepancies within each block before applying the block weights:

$$\mathcal{L}(\Theta) = \frac{\sum_b \omega_b K_b^{-1} \sum_{k=1}^{K_b} r_{bk}(\Theta)^2}{\sum_b \omega_b}. \tag{75}$$

Averaging within blocks prevents longer paths from mechanically receiving greater weight. Table D.2 lists the moment rows. The peak-date target imposes zero loss for 1930–1960 and a quadratic penalty outside that interval.

Table D.2: Calibration moment blocks

Block	Rows	Measurement
Sectoral employment paths	48	Three sectors at 16 benchmark years
Manufacturing peak share	1	Maximum manufacturing employment share
Manufacturing peak timing	1	Interval target, 1930–1960
Fertility path	16	Aggregate fertility at benchmark years
Sectoral fertility ratio	1	Agriculture relative to services
GDP-per-capita path	18	U.S. series normalized to one in 1850
Sectoral VA productivity growth	3	Annual means, 1970–2010
Aggregate skilled share	1	Employment share, 2019
Sectoral HSPS shares	3	Means over 2015–2020
Aggregate HSPS share	1	Sector-weighted accounting aggregate
Within-VA skilled shares	3	Means over 2015–2020
Skilled earnings premium	1	Skilled/unskilled earnings, 2019
Sectoral firm size	3	Employees per employer firm, 2019
Total	100	

Producer-service employment. Producer and consumer services are distinguished by use, not the supplier's industry. I combine 2019 IPUMS USA microdata with the 2019 BEA Summary Use Table, separating occupations within agriculture, manufacturing, and consumer services into value-added work and organizational or technical producer-service work. Producer-service workers are classified as high skill if they have at least a bachelor's degree. Employment in industries primarily supplying intermediate services is allocated to destination sectors in proportion to BEA intermediate-use expenditures; the household share is assigned to consumer-service value added. The replication files contain the IPUMS–BEA crosswalk and occupational classification.

This allocation assumes that each supplier industry's skill composition and labor requirement per dollar of output are invariant across buyers, because input-output tables do not link transactions to workers' characteristics. Applying the same rule to model and data produces the three HSPS shares and three within-value-added skill shares. The aggregate HSPS moment uses common sector weights and is an accounting check rather than an independent fourth source of variation.

Firm scale. The 2019 Census Statistics of U.S. Businesses targets are employees per employer *firm*: 7.435 in agriculture, 49.694 in manufacturing, and 20.454 in broad services. Accordingly, the model's producing unit is interpreted as an employer firm. Including all service industries makes the service target conservative relative to a narrower employer-sector definition. Historical initial scale is anchored to the 1880 Census of Manufactures. Modern scale moments primarily discipline the three entry costs; entry-supply curvature and ability dispersion are assigned externally.

Productivity, skill, and demand. Sectoral productivity targets are mean annual value-added-per-worker growth over 1970–2010. GDP per capita is normalized to one in 1850, so its path identifies growth rather than a level unit. The terminal skill target is the college-educated employment share; the earnings premium is skilled relative to unskilled earnings. Employment shares at 16 dates over 1850–2010 and the manufacturing peak discipline nonhomothetic demand.

D.4.2 Assigned parameters and normalizations

Table D.3 separates unit choices, external anchors, and maintained benchmarks fixed outside the 19-dimensional calibration. Evidence on related objects anchors empirical magnitudes without supplying direct estimates. Generation length is 20 years (Lagerlöf, 2006), final-good substitution is 0.30 (Comin et al., 2021), and both skill-substitution elasticities are 1.20 (Krusell et al., 2000; Acemoglu and Autor, 2011). Input-output coefficients move gradually from value-added production in 1850 to modern BEA coefficients in 2025, allowing intermediate producer services to emerge endogenously over the transition.

With $\bar{N} = 25$, recurring coordination absorbs 5.42 percent of agricultural employment, 5.04 percent of manufacturing employment, and 5.19 percent of consumer-service employment in 2019. This is comparable in magnitude to the organization-capital evidence in Atkeson and Kehoe (2005), which anchors recurring organizational inputs rather than directly estimating $\bar{N}$.

The diffusion parameter. Let E_t denote the employment-weighted exposure to privately created routines that have not yet entered the public stock. The maintained accumulation law is

$$K_{t+1} = K_t + \min\{1 - K_t, \delta_K E_t\}. \tag{76}$$

Thus δ_K converts current exposure into an annual addition to the stock of publicly usable organizational knowledge.

Diffusion evidence provides an approximate annual scale. Across 12 process innovations in four industries, Mansfield (1961) reports half-adoption times of 0.9–15 years, averaging 7.8 years. Under a constant annual closing rate, $d(T) = 1 - 2^{-1/T}$, so $d(7.8) = 0.085$ and $d(15) = 0.045$. Greenstein (1994) finds that average mainframe-computing practice advanced 14.4 percent annually over 1968–1983 and lagged best practice by at most six to seven years. Slow and fast annual benchmarks of 0.04 and 0.14 imply the midpoint $\delta_K = 0.09$, also close to Mansfield's mean-based rate.

These studies measure adoption of capital-embodied process innovations and installed computer performance, respectively, whereas the model tracks publicly available organizational routines. They therefore anchor the annual scale of diffusion rather than directly estimating δ_K. Section D.4.6 varies δ_K over 0.01–0.15 and changes the accumulation law.

The remaining assigned parameters are held fixed before the joint search. The calibration chooses

$$\Theta = \{\tau, \tau_a^n, \phi_a, \phi_m, \phi_c, g_{GO}, c_{\text{col}}, \alpha_a, \alpha_m, \alpha_c, \epsilon_c, w_a, w_c, R, \xi_a, \xi_c, F_a, F_m, F_c\}. \tag{77}$$

Identification is joint rather than one parameter per moment. Fertility disciplines τ and τ_a^n; GDP and productivity paths discipline ϕ_j and g_{GO}; skill shares and the earnings premium discipline c_{col} and α_j; structural change disciplines demand; HSPS shares discipline R, ξ_a, and ξ_c; and firm size disciplines F_j. In particular, ξ_c is calibrated, not normalized.

D.4.3 Numerical implementation

The model is solved annually over 1790–2289, extending beyond reported outcomes to limit terminal-condition effects on 2019 and policy transitions. Exact calibration uses nine productivity types and 101 organizational-task grid points, jointly solving entry, scale, education, formalization, and public knowledge.

Table D.3: Assigned parameters, external anchors, and normalizations

Parameter	Value	Interpretation or source
A. Units and reference normalizations		
p_m; h_u; β_π	1; 1; 1	Manufacturing-price, unskilled-efficiency, and skill-index units
$\log A_{j0}$	0	Initial value-added productivity normalization
τ_e^u	1	Education-time unit
ϵ_m; w_m	1; 2.5	Manufacturing-demand reference values
κ; ξ_m	1; 1	Task-complexity scale and manufacturing-adaptability normalization
B. Parameters disciplined by external evidence		
β	0.96	Annual discount factor; Kydland and Prescott (1982)
δ_f	0.04	Annual establishment exit rate; Asturias et al. (2023)
η	0.85	Span-of-control curvature; Atkeson and Kehoe (2005)
$\bar{N}$	25	Organizational-scale benchmark; recurring coordination absorbs about 5 percent of employment; Atkeson and Kehoe (2005)
σ_a	0.16	Dispersion of firm ability, calculated from World Bank Enterprise Survey data
ϕ_E	1.552	Entry-supply curvature; Gutiérrez et al. (2021)
σ	0.30	Elasticity of substitution across final goods; Comin et al. (2021)
σ_L; σ_P	1.20; 1.20	Substitution across skill types; Krusell et al. (2000); Acemoglu and Autor (2011)
Δ	20	Generation length in years; Lagerlöf (2006)
λ_A	0.015	Productivity-growth scale; Fernald (2015)
h_s	1.60	Skilled-efficiency scale, benchmarked to the U.S. college earnings premium; Acemoglu and Autor (2011)
δ_K	0.09	Annual diffusion benchmark; Mansfield (1961); Greenstein (1994)
C. Model-specific maintained benchmarks		
σ_{skill}	0.10	Scale of the smooth education-choice margin; McFadden (1974)
τ_e^s	0.50	Skill–education cost complementarity[a]; Cunha and Heckman (2007).
τ_m^n; τ_s^n	1; 1	Common child-time cost outside agriculture[b]
ρ	0.40	Adaptation cost relative to creation cost; sensitivity reported below
γ	0.05	Child-preference weight, benchmarked to an education-expenditure share of approximately 5 percent
ϵ_a	0.01	Near-subsistence agricultural-demand benchmark; Comin et al. (2021)

Notes: Panel A contains unit choices and reference normalizations. Panel B contains parameters fixed before the joint calibration using external empirical evidence. Panel C contains model-specific benchmark values that are not directly estimated from the calibration moments. The citations in Panel C motivate the relevant functional form or empirical scale and are not interpreted as one-for-one estimates of the reported parameter.
The ability dispersion σ_a is calculated from the pooled, employment-weighted dispersion of firm employment in the World Bank Enterprise Surveys. The knowledge parameter δ_K governs the conversion of exposure to privately created routines into additions to the public stock of organizational knowledge; it is not a physical depreciation rate.
[a] Cunha and Heckman (2007) motivate the education-environment complementarity. I set $\tau_e^s = 0.5$ as a conservative benchmark: at the calibrated skilled share of $S \simeq 0.34$, it implies that the time cost of education is only about 17 percent lower than in a fully unskilled environment.
[b] Setting $\tau_m^n = \tau_s^n = 1$ shuts down systematic differences in the fixed time cost of children between manufacturing and service households. This is consistent with the absence of a persistent manufacturing–service fertility gap in the U.S. historical data and with the developing-country evidence discussed by Peters et al. (2026).

Strictly positive parameters are searched in logs using multiple starts and damped local updates, followed by exact-grid evaluation. Candidates must converge on the exact grid and keep every 2019 sectoral firm-size error below three percent. The selected vector has $\mathcal{L} = 0.011878$ and maximum firm-size error 0.08 percent. The cold-start fixed point is accepted after 30 iterations; the last discrete-formalization

update is 0.00646, below the three-grid-cell epsilon-equilibrium tolerance of 0.03, and the 2019 update is 0.00171. Education-budget and size-accounting residuals are zero to machine precision.

All 12 independent cold starts converge to the same exact-grid equilibrium within solver tolerance. Calibration and cross-country counterfactuals are also solved independently on the exact grid. Finite differences perturb the converged benchmark, using its solution as the common initialization for nearby parameters.

D.4.4 Detailed fit

Figure D.1 compares targeted HSPS shares, firm size, and value-added productivity growth. Agricultural, manufacturing, and service HSPS shares are 0.325, 0.461, and 0.399 in both model and data at reported precision; the maximum discrepancy is 0.04 percentage points. Model firm sizes of 7.44, 49.72, and 20.45 workers are within 0.08 percent of their targets. Annual productivity growth is 3.85, 2.98, and 1.31 percent, versus 3.71, 3.17, and 1.42 percent in the data.

Untargeted employment-weighted standard deviations of formalization are 0.176 in agriculture, 0.187 in manufacturing, and 0.049 in services, versus data-based proxies of 0.229, 0.205, and 0.211. Without sector-specific dispersion parameters, the model reproduces 76.8, 91.1, and 23.1 percent of the respective dispersions.

D.4.5 Local identification and uncertainty

Let $r(\theta)$ denote the vector of weighted residuals and let $\theta = \log \Theta$ denote the 19-dimensional vector of calibrated parameters. I compute the Jacobian

$$J = \frac{\partial r(\theta)}{\partial \theta'} \tag{78}$$

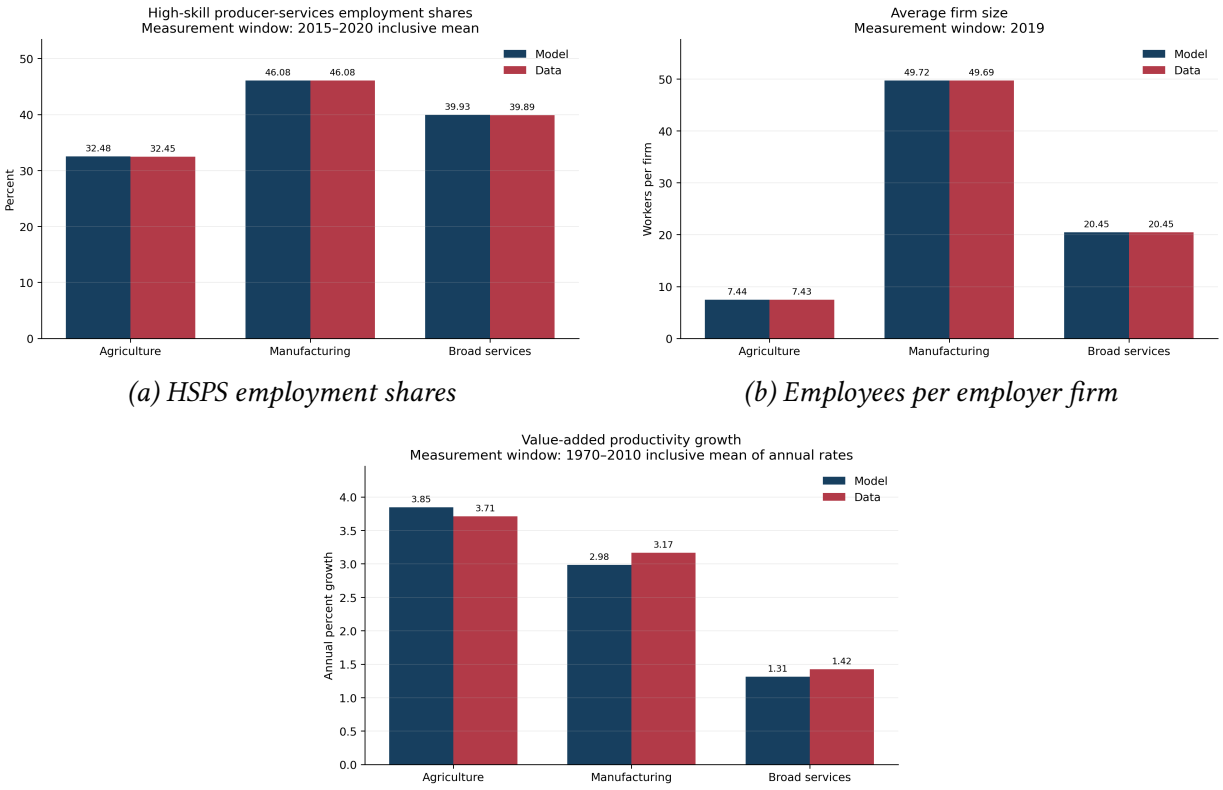
using exact-grid central differences around the frozen benchmark equilibrium. Every perturbation uses the same benchmark-equilibrium anchor. For parameter p and step h, the numerical derivative is

$$J_{\cdot p}(h) = \frac{r(\theta + he_p) - r(\theta - he_p)}{2h}. \tag{79}$$

I use $h = 0.025$ and $h = 0.05$. All 77 required exact simulations—one baseline and four perturbations for each of 19 parameters—converge.

The 100-by-19 objective Jacobian has full column rank at both steps. At the preferred five-percent step, its condition number is 952.1 and its effective rank is 17 at a one-percent relative singular-value cutoff and 19 at a 0.1-percent cutoff. Removing the locally flat peak-timing penalty, aggregate-HSPS identity, and one redundant sectoral employment share per date gives a smooth 82-by-19 Jacobian with full column rank, condition number 789.9, and the same effective ranks. Thus two local directions are weak, but no calibrated parameter is exactly redundant. Each column retains at least 14.0 percent of its norm after projection on the other 18. The identified subspace is stable across step sizes except in the weakest direction.

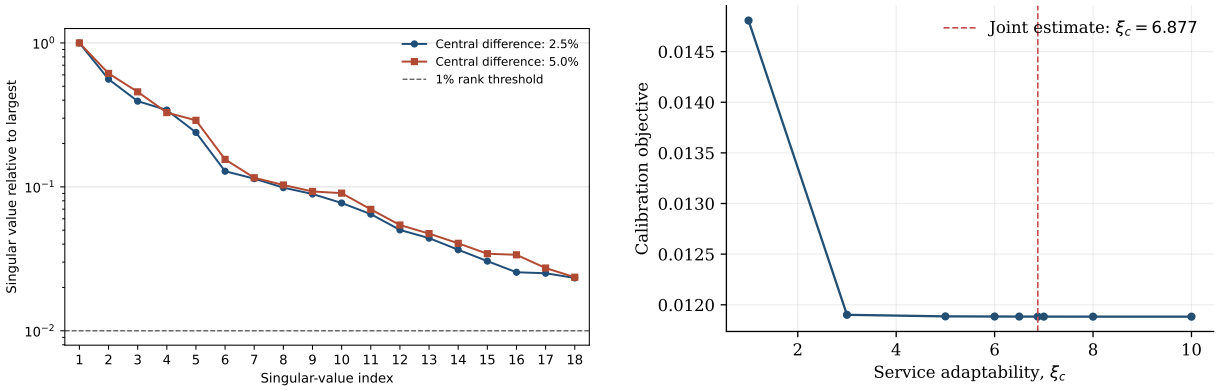
Service adaptability illustrates the difference between identification and precision. The exact objective falls from 0.014807 at $\xi_c = 1$ to 0.011901 at $\xi_c = 3$ and 0.0118833 at the calibrated $\xi_c = 6.877$. It is



(a) HSPS employment shares *(b) Employees per employer firm*

(c) VA productivity growth

Figure D.1: Targeted calibration fit. Bars compare the calibrated model with the corresponding U.S. targets. HSPS shares are means over 2015–2020, firm size is measured in 2019, and productivity growth is the annual average over 1970–2010.



(a) Singular spectrum *(b) Objective slice for ξ_c*

Figure D.2: Identification diagnostics. Panel (a) reports singular values of the exact weighted-moment Jacobian. Panel (b) evaluates the exact objective over ξ_c while holding the other 18 calibrated parameters at their reported values.

nearly flat in the saturation region: 0.0118837 at $\xi_c = 6.5$, 0.0118832 at 7, and 0.0118829 at 8. Service-formalization moments reject low adaptability but do not precisely locate ξ_c within this region, explaining its large local standard error.

Table D.5 reports diagnostics from the smooth, nonredundant system. The unique share divides a Jacobian column's norm after projection on other columns by its original norm. HC1 residual-based minimum-distance standard errors are expressed as percentages of estimates. They summarize local curvature and residual fit conditional on the target vector, not sampling confidence intervals for the heterogeneous data sources.

For formal SMM inference, let Ω denote the sampling covariance matrix of the empirical moments and W the calibration weight matrix. The corresponding sandwich covariance is

$$\mathrm{Avar}(\widehat{\theta}) = (J'WJ)^{-1}J'W\Omega WJ(J'WJ)^{-1}. \tag{80}$$

Estimating Ω must preserve serial dependence and cross-moment covariance within each historical, administrative, or microdata source. The rank, objective-slice, and residual-based exercises establish local numerical identification of the 19 parameters without treating heterogeneous moment rows as a common random sample.

D.4.6 Robustness and Sensitivity

Two exercises assess the manufacturing–development relationship: 18 alternative specifications and a finer grid for knowledge diffusion δ_K. Parameters not varied remain calibrated, without recalibration. Slopes use the same five economies, differing only in manufacturing-demand weights. Every specification preserves the association of a higher manufacturing peak with higher GDP per capita, faster GDP-per-capita growth, greater tertiary-education and HSPS shares, and lower population growth. Magnitudes respond most to diffusion and firm scale, and less to adaptation costs, producer-service substitution, and the accumulation law.

Table D.6 reports slope changes relative to the benchmark. All 90 slopes across 18 specifications retain their signs. Median absolute changes are 2.9 percent for GDP levels, 8.5 percent for GDP growth, 10.2 percent for tertiary education, 2.4 percent for HSPS, and 4.3 percent for population growth. Excluding zero diffusion and equal sector weights, the largest absolute changes are 15.1, 17.5, 27.3, 16.4, and 13.7 percent, respectively.

Removing diffusion attenuates GDP-level and GDP-growth slopes by 30.4 and 33.8 percent, tertiary education by 77.9 percent, HSPS by 56.6 percent, and the magnitude of population growth by 34.3 percent. Equal sector weights reduce education and HSPS slopes by 38.5 and 24.0 percent and GDP levels and growth by 16.7 and 13.4 percent. Diffusion therefore matters particularly for education and organizational formalization, although preserved signs without diffusion show that spillovers are not the sole source of these relationships.

Firm-scale assumptions change magnitudes without reversing signs. Varying span of control changes GDP-level slopes by -10.4 to 15.1 percent and education slopes by -12.7 to 27.2 percent. Lower ability dispersion raises GDP-growth and education slopes by 17.5 and 27.3 percent. Reducing routine repetition by 25 percent raises the growth slope by 16.3 percent but lowers HSPS by 16.4 percent. Initial knowledge $K_0 = 0.10$ attenuates education by 24.4 percent, GDP levels by 5.0 percent, and HSPS by 3.1 percent. Education is thus more sensitive than GDP levels to several organizational assumptions.

In $\tau_e(S) = 1 - S(1 - \tau_e^s)$, lower τ_e^s strengthens the fall in education costs with skill. Relative to the benchmark 0.5, setting $\tau_e^s = 0.25$ lowers the education slope by 11.6 percent and the population-growth magnitude by 9.1 percent; setting it to 0.75 raises them by 8.8 and 8.3 percent. GDP-level, GDP-growth,

and HSPS slopes change by at most 3.5 percent. This complementarity chiefly affects education and demographics over the tested range.

Changing adaptation costs by 25 percent moves every slope by less than 4 percent and GDP levels by less than 1 percent. Producer-service substitution of 0.90 or 1.50 changes slopes by at most 1.3 percent. Replacing (76) with an exponential-hazard law having the same first-order flow changes every slope by less than 0.7 percent. Results are therefore insensitive to the benchmark's hard cap and only modestly sensitive to adaptation costs over these ranges.

The finer diffusion exercise varies δ_K from 0.01 to 0.15 in steps of 0.01. Table D.7 reports slopes for a 10-percentage-point increase in peak manufacturing employment. Responses are nonmonotonic, with several slopes largest near the benchmark. Very slow diffusion chiefly attenuates skill and HSPS relationships.

Over the empirically motivated interval $[0.04, 0.14]$, slope changes relative to $\delta_K = 0.09$ range from -13.8 to 0.6 percent for GDP levels, -6.9 to 5.0 percent for GDP growth, -27.8 to 4.1 percent for tertiary education, -22.4 to 1.6 percent for HSPS, and -16.2 to 0.6 percent for the population-growth magnitude. Over $[0.01, 0.15]$, maximum attenuations are 28.2, 30.1, 73.7, 49.8, and 32.8 percent, respectively. Every sign is preserved; slow diffusion weakens education and organizational formalization more than the GDP relationship.

Table D.4: Parameter estimates and reported calibration fit

Panel A: Jointly estimated parameters and principal disciplines

Parameter	Estimate	Principal empirical discipline
Child-rearing time cost, τ	0.03236	Aggregate fertility path
Agricultural fertility shifter, τ_a^n	0.44017	Agriculture-to-services fertility ratio
Agricultural growth elasticity, ϕ_a	0.99840	Agricultural VA productivity growth
Manufacturing growth elasticity, ϕ_m	0.91919	Manufacturing VA productivity growth
Service growth elasticity, ϕ_c	0.69046	Service VA productivity growth
Gross-output productivity growth, g_{GO}	0.00318	GDP-per-capita path and sectoral growth
Skill-acquisition cost, c_{col}	0.56821	Aggregate skilled-employment share
Agricultural skill weight, α_a	0.24338	Skilled share within agricultural VA
Manufacturing skill weight, α_m	0.33001	Skilled share within manufacturing VA
Service skill weight, α_c	0.40995	Skilled share within service VA
Service nonhomotheticity, ϵ_c	1.01048	Sectoral employment paths
Agricultural demand weight, w_a	0.65105	Sectoral employment paths
Service demand weight, w_c	3.82326	Sectoral employment paths
Discounted repetitions, R	22.78696	Sectoral HSPS employment shares
Agricultural adaptability, ξ_a	3.56402	Agricultural HSPS employment share
Service adaptability, ξ_c	6.87739	Service HSPS employment share
Agricultural setup-cost shifter, F_a	25.872	Agricultural firm size
Manufacturing setup-cost shifter, F_m	187.625	Manufacturing firm size
Service setup-cost shifter, F_c	85.117	Service firm size

Panel B: Reported scalar targets and model outcomes

Moment	Data	Model
Agricultural HSPS share	0.325	0.325
Manufacturing HSPS share	0.461	0.461
Service HSPS share	0.399	0.399
Agricultural firm size	7.435	7.44
Manufacturing firm size	49.694	49.72
Service firm size	20.454	20.45
Agricultural VA productivity growth (%)	3.71	3.85
Manufacturing VA productivity growth (%)	3.17	2.98
Service VA productivity growth (%)	1.42	1.31
Manufacturing peak employment share (%)	28.28	24.93
Manufacturing peak year	1950	1928

Notes: Panel A reproduces the 19 estimates in Table 2 and their principal disciplines; the parameters are estimated jointly, so these are not one-parameter–one-moment mappings. Panel B collects the numerical fit values reported in the main text. HSPS shares are fractions and use 2015–2020 means; firm size is employees per employer firm in 2019; productivity growth is the annual mean over 1970–2010. Entries retain the source's reported precision. The observed manufacturing peak is in 1950, while the estimation loss assigns zero timing penalty to dates from 1930 through 1960. Figures 5, 4, and D.1 report the corresponding paths and graphical fit. Other target blocks and their measurement are listed in Table D.2.

Table D.5: Parameter-level local identification diagnostics

Parameter	Estimate	Unique share (%)	HC1 relative SE (%)
τ	0.03236	17.92	4.86
τ_a^n	0.44017	30.66	5.92
ϕ_a	0.99840	15.49	10.15
ϕ_m	0.91919	14.04	10.38
ϕ_c	0.69046	15.97	10.86
g_{GO}	0.00318	61.33	24.31
c_{col}	0.56821	35.90	5.36
α_a	0.24338	81.42	2.81
α_m	0.33001	84.90	2.96
α_c	0.40995	82.84	2.62
ϵ_c	1.01048	24.87	30.15
w_a	0.65105	34.49	12.44
w_c	3.82326	25.57	39.53
R	22.78696	33.06	5.33
ξ_a	3.56402	64.17	88.56
ξ_c	6.87739	70.22	491.31
F_a	25.872	80.00	48.09
F_m	187.625	69.56	60.27
F_c	85.117	91.00	22.85

Notes: The unique shares use the exact, smooth, nonredundant 82-by-19 Jacobian. HC1 standard errors are computed in log-parameter space and translated to levels by the delta method.

Table D.6: Sensitivity of cross-country slopes

Specification	GDP level	GDP growth	Tertiary share	HSPS share	Population growth
Adaptation cost: -25%	0.8	1.4	3.3	1.8	0.6
Adaptation cost: $+25\%$	0.1	2.5	3.6	0.4	0.5
Knowledge diffusion: -25%	2.4	7.7	11.3	4.6	2.3
Knowledge diffusion: $+25\%$	-4.4	-11.2	-20.8	-8.4	-4.6
Routine repetition: -25%	-7.0	16.3	4.2	-16.4	-8.2
Routine repetition: $+25\%$	1.8	-11.1	-9.0	5.5	3.0
Span of control: 0.80	15.1	13.0	27.2	0.8	13.7
Span of control: 0.90	-10.4	-9.8	-12.7	2.6	-12.0
Producer-service substitution: 0.90	0.1	0.2	0.0	0.2	0.0
Producer-service substitution: 1.50	-0.4	-0.9	-1.3	-0.8	-0.4
Firm-ability dispersion: -25%	5.0	17.5	27.3	6.9	4.0
Firm-ability dispersion: $+25\%$	-2.3	-5.6	-4.4	-2.1	-1.6
Education complementarity: $\tau_e^s = 0.25$	3.5	-2.5	-11.6	-2.0	-9.1
Education complementarity: $\tau_e^s = 0.75$	-2.0	-0.5	8.8	1.6	8.3
Initial knowledge: $K_0 = 0.10$	-5.0	-9.2	-24.4	-3.1	-6.0
Equal sector weights	-16.7	-13.4	-38.5	-24.0	-20.1
No knowledge spillovers	-30.4	-33.8	-77.9	-56.6	-34.3
Exponential-hazard accumulation	-0.1	0.5	0.6	0.0	-0.2

Notes: Entries are percentage changes in the absolute magnitude of the cross-country slope relative to the benchmark. Positive entries indicate a stronger relationship; negative entries indicate attenuation. All slopes retain their benchmark sign. Outcomes are log GDP per capita in 2019, annual GDP-per-capita growth over 2010–2019, average tertiary-education and HSPS shares over 2010–2019, and cumulative log population growth over 1850–2019. Each specification uses the same five manufacturing-demand economies. Parameters not listed in a row remain fixed. The benchmark and 18 alternatives comprise 95 economies, all of which satisfy the numerical convergence criterion. This broader exercise uses a coarser numerical approximation than Table D.7; all percentage changes are calculated relative to the benchmark computed with the same approximation.

Table D.7: Cross-country slopes by the rate of knowledge diffusion

δ_K	Log GDPpc	GDPpc growth	College share	HSPS share
0.01	0.1312	0.00112	0.0119	0.0333
0.02†	0.1230	0.00105	0.0081	0.0291
0.03†	0.1322	0.00128	0.0164	0.0364
0.04†	0.1477	0.00139	0.0222	0.0450
0.05†	0.1604	0.00151	0.0281	0.0520
0.06	0.1676	0.00157	0.0308	0.0558
0.07	0.1677	0.00155	0.0308	0.0564
0.08	0.1725	0.00156	0.0320	0.0588
0.09	0.1714	0.00150	0.0307	0.0579
0.10	0.1708	0.00148	0.0299	0.0574
0.11	0.1693	0.00147	0.0303	0.0569
0.12	0.1682	0.00145	0.0297	0.0564
0.13	0.1667	0.00143	0.0287	0.0558
0.14	0.1652	0.00140	0.0278	0.0553
0.15	0.1636	0.00137	0.0265	0.0541

Notes: Each entry is the OLS slope associated with a 10-percentage-point increase in the endogenous peak manufacturing employment share across five manufacturing-demand economies. All other parameters remain at their calibrated values. GDP per capita is measured in 2019; growth and education and HSPS shares refer to 2010–2019. Growth rates and shares are expressed as fractions. The population-growth slope, omitted from the table, is negative at every value of δ_K. †In one economy in each marked row, the discrete formalization decision continues to oscillate beyond the numerical tolerance, although the residuals for knowledge, employment, and establishment counts are below 8.5×10^{-4}. The reported values retain the same solution procedure across specifications. As an additional check, using a neighboring economy's solution as the starting point at $\delta_K = 0.02$ changes the four reported slopes to $(0.1188, 0.00105, 0.0082, 0.0291)$ and preserves all five signs. The precise magnitudes in the marked rows should therefore be interpreted with this numerical qualification.

D.4.7 Interpreting the manufacturing demand counterfactual

Varying manufacturing demand w_m changes manufacturing's relative pull during the transition and its endogenous employment peak, holding technology, fertility and education preferences, and production primitives fixed.

This counterfactual has an equivalent price-wedge representation. Final demand over agriculture, manufacturing, and consumer services implies relative demand terms of the form

$$K_a = \left(\frac{p_a}{p_m}\right)^{1-\sigma} Y^{(1-\sigma)(k_a-1)} \frac{w_a}{w_m^{k_a}}, \qquad K_c = \left(\frac{p_c}{p_m}\right)^{1-\sigma} Y^{(1-\sigma)(k_c-1)} \frac{w_c}{w_m^{k_c}},$$

where $k_j \equiv \varepsilon_j/\varepsilon_m$. Changing the manufacturing demand weight from w_m to $f_m w_m$ therefore changes the relative demand terms to

$$K_a(f_m) = K_a(1) f_m^{-k_a}, \qquad K_c(f_m) = K_c(1) f_m^{-k_c}.$$

Now consider an economy with the original w_m, but in which households face effective final-good prices

$$\widetilde{p}_{a,t} = (1+\tau_a) p_{a,t}, \qquad \widetilde{p}_{m,t} = p_{m,t}, \qquad \widetilde{p}_{c,t} = (1+\tau_c) p_{c,t}.$$

The wedge economy generates the same relative demand schedules as the $f_m w_m$ economy if

$$(1+\tau_a)^{1-\sigma} = f_m^{-k_a}, \qquad (1+\tau_c)^{1-\sigma} = f_m^{-k_c}.$$

Equivalently,

$$\boxed{1+\tau_a = f_m^{-k_a/(1-\sigma)}, \qquad 1+\tau_c = f_m^{-k_c/(1-\sigma)}.} \tag{81}$$

The w_m counterfactual can therefore be represented as either a manufacturing-demand shifter or effective final-good prices. This equivalence concerns household relative-demand schedules conditional on income and producer prices; it does not require a primitive preference interpretation. Table D.8 maps the selected economies in Figure 7 into price wedges.

Table D.8: Manufacturing-demand counterfactuals and equivalent final-consumption wedges

w_m	f_m	Peak manufacturing employment (%)	Agriculture wedge τ_a (%)	Consumer-services wedge τ_c (%)
$0.60^{\dagger}$	0.240	11.19	2.06	684.67
1.16	0.464	15.92	1.10	202.97
1.80	0.720	20.62	0.47	60.67
2.50 (benchmark)	1.000	24.93	0.00	0.00
2.56	1.024	25.27	-0.03	-3.37
3.49	1.396	29.88	-0.48	-38.22
4.69	1.876	34.56	-0.89	-59.67
6.10	2.440	38.80	-1.27	-72.41

Notes: The calibrated benchmark is $w_m^* = 2.5$, and $f_m \equiv w_m/w_m^*$. The wedges are computed from $1+\tau_j = f_m^{-k_j/(1-\sigma)}$, using $\sigma = 0.30$, $k_a = 0.0100$, and $k_c = 1.0105$. The normalization holds the effective manufacturing price fixed. Positive values raise the effective final-consumption price relative to manufacturing; negative values lower it. Peak manufacturing employment is endogenous. The wedges are demand-equivalent shadow prices, not statutory tax rates.

The asymmetry between the two wedges follows directly from the calibrated demand system. For agriculture, the relevant exponent is

$$-\frac{k_a}{1-\sigma} = -0.0143,$$

so even large changes in w_m imply small movements in the effective agricultural price. For consumer services, the corresponding exponent is

$$-\frac{k_c}{1-\sigma} = -1.4435.$$

The service-price mapping is therefore substantially more responsive and becomes nonlinear as f_m falls. For example, the $w_m = 1.16$ counterfactual implies $1 + \tau_c = 3.03$: under the normalization in the table, the effective consumer-service price is approximately three times its benchmark level. At $w_m = 1.80$, the corresponding multiplier is 1.61.

These shadow prices summarize combinations of market access, distribution costs, markups, regulation, product availability, relative quality, and other pecuniary or nonpecuniary forces producing the same relative-demand schedules. The counterfactuals introduce no government revenue or tax rebates, and the wedges play no independent calibration role. Endogenous manufacturing peaks discipline the variation; the wedge mapping interprets the reduced-form demand differences.

D.5 Policy extension

This subsection documents the LMIC calibration and the implementation of the policy experiments in Section 6. Parameters not listed below are inherited from the benchmark model.

D.5.1 Distortions and LMIC calibration

Before drawing permanent ability $a \sim G$, an entrant chooses setup intensity $h \in [\underline{h}, 1]$, which determines effective productivity $x = ha$. In manufacturing and consumer services, the productivity-correlated distortion is

$$1 - \tau_{i,t} = \kappa_{j,t} x_{i,t}^{-d_j}, \qquad x_{i,t} = h_{i,t} a_i, \qquad j \in \{m, c\}, \tag{82}$$

where $\kappa_{j,t}$ sets the revenue-weighted mean of $\tau_{i,t}$ to zero. Setup intensity requires

$$e(h, d_j) = \exp\{-\lambda_E d_j (1 - h)\} \tag{83}$$

units of the benchmark setup input. We set $d_m = d_c = 0.41$, following the excess productivity-correlated distortion measured by Bento and Restuccia (2017, 2021), and $\lambda_E = 12$. The permanent-reform experiment reduces d_m and d_c linearly during 2026–2035 and sets them to zero thereafter.

The joint LMIC parameter vector contains the organizational-scale normalization, three sectoral entry-cost shifters, three sectoral productivity-growth parameters, the manufacturing-demand shifter, and the skill-acquisition cost. The selected vector has $\bar{N} = 3$ and retains the externally disciplined LMIC productivity dispersion of 0.16. The historical establishment-count initialization from the benchmark transition is retained. Table D.9 reports the moments used in the calibration and their model counterparts.

Table D.9: Calibration of the LMIC policy economy

Block	Target moment	Data	Model
Productivity	Agricultural VA productivity growth, 1990–2018 (%)	2.620	2.716
	Manufacturing VA productivity growth, 1990–2018 (%)	2.013	2.084
	Services VA productivity growth, 1990–2018 (%)	1.111	1.145
Firm scale	Agricultural average establishment size, 2019	2.220	2.123
	Manufacturing average establishment size, 2019	6.238	5.864
	Services average establishment size, 2019	1.883	1.714
Education	Tertiary-skilled employment share, 2019 (%)	12.000	11.980
Demand	Peak manufacturing employment, 1990–2018 (%)	11.713	10.442
Organization	High-skill producer-service employment, 2019 (%)	1.326	1.352

Notes: Productivity growth and peak manufacturing employment use the GGDC/UNU–WIDER Economic Transformation Database. The firm-size targets use the LMIC target construction based on Bento and Restuccia (2021); the agricultural entry is a low-scale proxy because harmonized agricultural establishment employment is unavailable for the same country sample. The high-skill producer-service target is the unweighted median employment share of university-educated workers in ISCO groups 1–4 and the maintained producer-service industries across the ten available LMIC censuses in the IPUMS sample. The denominator is total employment. The model counterpart is total high-skill producer-service employment.

D.5.2 Policy implementation

Four programs operate during 2026–2045. The college scholarship reimburses part of the private cost of acquiring skill. Firm-formalization support reimburses part of the one-time cost of creating an organizational procedure. The mixed package applies both reimbursements at equal statutory rates. The manufacturing wage subsidy reimburses part of the skilled and unskilled manufacturing wage bill. Table D.10 reports the rates used in each distortion environment.

The scholarship operates through the existing education-choice margin. Let c_t be the calibrated cost of acquiring skill, τ_t^e the scholarship rate, and p_t^e the skilled share of the entering cohort. The calibrated cost is treated as pecuniary tuition, so

$$p_t^e = \Lambda\left(\frac{\beta_\pi \log(w_t^s/w_t^u) - (1-\tau_t^e)c_t}{\sigma_e}\right). \tag{84}$$

To convert the choice-index cost into monetary resources, one unit of c_t is assigned $\kappa_{\text{edu}} = 1.696$ annual unskilled-wage units. The conversion uses the tuition coefficient in Cordeiro and Cox (2023), education-choice dispersion $\sigma_e = 0.10$, eight semester payments, and mean earnings for workers with completed secondary education (Instituto Brasileiro de Geografia e Estatistica, 2020):

$$\kappa_{\text{edu}} = \frac{1000 \sum_{j=0}^{7} 0.96^{j/2}}{0.10 \times 2.049 \times (12 \times 1788)} = 1.696. \tag{85}$$

With cohort mass L_t/g and $g = 20$, government scholarship spending is

$$G_t^e = \tau_t^e \kappa_{\text{edu}} c_t w_t^u \left(\frac{L_t}{g}\right) p_t^e. \tag{86}$$

All eligible skilled entrants receive the reimbursement, including inframarginal entrants. At zero subsidy, tuition is already included in calibrated household final expenditure. Under the policy, the government pays the subsidized share and finances it with a contemporaneous lump-sum household tax. The welfare calculation therefore does not count private tuition twice.

Table D.10: Temporary-policy rates

Policy	Distortions remain	Full reform
College scholarship	32.0%	29.0%
Firm-formalization support	90.0%	90.0%
Mixed package: scholarship/formalization	36.0%/36.0%	27.5%/27.5%
Manufacturing wage subsidy	6.702%	6.702%

Notes: The mixed-policy rates apply separately to tuition and procedure-creation costs; equal rates do not imply equal realized spending. Rates are chosen using equilibrium take-up to keep present-value expenditure near 0.6 percent of discounted policy-economy GDP. Formalization support is evaluated at the highest stable reimbursement rate used in the exercise.

Firm-formalization support reimburses the stated share of the one-time procedure-creation cost; firms continue to pay for the producer-service labor used to operate installed procedures. The manufacturing subsidy applies to the contemporaneous skilled and unskilled manufacturing wage bill. All government outlays are financed by contemporaneous lump-sum household taxes. Present-value fiscal cost is discounted government spending divided by discounted GDP during 2026–2045, and consumption-equivalent variation is evaluated over 2026–2089.

D.5.3 Withdrawal and outcome construction

Withdrawal is unexpected. Agents first solve the 2026–2045 equilibrium believing that the policy will continue. At the start of 2046, the policy rate is set to zero and the continuation begins from the inherited distribution of firms, setup intensities, installed procedures, private localized knowledge, and public organizational knowledge. The permanent-policy and withdrawal solutions are identical through 2045. Each policy path is compared with a no-temporary-policy path under the same permanent-distortion environment.